

# Flip-width: Cops and Robber on dense graphs

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**Abstract**—We define new graph parameters, called *flip-width*, that generalize treewidth, degeneracy, and generalized coloring numbers for sparse graphs, and clique-width and twin-width for dense graphs. The flip-width parameters are defined using variants of the Cops and Robber game, in which the robber has speed bounded by a fixed constant  $r \in \mathbb{N} \cup \{\infty\}$ , and the cops perform flips (or perturbations) of the considered graph. We then propose a new notion of tameness of a graph class, called *bounded flip-width*, which is a dense counterpart of classes of bounded expansion of Nešetřil and Ossona de Mendez, and includes classes of bounded twin-width of Bonnet, Kim, Thomassé, and Watrigant. This unifies Sparsity Theory and Twin-width Theory, for the first time providing a common language for studying the central notions of the two theories, such as weak coloring numbers and twin-width – corresponding to winning strategies of one player – or dense shallow minors, rich divisions, or well-linked sets, corresponding to winning strategies of the other player. To demonstrate the robustness of the introduced notions, we prove that boundedness of flip-width is preserved by first-order interpretations, or transductions, generalizing previous results concerning classes of bounded expansion and bounded twin-width. We also show that the considered notions are amenable to algorithms, by providing an algorithm approximating the flip-width of a given graph, which runs in slicewise polynomial time (XP) in the size of the graph. Finally, we propose a more general notion of tameness, called *almost bounded flip-width*, which is a dense counterpart of nowhere dense classes. We conjecture, and provide evidence, that classes with almost bounded flip-width coincide with monadically dependent (or monadically NIP) classes, introduced by Shelah in model theory. We also provide evidence that classes of almost bounded flip-width characterise the hereditary graph classes for which the model-checking problem is fixed-parameter tractable, which is of central importance in structural and algorithmic graph theory.

## I. INTRODUCTION

A recent focus of algorithmic and structural graph theory, and of finite model theory, is to find graph parameters that extend the parameters used in the context of sparse graphs, to the dense setting. More generally, the goal is to extend the Sparsity theory of Nešetřil and Ossona de Mendez [NdM12] to dense graph classes. Two central parameters used in the sparse setting are treewidth and degeneracy; both have found numerous applications in algorithms and combinatorics. Whereas the dense analogue of treewidth – clique-width, or rank-width – is well-understood, there is not even a clear

candidate for the dense analogue of degeneracy<sup>1</sup>. Recall that a graph has degeneracy at most  $d$  if its vertices can be totally ordered so that every vertex has at most  $d$  neighbors before it in the order. Generalized coloring numbers are related parameters, specified by a radius  $r$ , which impose restrictions on neighborhoods of radius  $r$ , degeneracy being the case of radius  $r = 1$ . Sparsity theory is a very successful theory studying classes of sparse graphs in which degeneracy and generalized coloring numbers play a central role. The fundamental notions of this theory are two tameness conditions for graph classes: *bounded expansion* and *nowhere denseness*. A class of graphs has bounded expansion if and only if each generalized coloring number is bounded by a constant (depending on the radius), on all graphs in the class. Examples include every class with bounded maximum degree, the class of planar graphs, classes of bounded treewidth, and every class that excludes some graph as a minor or as a topological minor. The more general nowhere dense classes are characterized analogously, with the constant bound on the generalized coloring numbers of  $n$ -vertex graphs replaced with  $O(n^\epsilon)$ , for any fixed  $\epsilon > 0$ . In particular, nowhere dense classes are sparse – every  $n$ -vertex graph in such a class has  $O(n^{1+\epsilon})$  edges for any fixed  $\epsilon > 0$ , and every nowhere dense class excludes some biclique  $K_{t,t}$  as a subgraph. Classes with bounded expansion and nowhere dense classes can be characterized in many other ways, in terms of their remarkable combinatorial, algorithmic, and logical properties, yielding multiple applications in those areas.

One of the driving questions in this line of work, on the algorithmic side, is to characterize those graph classes for which the model-checking problem for first-order logic is fixed-parameter tractable: there is an algorithm that determines whether a given graph  $G$  from the class satisfies a given first-order sentence  $\varphi$  in time  $f(\varphi) \cdot |G|^c$ , for some constant  $c$  and function  $f$  that depend only on the class. Such a characterization is known in the special case of *monotone* graph classes, that is, graph classes that are closed under removing vertices and edges. In a landmark result, Grohe, Kreutzer and Siebertz [GKS17] proved that a monotone graph class has fixed-parameter tractable model-checking if and only if it is nowhere dense (under a complexity-theoretic assumption,  $\text{FPT} \neq \text{AW}[*]$ ).

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<sup>1</sup>We are only aware of the recent notion of graph functionality [AAL21] as a possible candidate; see Sec. V-E.

There is an ongoing effort to lift Sparsity theory to *hereditary* graph classes, that is, classes that are closed under removing vertices. For example, the class of cliques is hereditary and combinatorially and logically very simple, but lies outside of the realm of Sparsity theory, which is only suited to the study of monotone graph classes. Indeed, all notions studied in Sparsity theory – generalized coloring numbers, bounded expansion, nowhere denseness, etc. – are monotone under edge removals. There are many other known hereditary graph classes that are well-behaved in a similar way to classes with bounded expansion and nowhere dense classes, but are not monotone, and are not sparse (e.g. contain arbitrarily large cliques). Those include for instance classes of bounded clique-width, the class of unit interval graphs, or proper hereditary classes of permutation graphs – which are all subsumed by the recently introduced classes of bounded twin-width [BKTW20] (see below) – as well as *structurally nowhere dense* classes [DMS23] – classes of graphs that can be obtained from a nowhere dense graph class by redefining the edges using a fixed first-order formula  $\varphi(x, y)$  – for instance, the edge-complements of graphs from a nowhere dense class, or the squares of graphs from a nowhere dense class.

The developments in Structural graph theory – where many results concerning graph classes of bounded treewidth are extended to the setting of classes of bounded clique-width – serves as an inspiration in attempts of lifting the results of Sparsity theory from the sparse (monotone) setting to the dense (hereditary) setting. It is expected that the fundamental notions of Sparsity theory – bounded expansion and nowhere denseness – should extend to more general tameness conditions for graph classes that are possibly dense, similarly as treewidth extends to clique-width. However, currently, even the most fundamental questions remain unanswered: What is the dense analogue of degeneracy? Of generalized coloring numbers? The pursuit after such notions has been a driving factor, and a major open problem in the area (see *Related Work* below). To date, no such combinatorial notions, with compelling evidence of their utility, have been proposed.

The recent and already very successful Twin-width theory, developed by Bonnet, Thomassé, and coauthors, [BKTW20], [BGK<sup>+</sup>21a], [BGK<sup>+</sup>21b], [BGOdM<sup>+</sup>22], [BGOdMT23], [BKRT22], [BGT22], [BCK<sup>+</sup>22], [GPT22], [GPPT22], provides a robust tameness condition for graph classes that are not necessarily sparse, classes of *bounded twin-width*. Those include many studied sparse graph classes, such as classes that exclude a fixed minor, as well as dense graph classes, such as unit interval graphs, proper hereditary classes of permutation graphs, or posets of bounded width. However, some very simple classes of bounded expansion, such as the class of subcubic graphs (with maximum degree three),

have unbounded twin-width. Thus, the scopes of Twin-width theory and Sparsity theory are incomparable. This motivates the quest of finding a unified theory that encompasses both Sparsity theory and Twin-width theory, and provides a common framework for studying the fundamental notions of the two theories.

Both Sparsity theory and Twin-width theory have found multiple combinatorial and algorithmic applications, and the same is expected of a theory unifying the two. As a concrete application, and a motivation for our pursuit, the sought theory is expected ultimately to resolve one of the central open problems in structural and algorithmic graph theory – of characterizing those hereditary graph classes for which the model-checking problem is fixed-parameter tractable [Gro07, Sec. 8.2] [GKS17, Sec. 9].

It is conjectured<sup>2</sup> (see [war16], [GHO<sup>+</sup>20]) that first-order model-checking is fixed-parameter tractable on a hereditary graph class  $\mathcal{C}$  if and only if  $\mathcal{C}$  is *monadically dependent* (also called *monadically NIP*). This notion, formulated in logical terms, originates in model theory, and was introduced by Shelah [She86] (see also Braunfeld and Laskowski [BL21]) in his momentous classification program of logical theories. Intuitively, a hereditary class  $\mathcal{C}$  is monadically dependent if for any fixed first-order formula  $\varphi(x, y)$ , there is some graph  $H$  that cannot be represented in any graph  $G \in \mathcal{C}$ , using the formula  $\varphi(x, y)$  to define the edges of  $H$  in  $G$ .

**Conjecture I.1.** *Let  $\mathcal{C}$  be a hereditary class of graphs. Then the model-checking problem for first-order logic is fixed-parameter tractable on  $\mathcal{C}$  if and only if  $\mathcal{C}$  is monadically dependent.*

Monadically dependent graph classes include all the graph classes mentioned above, and are considered (see [AA14], [BL21], [NdMP<sup>+</sup>21], [GPT22]) to be the dense counterpart of nowhere dense classes, as expressed e.g. by Conjecture I.1. For instance, nowhere dense classes are exactly those monadically dependent classes that exclude some biclique as a subgraph. However, to date, no *combinatorial* characterization of monadically dependent graph classes – akin to the multiple characterizations of nowhere dense classes – is known. As a consequence, in general, monadically dependent classes are currently beyond the reach of algorithmic methods.

*a) Contribution:* We propose a new family of graph parameters, called *flip-width* of radius  $r$ , for  $r \in \mathbb{N} \cup \{\infty\}$ , that are based on new *flipper games*. Those games are similar to the Cops and Robber game considered by Seymour and Thomas to characterize treewidth, in

<sup>2</sup>The conjecture has been circulating in the community since around 2016. As far as we know, it has first been stated explicitly during the open problem session of [war16]. There, dependent (or NIP) classes were considered instead of monadically dependent classes, but those two notions coincide for hereditary classes, by a result of Braunfeld and Laskowski [BL22].

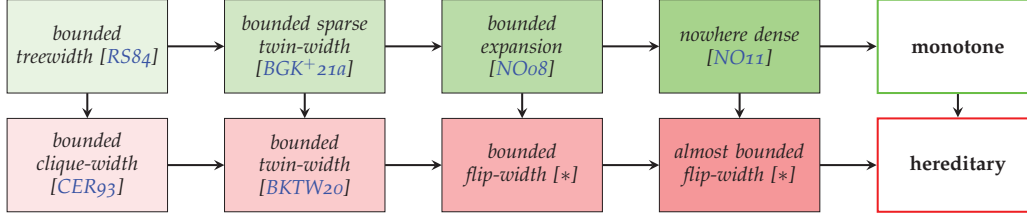


Fig. 1: Properties of graph classes, and implications among them. Each property in the lower row, restricted to monotone graph classes, yields the property above it. The properties marked [\*] are introduced in this paper. In this figure, *almost bounded flip-width* could be replaced by *monadically dependent* [She86]; we conjecture that those properties are equivalent.

their classic paper [ST93]. Variants of our game can be used to characterize all the aforementioned parameters: treewidth, degeneracy, and generalized coloring numbers used in the context of sparse classes, as well as clique-width and twin-width in the context of dense graph classes. More importantly, they provide generalizations of degeneracy, and of generalized coloring numbers, to the setting of graphs that are not necessarily sparse, and offer a compelling dense counterpart of classes of bounded expansion – called classes of bounded *flip-width* – and of nowhere dense classes – called classes of *almost bounded flip-width*. Classes of bounded flip-width include classes of bounded expansion as well as classes of bounded twin-width, and provide a unified framework for understanding those fundamental notions (see Figure 1.) We conjecture, and confirm in many special cases, that classes of *almost bounded flip-width* coincide with monadically dependent classes. If true, this would give a combinatorial, quantitative characterization of monadic dependence, analogous to the characterization of nowhere dense classes in terms of generalized coloring numbers. Moreover, we verify that classes of almost bounded flip-width include most known examples of hereditary graph classes that are known to have a fixed-parameter tractable model-checking problem (we could not confirm this only for classes with structurally bounded local cliquewidth).

On a high level, the main contribution of this paper is to lay new foundations of a theory of structurally tractable graph classes, extending Sparsity theory to the dense setting, and unifying it with Twin-width theory. In this paper, we define the fundamental notions of our theory, and provide overwhelming evidence that they are the sought generalizations of the fundamental notions of Sparsity theory to the dense setting. This evidence is provided by multiple results, demonstrating that various central concepts of structural graph theory can be uniformly explained in terms of our notions. In the next section, we present an overview of our main results.

## II. OVERVIEW

In this section, we give a high-level overview of the main results of this paper. These results are discussed

systematically in greater detail, with many illuminating examples and proof sketches, in the following sections. Many of the statements recounted in this section are simplified versions of the more precise statements given in the next sections. The full statement of each result is referenced in parentheses. All relevant notions are defined in Section III.

a) *Cop-width*: Our starting point is the – apparently new – observation that classes with bounded expansion can be characterized in terms of a variant of the Cops and Robber game considered by Seymour and Thomas [ST93]. Recall that in this game,  $k$  cops and a robber are occupying the vertices of a graph. In each round, some of the cops move to new positions in helicopters – that is, not necessarily along edges in the graph – whereas the robber may traverse any path in the graph which avoids the vertices occupied by the cops that remain on ground. The minimal number  $k$  of cops needed to capture the robber is equal to one plus the treewidth of the graph.

In Section IV we consider a variant of the Cops and Robber game, in which the robber runs at speed  $r$ , for some fixed  $r \in \mathbb{N} \cup \{\infty\}$ : they may traverse a path of length at most  $r$  that does not run through a cop. We call the parameter  $r$  the *radius* of the game, while the number of cops is called the *width* of the game. The *radius- $r$  cop-width* of  $G$ , denoted  $\text{copwidth}_r(G)$ , is the least number  $k$  such that  $k$  cops win the game with radius  $r$ .

Thus, we obtain a family of graph parameters, one for each  $r \in \mathbb{N} \cup \{\infty\}$ . As we observe, variants of the above game characterize the graph parameters mentioned earlier: treewidth (for  $r = \infty$ ), degeneracy (for  $r = 1$ ), and generalized coloring numbers (for  $1 \leq r < \infty$ ). Namely, the result of Seymour and Thomas can be phrased as follows.

**Theorem II.1** ([ST93]). *For every graph  $G$ ,*

$$\text{copwidth}_\infty(G) = \text{treewidth}(G) + 1.$$

On the other extreme, for radius  $r = 1$ , we have the following:

**Theorem II.2** (Thm. IV.4). *For every graph  $G$ ,*

$$\text{copwidth}_1(G) = \text{degeneracy}(G) + 1.$$

For higher radii, we prove a correspondence of the cop-width parameters with generalized coloring numbers. Recall that a graph class has bounded expansion if and only if each generalized coloring number is bounded by a constant on all graphs from the class.

**Theorem II.3** (Cor. IV.6). *A graph class  $\mathcal{C}$  has bounded expansion if and only if  $\text{copwidth}_r(\mathcal{C}) < \infty$  for every fixed  $r \in \mathbb{N}$ .*

Here and later on, for a graph parameter  $f$  and graph class  $\mathcal{C}$ , we denote the supremum of  $f(G)$ , for  $G \in \mathcal{C}$ , by  $f(\mathcal{C})$ . In Theorem II.2, we show that a witness to having degeneracy  $d$  – an ordering of the vertices such that every vertex has at most  $d$  vertices before it – yields a winning strategy for the Cops to win the game of width  $d + 1$  and radius 1, in which the Cops move to the current position of the robber and all neighbors before it, thus forcing the robber to move upwards in the order. Conversely, a witness to having degeneracy larger than  $d$  – a set  $U$  of vertices such that every vertex in  $U$  has more than  $d$  neighbors in  $U$  – yields a winning strategy for the robber, allowing him to always remain in the set  $U$ . Similarly, to prove Theorem II.3, we consider the Cops and Robber game for higher radii  $1 \leq r < \infty$ , and rely on central duality theorems of Sparsity theory, which describe combinatorial obstructions to having a small *weak coloring number*, in terms of a *dense shallow minor* [NO08]. We show that a witness to having a small weak coloring number yields a winning strategy for the Cops, while a dense shallow minor yields a winning strategy for the Robber in the Cops and Robber game of a fixed, finite radius. Those observations are analogues to the central duality result of Seymour and Thomas [ST93], which shows that a witness to having small treewidth yields a winning strategy for the Cops, while the dual obstruction to small treewidth, called a *haven* or *bramble*, yields a winning strategy for the Robber in the Cops and Robber game of infinite radius.

Parallel to Theorem II.3, we obtain a characterization of nowhere dense classes:

**Theorem II.4** (Cor. IV.7). *A hereditary graph class  $\mathcal{C}$  is nowhere dense if and only if for every fixed radius  $r \in \mathbb{N}$  and  $\varepsilon > 0$ , and for all  $n$ -vertex graphs  $G \in \mathcal{C}$ , we have that  $\text{copwidth}_r(G) \leq O(n^\varepsilon)$ .*

As is made apparent by the results above (and discussed in Section IV) the cop-width parameters capture a substantial part of the fundamental notions of Sparsity theory. However, being monotone under edge removals, they are only suited to the study of sparse graphs.

*b) Flip-width:* In Section V, we introduce a variant of the Cops and Robber game that is suited for dense graphs, dubbed the *flipper game*. This is similar to the recent development [GHN<sup>+</sup>12], [GMM<sup>+</sup>23], where in some contexts, it is shown that vertex removals in sparse graphs correspond to *flips* in dense graphs, and a differ-

ent variant of the flipper game is considered (see *Related work* below).

*Flipping* a pair of sets  $X, Y \subseteq V(G)$  in a graph  $G$  results in the graph  $G'$  obtained from  $G$  by inverting the adjacency of every pair  $(x, y)$  of vertices with  $x \in X$  and  $y \in Y$ : every such pair that is adjacent in  $G$  becomes non-adjacent in  $G'$ , and vice-versa. A  $k$ -flip of a graph  $G$  is obtained by partitioning  $V(G)$  into  $k$  parts, and then performing flips between some pairs  $X, Y$  of parts of the partition (possibly with  $X = Y$ ).

To motivate the flipper game, observe that in the Cops and Robber game of width  $k$ , we may think of the Cops as *isolating* at most  $k$  vertices, instead of placing the Cops on those vertices, where isolating a vertex amounts to (temporarily) removing all edges that are incident to it. Isolating a single vertex  $v$  in a graph  $G$  can be achieved by performing a 3-flip of  $G$ : partition the vertices into  $\{v\}$ , the set of neighbors of  $v$ , and the remaining vertices, and flip the first two sets. Similarly, a graph  $G$  obtained by isolating  $k$  vertices  $v_1, \dots, v_k$  is a  $(k + 2^k)$ -flip of  $G$ , by taking the common refinement of the partitions used for isolating each vertex  $v_i$  separately. In the flipper game defined below, we will allow one player to perform flips, significantly extending their power comparing to the cops in the Cops and Robber game.

The *flipper game* of radius  $r$  and width  $k$  on a graph  $G$  is a game played by two players, *flipper* and *runner*, proceeds as follows. Initially, the runner picks any vertex  $v_0$  of  $G$ . In the  $i$ th round, the flipper announces a  $k$ -flip  $G_i$  of the original graph  $G$ . The runner can move from their previous position  $v_{i-1}$  to a new position  $v_i$ , by traversing a path of length at most  $r$  in the *previous*  $k$ -flip  $G_{i-1}$  of  $G$  (where  $G_0 = G$ ). The game is won by the flipper if the runner's new position  $v_i$  is isolated in the announced  $k$ -flip  $G_i$ .

The *flip-width* of radius  $r$  of a graph  $G$ , denoted  $\text{fw}_r(G)$ , is the smallest number  $k$  for which the flipper has a winning strategy in the game described above. We say that a graph class  $\mathcal{C}$  has *bounded flip-width* if  $\text{fw}_r(\mathcal{C}) < \infty$  for all  $r \in \mathbb{N}$ .

Note that if  $H$  is an induced subgraph of  $G$  then  $\text{fw}_r(H) \leq \text{fw}_r(G)$ , for all  $r$ . In particular, a graph class  $\mathcal{C}$  has bounded flip-width if and only if its *hereditary closure* – consisting of all induced subgraphs of graphs in  $\mathcal{C}$  – has bounded flip-width.

As we argue below, the flip-width parameters are the sought dense analogues of the generalized coloring numbers, while classes of bounded flip-width are the dense analogues of classes of bounded expansion. Additionally, we show that variants of the flipper game can be used to uniformly characterize several among the most important graph parameters studied in structural graph theory: degeneracy, treewidth, and generalized coloring numbers, as well as clique-width and twin-width. For the first time, this provides a common perspective on parameters such as degeneracy and twin-width, or generalized

coloring numbers and clique-width – all of which can be seen as representations of winning strategies in the flipper game. This common perspective also allows to view in a unified way various key combinatorial notions representing obstructions – namely *havens* or *brambles* studied in the context of treewidth [ST93], *well-linked sets* studied in the context of clique-width [iOs06], *dense shallow minors* studied in the context of bounded expansion classes [NO08], or *rich divisions* studied in the context of twin-width [BGOdM<sup>+</sup>22]. Those notions lie at the core of duality theorems for the related parameters, and also have important algorithmic applications. As we will see, those notions can be seen as representations of winning strategies for the runner in the flipper game.

We now give a summary of our results concerning flip-width.

c) *Relating flip-width to other notions:* We start with comparing classes of bounded flip-width with the fundamental notions of Sparsity theory and Twin-width theory.

First, regarding the sparse graph parameters – degeneracy, treewidth, and generalized coloring numbers – we show that those correspond to the parameters  $\text{fw}_1$ ,  $\text{fw}_\infty$  and  $\text{fw}_r$ , for  $1 \leq r < \infty$ , when considering *weakly sparse* graph classes. A graph class is weakly sparse if it excludes some fixed biclique as a subgraph.

**Theorem II.5.** *Let  $\mathcal{C}$  be a graph class. Then:*

- 1) (Thm. VI.1)  $\mathcal{C}$  has bounded degeneracy if and only if  $\mathcal{C}$  is weakly sparse and  $\text{fw}_1(\mathcal{C}) < \infty$ ,
- 2) (Cor. VI.2)  $\mathcal{C}$  has bounded treewidth if and only if  $\mathcal{C}$  is weakly sparse and  $\text{fw}_\infty(\mathcal{C}) < \infty$ ,
- 3) (Thm. VI.3)  $\mathcal{C}$  has bounded expansion if and only if  $\mathcal{C}$  is weakly sparse and  $\mathcal{C}$  has bounded flip-width.

This shows that the flipper game of radius  $r$  is a sensible generalization of the Cops and Robber game of radius  $r$  discussed earlier, since we already know that the above notions (degeneracy, treewidth, bounded expansion) can be characterized in terms of the latter game. This fact only gives the rightwards implications in the above statements. To prove the converse implications, we rely on characterizations of  $K_{t,t}$ -free graphs with small degeneracy/treewidth/expansion in terms of forbidden induced subgraphs. Such induced subgraphs can be exploited by the runner to win the flipper game.

We then move to studying the flip-width parameters in graph classes that are not necessarily sparse. As a first case study, we verify that the flipper game with radius  $\infty$  corresponds to clique-width, similarly as the Cops and Robber game corresponds to treewidth. Namely, we prove:

**Theorem II.6** (Thm. V.17). *A class  $\mathcal{C}$  of graphs has bounded clique-width if and only if  $\text{fw}_\infty(\mathcal{C}) < \infty$ .*

This gives the first, to our knowledge, game characterization of classes of bounded clique-width, analogous to

the classic characterization of treewidth by Seymour and Thomas. To prove the rightwards implication, we view a clique-width expression (used to construct a graph of bounded clique-width) as a description of a winning strategy for the flipper in the radius- $\infty$  flipper game. For the converse implication, again we show that obstructions for bounded clique-width – called *well-linked sets* – yield winning strategies for the runner.

As clique-width is the dense analogue of treewidth, Theorem II.6 is another indication that our flipper game is an adequate generalization of the Cops and Robber game for the study of dense graph classes. In Section VII we provide further evidence of this, by demonstrating that another crucial graph parameter studied in the context of dense graphs, twin-width, can be characterized in terms of flip-width. First, we show that classes of bounded flip-width include all classes of bounded twin-width. Those include, for example, the class of unit interval graphs, or every proper hereditary class of permutation graphs.

**Theorem II.7** (Thm. VII.1). *Every class of bounded twin-width has bounded flip-width.*

The proof of this result views *contraction sequences*, which are recursive decompositions of graphs of bounded twin-width, as strategies for the flipper in the flipper game.

By the results above, classes of bounded flip-width include all classes with bounded expansion as well as classes of bounded twin-width. As those two notions are incomparable, classes of bounded flip-width strictly extend each of them. In particular, the converse to Theorem II.7 fails.

To characterize twin-width using flip-width, we naturally extend the flip-width parameters to arbitrary structures equipped with one or more binary relation, such as *ordered graphs* – graphs equipped with a total order. We prove that for classes of ordered graphs, bounded flip-width and bounded twin-width coincide:

**Theorem II.8** (Thm. VII.3). *A class of ordered graphs has bounded twin-width if and only if it has bounded flip-width.*

The rightwards implication follows from Theorem II.7. To prove the converse, we view the combinatorial obstructions to small twin-width, called *rich divisions* [BGOdM<sup>+</sup>22], as descriptions of winning strategies for the runner in the flipper game.

It is known [BGOdM<sup>+</sup>22] that a class of graphs has bounded twin-width if and only if it can be obtained from a class of ordered graphs of bounded twin-width, by forgetting the order. Thus, classes of bounded twin-width are exactly classes of ordered graphs of bounded flip-width, with the order forgotten.

Reassuring, variants of our flipper game capture degeneracy, treewidth, bounded expansion, clique-width,

and twin-width, all of which are of central importance in structural and algorithmic graph theory. Moreover, structural results can be employed to describe winning strategies for the flipper, while the dual combinatorial obstructions can be used to obtain strategies for the runner.

d) *Closure properties*: Another main contribution of this paper are results showing that classes of bounded flip-width enjoy good closure properties, in particular, closure under first-order interpretations (or transductions, see Theorem VIII.2). More precisely, for a first-order formula  $\varphi(x, y)$  and graph  $G$ , define the graph  $\varphi(G)$  with vertices  $V(G)$  and edges  $uv$  with  $u \neq v$ , such that  $\varphi(u, v) \vee \varphi(v, u)$  holds in  $G$ . For a graph class  $\mathcal{C}$ , let  $\varphi(\mathcal{C}) := \{\varphi(G) \mid G \in \mathcal{C}\}$ . For example, if  $\varphi(x, y)$  is the formula  $\exists z. E(x, z) \wedge E(z, y)$  (where  $E$  denotes adjacency), then  $\varphi(G)$  is the square of  $G$ . In Section VIII, we prove the following.

**Theorem II.9** (Thm. VIII.2). *Let  $\varphi(x, y)$  be a first-order formula and  $\mathcal{C}$  be a graph class. If  $\mathcal{C}$  has bounded flip-width then  $\varphi(\mathcal{C})$  has bounded flip-width.*

This generalizes a prior analogous result for classes of bounded twin-width [BKTW20] and a result concerning classes of bounded expansion. Also, this result provides further examples of classes of bounded flip-width, for instance, classes of *structurally bounded expansion* [GKN<sup>+</sup>20], that is, classes of the form  $\varphi(\mathcal{C})$ , where  $\mathcal{C}$  has bounded expansion and  $\varphi(x, y)$  is a first-order formula. Moreover, Theorem II.9 implies that classes of bounded flip-width are monadically dependent, a notion that conjecturally delimits the tractability frontier for the model checking problem (see Conjecture I.1).

The proof of Theorem II.9 uses a tool from finite model theory – namely, locality of first-order logic – that has so far been used mostly in the context of sparse graph classes, but more recently has also been successfully applied in the context of dense graph classes [BKTW20], [BDG<sup>+</sup>22], [DMS23].

e) *Approximation*: Determining the exact flip-width of radius  $r$  of a given graph  $G$  seems computationally difficult (the naive approach is exponential in the size of  $G$ ). As our next contribution, in Section IX, we obtain a slicewise polynomial (XP) algorithm that approximates the flip-width of a given graph  $G$ , which runs in time polynomial in the size of  $G$ , when the flip-width is considered fixed.

**Theorem II.10** (Thm. IX.7). *There is a constant  $C > 0$  and an algorithm that, given as inputs an  $n$ -vertex graph  $G$  and numbers  $r, k \in \mathbb{N}$ , runs in time  $n^{O(k)} \cdot 2^{O(2^k)}$ , and either concludes that  $\text{fw}_r(G) \leq 2^k$ , or concludes that  $\text{fw}_{5r}(G) \geq C \cdot k^{1/3}$ .*

Note that no XP algorithm approximating twin-width is known. This suggests that flip-width might be easier

to approximate than twin-width, even if it defines a more general notion.

To prove Theorem II.10, we define a variant of the flipper game, called the *definable flipper game*, in which the partitions used by the Cops for defining flips are *definable*: they partition the vertex set according to their adjacency in a set of vertices of bounded size. The advantage of this game variant is that it has polynomially many (in terms of the graph size) possible configurations, as opposed to the flipper game, which has exponentially many configurations. Although not every  $k$ -flip of a graph  $G$  can be defined by using a partition definable using a vertex set of bounded size, we prove that nevertheless the graph parameters defined by the definable flipper game can still be bounded in terms of the flip-width parameters. The proof uses tools from Vapnik-Chervonenkis theory.

f) *Almost bounded flip-width*: As our final contribution, in Section X we introduce and study classes of almost bounded flip-width, as a candidate dense counterpart of nowhere dense classes. A graph class  $\mathcal{C}$  has *almost bounded flip-width* if for every  $\varepsilon > 0$  and  $r \in \mathbb{N}$ , we have  $\text{fw}_r(G) \leq O(n^\varepsilon)$  for every  $n$ -vertex graph  $G$  which is an induced subgraph of a graph in  $\mathcal{C}$ . Thus, every class of bounded flip-width has almost bounded flip-width, but the converse does not hold.

We provide substantial evidence that classes of almost bounded flip-width coincide with monadically dependent classes (cf. Conjecture I.1), and also that the model checking problem is fixed-parameter tractable for those classes. This is corroborated by the following results, which examine this notion in special cases.

First, in the setting of weakly sparse classes, we prove:

**Theorem II.11** (Thm. X.9). *Let  $\mathcal{C}$  be a weakly sparse graph class. The following conditions are equivalent:*

- 1)  $\mathcal{C}$  has almost bounded flip-width,
- 2)  $\mathcal{C}$  is nowhere dense,
- 3)  $\mathcal{C}$  is monadically dependent.

The equivalence of the second and third conditions was known; the new result is the equivalence with the first condition.

In the setting of classes of ordered graphs, we prove an analogous statement, with nowhere denseness replaced by bounded twin-width. Moreover, in this case, almost bounded flip-width collapses to bounded flip-width. This collapse is, ultimately, a consequence of the Marcus-Tardos theorem/Stanley-Wilf conjecture [MT04] from enumerative combinatorics.

**Theorem II.12** (Thm. X.17). *Let  $\mathcal{C}$  be a class of ordered graphs. The following conditions are equivalent:*

- 1)  $\mathcal{C}$  has almost bounded flip-width,
- 2)  $\mathcal{C}$  has bounded flip-width,
- 3)  $\mathcal{C}$  has bounded twin-width,
- 4)  $\mathcal{C}$  is monadically dependent.

The equivalence of conditions 3,4 was known previously [BGOdM<sup>+</sup>22], and their equivalence with condition 2 is by Theorem II.8. The implication 2→3 is immediate. We prove the implication 1→3 using another characterisation from [BGOdM<sup>+</sup>22].

In particular, in the two settings considered in the theorems above, the model checking problem is fixed-parameter tractable, by the results of [GKS17] and [BGOdM<sup>+</sup>22], respectively.

We now move to a more general setting than that of weakly sparse classes, namely of *stable* classes. A class  $\mathcal{C}$  is *edge-stable* if there is a number  $k$  such that no graph  $G \in \mathcal{C}$  contains vertices  $a_1, \dots, a_k, b_1, \dots, b_k$  such that  $E(a_i, b_j) \iff i < j$  for  $i, j \in \{1, \dots, k\}$  (see Fig. 2). Edge-stable, monadically dependent graph classes coincide with *monadically stable* classes [BS85], [NdMP<sup>+</sup>21], an important subfamily of monadically dependent classes. Examples include all *structurally nowhere dense classes*, that is, classes of the form  $\varphi(\mathcal{C})$ , for some first-order formula  $\varphi(x, y)$  and nowhere dense class  $\mathcal{C}$ . It is conjectured [NdMP<sup>+</sup>21, Conjecture 6.1] that all monadically stable graph classes are structurally nowhere dense. By a recent result [DMS23], the model-checking problem is fixed-parameter tractable for all structurally nowhere dense classes, thus significantly extending the result of [GKS17] concerning model-checking on nowhere dense classes. We prove the following.

**Theorem II.13** (Thm. X.12). *Every structurally nowhere dense class has almost bounded flip-width.*

In the proof, we use the recent description of structurally nowhere dense classes [DGK<sup>+</sup>22a], which provides certain treelike decompositions for such classes, that can be used to produce a winning strategy for the flipper. As a weak converse, we prove:

**Theorem II.14** (Thm. X.15). *Every edge-stable class of almost bounded flip-width is monadically stable.*

The proof relies on a recent characterization of obstructions to monadic stability [GMM<sup>+</sup>23], which can be used to produce a winning strategy for the runner.

In particular, if all monadically stable classes are structurally nowhere dense as conjectured in [NdMP<sup>+</sup>21], the two results imply that among edge-stable graph classes, almost bounded flip-width coincides with monadic stability (and thus with monadic dependence). Furthermore, by the result of [DMS23], the conjecture would imply fixed-parameter tractability for all edge-stable classes of almost bounded flip-width.

Those results provide substantial evidence indicating that classes of almost bounded flip-width form the dense counterpart of nowhere dense classes, that they coincide with monadically dependent classes, and that they admit fixed-parameter tractable model-checking. In

fact, almost all<sup>3</sup> hereditary graph classes having fixed-parameter tractable model-checking that we are aware of [GKS17], [BKTW20], [DMS23], [BCK<sup>+</sup>22], have almost bounded flip-width. Based on this, we believe that classes of almost bounded flip-width will play a key role in combinatorial and algorithmic approaches to the analysis of monadically dependent classes.

We supplement our results with a discussion, in Section XI, providing evidence for various stated conjectures, and outline a potential approach towards an algorithmic and combinatorial understanding of the classes introduced in this paper.

g) *Related work:* Our flipper game is inspired by the paper [GMM<sup>+</sup>23]. There, another game based on performing flips is introduced, and is also called *flipper game*. We consider both games to be variants of a broader family of a flipper games, similarly as there are many variants of the Cops and Robber game. As far as we know, only two variants of the flipper game have been considered so far, but we anticipate that more variants might emerge. To distinguish the two variants, we may call the variant introduced in this paper the *open flipper game*, and the variant from [GMM<sup>+</sup>23] the *confined flipper game of bounded duration*. The latter game is an analogue of the *splitter game*, which characterizes nowhere dense classes [GKS17]. Similarly, the flipper game from [GMM<sup>+</sup>23] characterizes monadically stable graph classes, a strict subset of monadically dependent classes. It differs from the open flipper game in two major ways. Firstly, the duration of the game is bounded, whereas in our variant, it is unbounded. Furthermore, in each round of the confined flipper game, the fugitive is confined to the radius- $r$  ball around his present position in the current flip of the graph in all future rounds, not just the next round as in the open game. The confined flipper game of bounded duration is used in [GMM<sup>+</sup>23] to characterize monadically stable classes, which are incomparable with classes of bounded flip-width, and are likely to be strictly contained in classes of almost bounded flip-width (see Section X).

There are multiple variations of the Cops and Robber game, and some of them are similar to the one considered above for sparse graphs. In some of those variants, the robber has bounded speed, as in our games. See [FKL12], [FGK<sup>+</sup>10], [AM15] and [FT08] for a survey. The cop-width parameters that we study for sparse graphs are closely related to the parameters studied in [RT08] and [LPPT20], which are defined in terms of a similar game, as we discuss in Section IV and in [Tor23].

Some attempts at defining graph classes that are dense analogues of classes of bounded expansion were made in [KPS17], [GKN<sup>+</sup>18], [NORS21], [NdMP<sup>+</sup>21], [GPT22], [BNdMS22b]. In [KPS17], [NORS21], [NdMP<sup>+</sup>21], the

<sup>3</sup>One exception are classes with structurally bounded local clique-width [BDG<sup>+</sup>22], which are tractable and monadically dependent, but we do not know whether they have almost bounded flip-width.

property of having low rankwidth covers is proposed as the dense analogue of bounded expansion. However, this notion does not include<sup>4</sup> classes of bounded twin-width, and it is not known whether this notion is closed under transductions. It has been proved recently that those classes are monadically dependent [BNdMS22a]. In [GPT22, Section 4] and [BNdMS22b, Section 8], an attempt at formalizing an abstract notion of a dense analogue of classes with bounded expansion was made, but those papers do not propose any workable combinatorial definition.

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*i) Organization:* The organization of this paper is as follows. Results marked with (\*) are proved in the full version of the paper [Tor23].

Section III introduces some standard notation, and recalls some basic notions and results, e.g. from Sparsity theory and Vapnik-Chervonenkis theory.

In Section IV we introduce the cop-width parameters, and demonstrate they characterize degeneracy, treewidth, and generalized coloring numbers. Using those parameters, we characterize classes of bounded expansion and nowhere dense classes.

In Section V we introduce our two main notions: the flip-width parameters, and classes of bounded flip-width. We give many examples, and derive some combinatorial properties. We show that radius- $\infty$  flip-width is equivalent to clique-width.

In Section VI we study the behavior of the flip-width parameters in the case of  $K_{t,t}$ -free graphs. We show that radius-one flip-width corresponds to degeneracy, and that for higher radii, the flip-width parameters correspond to the generalized coloring numbers. Consequently, a weakly sparse graph class has bounded flip-width if and only if it has bounded expansion.

In Section VII we study the relationship between twin-width and flip-width. We prove that every class of bounded twin-width has bounded flip-width. We also prove that for classes of ordered graphs, bounded flip-width and bounded twin-width are equivalent.

In Section VIII we prove that classes of bounded flip-width are preserved by first-order transductions.

In Section IX we introduce a definable variant of flip-width, and prove its equivalence with flip-width. As a consequence, we get a slicewise polynomial (XP) approximation algorithm for the flip-width parameters.

In Section X we introduce our third main notion, classes of almost bounded flip-width. We prove that

they contain all structurally nowhere dense classes, and study their relationship with monadically stable and monadically dependent classes.

Finally, in Section XI, we discuss possible directions of further research, conjectures and questions.

<sup>4</sup>An example was provided by Jakub Gajarský (private communication)

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## III. PRELIMINARIES

We introduce basic notation in Section III-A. In Section III-B we recount the fundamental notions of Sparsity theory. In Section III-C we recall the notion of Vapnik-Chervonenkis dimension of a set system, of a graph, and of a binary relation. In Section III-D we recall basic notions from logic (structures, formulas), and a result characterizing nowhere dense classes in terms of the VC-dimension of certain set systems.

### A. Notation

$\mathbb{N} = \{0, \dots\}$  denotes the set of nonnegative integers. For two sets  $A$  and  $B$  their symmetric difference is denoted  $A \triangle B := (A - B) \cup (B - A)$ . We write  $O(n)$  (resp.  $\Omega(n)$ ) to denote a value that is bounded from above (resp. from below) by  $c \cdot n + d$ , for some reals  $c, d$  with  $c > 0$ . Sometimes we write  $O_p(n)$  (resp.  $\Omega_p(n)$ ), where  $p$  is a list of parameters, to indicate that the constants  $c$  and  $d$  above depend on the parameters  $p$ . For a function  $f: \mathbb{N} \rightarrow \mathbb{R}$ , we write  $o(f(n))$  to denote a function  $g: \mathbb{N} \rightarrow \mathbb{R}$  with  $\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = 0$ .

Graphs are finite, undirected and without self-loops. The set of vertices of a graph  $G$  is denoted  $V(G)$ , and the set of edges of  $G$  is denoted  $E(G)$ . An edge joining  $u$  and  $v$  is denoted  $uv$ . In particular,  $uv = vu$  and  $u \neq v$  for all  $uv \in E(G)$ . We write  $|G|$  for the number of vertices of  $G$ . For a vertex  $v$  of a graph  $G$  the (open) *neighborhood* of  $v$  in  $G$  is  $N_G(v) := \{u \mid uv \in E(G)\}$ , denoted  $N(v)$  if  $G$  is understood from the context. The set of vertices at distance at most  $r$  from  $v$  in  $G$  is denoted  $B_G^r(v)$ .

A graph  $H$  is a *subgraph* of  $G$  if  $H$  is obtained by removing vertices and/or edges from  $G$ , and is an *induced subgraph* of  $G$  if  $H$  is obtained by removing vertices from  $G$ , alongside with the edges incident to them. The subgraph of  $G$  induced by a set of vertices  $X \subseteq V(G)$  is the graph  $G[X]$  with vertices  $X$  and edges  $uv \in E(G)$  with  $u, v \in X$ . For  $X, Y \subseteq V(G)$ , the bipartite graph *semi-induced* by  $X$  and  $Y$  in a graph  $G$  has parts  $X$  and  $Y$  and edges  $xy$  such that  $x \in X, y \in Y$  and  $xy \in E(G)$ . Note that  $X$  and  $Y$  need not be disjoint in  $G$ ; in this case,  $G[X, Y]$  contains two copies of every vertex in  $X \cap Y$ . Two sets  $X, Y$  are *complete* in  $G$  if  $G[X, Y]$  is the complete bipartite graph, and *anti-complete* in  $G$  if  $G[X, Y]$  has no edges, and *homogeneous* if they are either complete or anti-complete.  $K_t$  denotes the complete graph on  $t$  vertices, and  $K_{s,t}$  denotes the complete bipartite graph with parts of sizes  $s$  and  $t$ .

A *graph class*  $\mathcal{C}$  is a set of graphs. A class  $\mathcal{C}$  is *hereditary* if it is closed under taking induced subgraphs. The *hereditary closure* of a class  $\mathcal{C}$  is the class of all induced subgraphs of graphs from  $\mathcal{C}$ .  $\mathcal{C}$  is *weakly sparse* if there is some  $t \in \mathbb{N}$  such that every  $G \in \mathcal{C}$  excludes the biclique  $K_{t,t}$  as a subgraph. A *graph parameter* is a function  $f$  assigning reals to graphs, which is invariant under graph isomorphism. For a graph class  $\mathcal{C}$  and graph parameter

$f$ , denote  $f(\mathcal{C}) := \sup_{G \in \mathcal{C}} f(G)$ , with  $f(\mathcal{C}) = \infty$  if  $f$  is unbounded on  $\mathcal{C}$ . Say that  $f$  is *bounded in terms of*  $g$  if there is a function  $\alpha: \mathbb{R} \rightarrow \mathbb{R}$  such that  $f(G) \leq \alpha(g(G))$  for all graphs  $G$ . Two functions  $f$  and  $g$  are *functionally equivalent* if each of them is bounded in terms of the other.

A  $k$ -colored graph is a graph  $G$  equipped with a function assigning a color from  $\{1, \dots, k\}$  to each vertex of  $G$ . A class  $\hat{\mathcal{C}}$  of  $k$ -colored graphs is a  $k$ -coloring of a graph class  $\mathcal{C}$ , if for every colored graph in  $\hat{\mathcal{C}}$ , the underlying uncolored graph belongs to  $\mathcal{C}$ , and for every graph  $G \in \mathcal{C}$ , some  $k$ -coloring of  $G$  belongs to  $\hat{\mathcal{C}}$ .

### B. Sparsity theory

We briefly recall the fundamental notions of Sparsity theory. See [NdM12] for more background.

A graph  $G$  is  $d$ -degenerate if there is a total order on the vertices of  $G$  such that every vertex has at most  $d$  neighbors before it in the order. The *degeneracy* of  $G$  is the least  $d$  such that  $G$  is  $d$ -degenerate.

An *exact  $r$ -subdivision* of a graph  $G$  is the graph obtained by replacing every edge of  $G$  by a path of length  $r + 1$ . If every edge is replaced by a path of length at most  $r + 1$ , the resulting graph is called a  $\leq r$ -subdivision of  $G$ . For a graph  $G$ , let  $\tilde{\nabla}_r(G)$  denote the maximum average degree,  $2|E(H)|/|V(H)|$ , of all graphs  $H$  whose  $\leq r$ -subdivision is a subgraph of  $G$  (called *shallow topological minors of  $G$  at depth  $r$* ). Note that  $\text{degeneracy}(G) \leq \tilde{\nabla}_0(G) \leq 2 \cdot \text{degeneracy}(G)$ .

**Definition III.1.** A graph class  $\mathcal{C}$  has *bounded expansion* if for every  $r \geq 1$  we have  $\tilde{\nabla}_r(\mathcal{C}) < \infty$ .

*Example III.2.* Classes of bounded expansion include many well-studied sparse classes: the class of planar graphs, every class of bounded maximum degree, classes of bounded tree-width, classes that exclude a fixed (topological) minor.

The weak coloring and admissibility numbers are two families of graph parameters that generalize the degeneracy number to higher radii  $r \geq 1$ , and are defined as follows. The  $r$ -weak coloring number of a graph  $G$ , denoted  $\text{wcol}_r(G)$ , is the smallest number  $k$  with the following property: There is a total order  $<$  on the vertices of  $G$  such that for every vertex  $v$ , there are at most  $k$  vertices  $w$  with  $w < v$  that are *weakly  $r$ -reachable* from  $v$ : there is a path of length at most  $r$  from  $v$  to  $w$ , in which  $w$  is the  $<$ -smallest vertex. On the other hand, the  $r$ -admissibility of a graph  $G$ , denoted  $\text{adm}_r(G)$ , is the smallest number  $k$  with the following property: There is a total order  $<$  on  $V(G)$  such that for every  $v \in V(G)$  one cannot find more than  $k$  paths of length at most  $r$  that start at  $v$ , end in some vertex  $w < v$ , and such that any two of the paths share only  $v$  as a common vertex.

Both parameters are of central importance in Sparsity theory. It is known that  $\text{wcol}_r(G)$  is bounded in terms

of  $\text{adm}_r(G)$ , and vice versa [Dvo13, Lemma 6]. More specifically, we have the following.

**Fact III.3.** For all graphs  $G$  and  $r \geq 1$  we have

$$\text{adm}_r(G) \leq \text{wcol}_r(G) \leq O(\text{adm}_r(G))^r. \quad (1)$$

The fundamental notion of Sparsity theory can be characterized using weak coloring numbers as follows:

**Fact III.4** ([Zhu09]). A class  $\mathcal{C}$  of graphs has *bounded expansion* if and only if  $\text{wcol}_r(\mathcal{C}) < \infty$  for every  $r \in \mathbb{N}$ .

By (1), we could replace  $\text{wcol}_r(\mathcal{C})$  by  $\text{adm}_r(\mathcal{C})$  in this characterization.

We use the following inequalities.

**Fact III.5.** For all graphs  $G$  and  $r \geq 1$  we have

$$\tilde{\nabla}_{r-1}(G)/2 < \text{adm}_r(G) \leq 6(\tilde{\nabla}_{r-1}(G))^3. \quad (2)$$

*Proof.* The upper bound below is [GKR<sup>+</sup>18, Theorem 3.1]. We prove the lower bound. Suppose  $G$  contains an  $\leq(r-1)$ -subdivision of a graph with average degree  $s$ . Then  $G$  contains an  $\leq(r-1)$ -subdivision of a graph  $H$  with minimum degree larger than  $s/2$ . Let  $U \subseteq V(G)$  consist of the principal vertices, corresponding to the vertices of  $H$ . Then for every vertex  $u \in U$  there are more than  $s/2$  paths of length  $r$  that start at  $u$ , end at vertices of  $U$ , and are vertex-disjoint apart from  $u$ . The set  $U$  witnesses that  $\text{adm}_r(G) > s/2$ .  $\square$

We now move to nowhere dense classes.

**Definition III.6.** A graph class  $\mathcal{C}$  is *nowhere dense* if for every  $r \geq 1$  there is some  $n \geq 1$  such that for all  $G \in \mathcal{C}$ , no  $\leq r$ -subdivision of  $K_n$  is contained as a subgraph of  $G \in \mathcal{C}$ .

It is immediate that every class with bounded expansion is nowhere dense, and there exist nowhere dense classes which do not have bounded expansion.

One of the central results of Sparsity theory is the following characterization of nowhere dense classes (recall that  $|G|$  is the number of vertices of  $G$ ).

**Fact III.7.** A hereditary graph class  $\mathcal{C}$  is nowhere dense if and only if for every  $r \geq 1$  and  $\varepsilon > 0$  there is a constant  $n_{r,\varepsilon}$  such that  $\text{wcol}_r(G) < |G|^\varepsilon$  for every  $G \in \mathcal{C}$  with  $|G| > n_{r,\varepsilon}$ .

The condition in Fact III.7 can be equivalently phrased as follows: for every  $r \geq 1$  and  $\varepsilon > 0$ ,  $\text{wcol}_r(G) \leq O_{r,\varepsilon}(|G|^\varepsilon)$  for every  $G \in \mathcal{C}$ . This can be written more concisely as  $\text{wcol}_r(G) \leq |G|^{o(1)}$ .

This fundamental result opens the door for multiple algorithmic applications of nowhere denseness, thanks to the existence of efficient algorithms approximating weak coloring numbers. In particular, the model-checking result [GKS17] relies on Fact III.7.

### C. Vapnik-Chervonenkis dimension

An important parameter measuring the complexity of graphs, and more generally, of set systems, is the *Vapnik-Chervonenkis dimension*, or *VC-dimension*.

A *set system* is a pair  $(X, \mathcal{F})$  with  $\mathcal{F} \subseteq 2^X$ . Its *VC-dimension* is the maximal size of a subset  $Y \subseteq X$  such that  $\{Y \cap F \mid F \in \mathcal{F}\} = 2^Y$ . We recall the fundamental Sauer-Shelah-Perles lemma [Sau72], [She72].

**Lemma III.8** (Sauer-Shelah-Perles lemma). *Let  $(X, \mathcal{F})$  be a set system of VC-dimension  $d$ . Then  $|\mathcal{F}| \leq O(|X|^d)$ .*

The VC-dimension of a graph  $G$ , denoted  $\text{VCdim}(G)$ , is defined as the VC-dimension of the set system  $(V(G), \{N(v) \mid v \in V(G)\})$ . More explicitly,  $\text{VCdim}(G)$  is the maximal size of a subset  $X \subseteq V(G)$  such that  $\{N(v) \cap X \mid v \in V(G)\} = 2^X$ .

For a binary relation  $R \subseteq V \times W$ , and elements  $a \in V$  and  $b \in W$ , denote  $\bar{R}(a) := \{w \in W \mid (a, w) \in R\}$  and  $\bar{R}(b) := \{v \in V \mid (v, b) \in R\}$ . The *VC-dimension* of  $R$  is the maximum of the VC-dimensions of the two set systems

$$(V, \{\bar{R}(a) \mid a \in V\}) \quad \text{and} \quad (W, \{\bar{R}(b) \mid b \in W\}).$$

### D. Logic

In this paper, we only consider *binary* signatures, that is signatures  $\Sigma$  consisting of unary relation symbols, binary relation symbols, and unary function symbols. Fix a binary signature  $\Sigma$ . A  $\Sigma$ -structure  $B$  consists of a set of vertices  $V(B)$ , and is equipped with interpretations for each of the symbols in  $\Sigma$ , as unary relations, binary relations, and unary functions on  $V(B)$ , respectively. The number of vertices of  $B$  is denoted  $|B|$ . A graph  $G$  is seen as a structure over the signature  $\Sigma$  consisting of one binary relation  $E$ , which is interpreted as adjacency in  $G$ .

*Terms* are defined inductively, as either a variable symbol, or a function symbol applied to a term, e.g.  $x$  and  $f(g(y))$  are terms if  $f, g$  are function symbols and  $x, y$  are a variable symbols. A *quantifier-free* formula over the signature  $\Sigma$  is a boolean combination of *atomic formulas* of the form  $U(t(x))$ , or  $t(x) = t'(y)$ , or  $R(t(x), t'(y))$ , where  $U \in \Sigma$  is a unary relation,  $R \in \Sigma$  is a binary relation, and  $t(x)$  and  $t'(y)$  are terms.

A *first-order formula* is built inductively: every atomic formula is a first-order formula, and if  $\varphi, \psi$  are first-order formulas and  $x$  is a variable, then so are  $\varphi \vee \psi$ ,  $\neg \varphi$ , and  $\exists x. \varphi$ . The notation  $\varphi \wedge \psi$ ,  $\varphi \rightarrow \psi$ , and  $\forall x. \varphi$  is used as syntactic sugar. The *quantifier rank* of a formula  $\varphi$  is the maximal nesting of quantifiers in it.

We may write  $\varphi(x_1, \dots, x_k)$  to indicate that the free-variables of  $\varphi$  are contained in  $\{x_1, \dots, x_k\}$ . For a formula  $\varphi(x_1, \dots, x_k)$ , structure  $A$  and elements  $a_1, \dots, a_k$  of  $A$ , we write  $A \models \varphi(a_1, \dots, a_k)$  to denote that the valuation of the variables  $x_1 \mapsto a_1, \dots, x_k \mapsto a_k$  satisfies the formula  $\varphi$  in  $A$ .

Say that a  $\Sigma$ -formula  $\varphi(x, y)$  is *symmetric* if  $G \models \varphi(u, v) \leftrightarrow \varphi(v, u)$  holds for every  $\Sigma$ -structure  $G$  and elements  $u, v$  of  $G$ .

The *Gaifman graph* of a  $\Sigma$ -structure  $B$  is the graph with vertices  $V(B)$  and edges  $uv$  with  $u \neq v$  such that  $(u, v) \in R$  or  $(v, u) \in R$  for some binary relation  $R \in \Sigma$ , or  $f(u) = v$  or  $f(v) = u$  for some unary function  $f \in \Sigma$ .

a) *Dependence*: Let  $\varphi(\bar{x}; \bar{y})$  be a first-order formula, whose set of free variables is partitioned into two disjoint sets  $\bar{x}$  and  $\bar{y}$ . For a structure  $H$  define the binary relation  $R_H^\varphi \subseteq V(H)^{\bar{x}} \times V(H)^{\bar{y}}$  as

$$R_H^\varphi = \{(\bar{u}, \bar{v}) \in V(H)^{\bar{x}} \times V(H)^{\bar{y}} \mid H \models \varphi(\bar{u}; \bar{v})\}.$$

Say that a class  $\mathcal{C}$  of  $\Sigma$ -structures is *dependent*, or *NIP* [SS71], [Adlo8] if for every first-order formula  $\varphi(\bar{x}; \bar{y})$  there is some  $k_\varphi \geq 1$  such that for every  $H \in \mathcal{C}$  the binary relation  $R_H^\varphi$  has VC-dimension at most  $k_\varphi$ . This is equivalent to saying that for every first-order formula  $\varphi(\bar{x}; \bar{y})$ , there is some bipartite graph  $G_\varphi$ , such that for all  $H \in \mathcal{C}$ , the bipartite graph with parts  $V(H)^{\bar{x}}$ ,  $V(H)^{\bar{y}}$  and edges  $(\bar{u}, \bar{v})$  such that  $H \models \varphi(\bar{u}; \bar{v})$ , does not contain  $G_\varphi$  as an induced subgraph.

The following fact is due to Podewski and Ziegler [PZ78] (see also [AA14] and [PST18]).

**Fact III.9.** *Every nowhere dense graph class  $\mathcal{C}$  is dependent. Conversely, every monotone, dependent graph class  $\mathcal{C}$  is nowhere dense.*

Hereditary, dependent graph classes are much more general than nowhere dense classes. The study of those classes is the main motivation of this paper. A closely related notion, of a *monadically dependent* class, is discussed in Section X. Monadically dependent classes are dependent, but in general, the converse implication does not hold. However, as shown by Braunfeld and Laskowski [BL22], for hereditary classes, the two notions coincide.

## IV. COP-WIDTH

We start with defining and analyzing the Cops and Robber game with finite radius, and the related cop-width parameters. To define the game, we invoke the original description of the Cops and Robber game by Seymour and Thomas [ST93]: “The robber stands on a vertex of the graph, and can at any time run at great speed to any other vertex along a path of the graph. He is not permitted to run through a cop, however. There are  $k$  cops, each of whom at any time either stands on a vertex or is in a helicopter (that is, is temporarily removed from the game). The objective of the player controlling the movement of the cops is to land a cop via helicopters on the vertex occupied by the robber, and the robber’s objective is to elude capture. (The point of the helicopters is that cops are not constrained to move along paths of the graph – they move from vertex to vertex arbitrarily.) The robber can see the helicopter approaching its landing

spot and may run to a new vertex before the helicopter actually lands.”

Seymour and Thomas proved that the least number of cops needed to catch a robber on a graph  $G$  is equal to one plus the treewidth of  $G$ . To this end, they proved a *min-max theorem*: either the cops have a winning strategy of a particularly simple, *monotone* form, which can be described by a tree decomposition of the graph, or otherwise, the robber has a winning strategy of a particularly simple form, called a *haven*.

In our variant of the game the robber runs at speed  $r$ , for some fixed  $r \in \mathbb{N} \cup \{\infty\}$ . That is, in each round, after the cops have taken off in their helicopters to their new positions, which are known to the robber, and before the helicopters have landed, the robber may traverse a path of length at most  $r$  that does not run through a cop that remains on the ground (he may also stay put). We call this game the *Cops and Robber game* with radius  $r$  and width  $k$ , if there are  $k$  cops, and the robber can run at speed  $r$ .

**Definition IV.1.** The radius- $r$  cop-width of  $G$ , denoted  $\text{copwidth}_r(G)$ , is the least number  $k$  such that the cops win the Cops and Robber game with radius  $r$  and width  $k$ .

Note that there is a graph parameter called the *cop-number* of a graph [AF84], which is equal to the number of cops needed to catch the robber in a game where the cops and the robber move at speed one in each turn. It could seem that  $\text{copwidth}_1(G)$  is upper bounded by the cop-number of  $G$ . However, there is a crucial difference with our notion: the cops do not announce their moves in advance. So for instance, every graph with a universal vertex (a vertex adjacent to all other vertices) has cop-number equal to one (such graphs are called *cop-win graphs*). On the other hand,  $\text{copwidth}_1(G)$  can be arbitrarily large on such graphs. Indeed, it is easy to see that for each  $r \in \mathbb{N} \cup \{\infty\}$ , the parameter  $\text{copwidth}_r(G)$  is monotone with respect to the subgraph relation: if  $H$  is a subgraph of  $G$  then  $\text{copwidth}_r(H) \leq \text{copwidth}_r(G)$ .

**Example IV.2.** Let  $G$  be a graph with maximum degree  $d$ . Then  $\text{copwidth}_r(G) < d^{r+1}$ , for all  $r \in \mathbb{N}$ . To see this, consider the following strategy for the cops: if the robber is initially placed on a vertex  $v$ , direct the cops to all (less than  $d^{r+1}$ ) vertices that are at distance at most  $r$  from  $v$ . Before the cops land at those locations, the robber traverses a path of length at most  $r$  to one of those vertices, and is caught by the landing cop.

**Example IV.3.** Let  $T$  be a rooted tree. Then  $G$  has tree-width at most one, so by the Seymour-Thomas result,  $\text{copwidth}_\infty(G) \leq 2$ . The strategy with two cops is as follows. In the first round, land one cop on the root of the tree, leaving the other cop in his helicopter. In each subsequent round, direct the cop that is further from the robber, to the child of the other cop’s position which is closest to the robber’s position. With this strategy, in

round  $i$  the robber will be at distance at least  $i$  from the root, so they will be caught after at most as many rounds as the height of  $T$ .

We start with comparing the most fundamental parameter of Sparsity theory, degeneracy, with the cop-width parameter for radius 1.

**Theorem IV.4.** For every graph  $G$ ,

$$\text{copwidth}_1(G) = \text{degeneracy}(G) + 1.$$

The simple proof of this fact relies on the fundamental duality result concerning degeneracy: for every  $d \in \mathbb{N}$ , every graph  $G$  is either  $d$ -degenerate, or it has a subgraph  $H$  in which every vertex has degree larger than  $d$ . This duality theorem is nothing else than a min-max theorem for the radius-1 Cops and Robber game, quite analogous to the min-max theorem in the eponymous paper of Seymour and Thomas. Indeed, a  $d$ -degeneracy order can now be viewed as a compact representation of a winning strategy for the cops involving  $d + 1$  cops, in the Cops and Robber game with radius 1: when robber is on a vertex  $v$ , place  $d + 1$  cops on  $v$  and the neighbors of  $v$  before  $v$ . Then the robber needs to move rightwards in the order, and eventually loses. This proves  $\text{copwidth}_1(G) \leq \text{degeneracy}(G) + 1$ . Dually, a subgraph  $H$  of  $G$  whose all vertices have degree larger than  $d$ , can be seen as a haven for the robber: they can forever evade  $d$ -cops by always moving to an unoccupied vertex of  $H$  (or remaining in place). This proves  $\text{copwidth}_1(G) \geq \text{degeneracy}(G) + 1$ , thus proving Theorem IV.4.

Next we observe that for higher radii  $r$ , the parameter  $\text{copwidth}_r(G)$  is closely related to the generalized coloring numbers: the weak coloring number  $\text{wcol}_r(G)$  and the admissibility numbers  $\text{adm}_r(G)$  (see Section III-B). Recall (see Fact III.3) that the two parameters are functionally equivalent, and that  $\text{adm}_r(G) \geq \tilde{\nabla}_{r-1}(G)$ .

We prove the following:

**Theorem IV.5.** For  $r \in \mathbb{N}$ ,

$$\text{adm}_r(G) + 1 \leq \text{copwidth}_r(G) \leq \text{wcol}_{2r}(G) + 1.$$

In particular, by Fact III.4, this gives the first, arguably very natural, characterization of classes with bounded expansion, in terms of a game:

**Corollary IV.6.** A graph class  $\mathcal{C}$  has bounded expansion if and only if for every  $r \in \mathbb{N}$  we have that  $\text{copwidth}_r(\mathcal{C}) < \infty$ .

Thus, all the classes  $\mathcal{C}$  from Example III.2 satisfy  $\text{copwidth}_r(\mathcal{C}) < \infty$ , for all  $r \in \mathbb{N}$ . Similarly, by Fact III.7 we get a new characterization of nowhere dense classes.

**Corollary IV.7.** A hereditary graph class  $\mathcal{C}$  is nowhere dense if and only if for every  $r \in \mathbb{N}$  and  $\varepsilon > 0$  we have that  $\text{copwidth}_r(G) \leq O_{r,\varepsilon}(|G|^\varepsilon)$  for  $G \in \mathcal{C}$ .

We believe that the proof of Theorem IV.5, which we now give, sheds a new light on the fundamental notions

of Sparsity theory. Namely, the total order  $<$  that appears in the definition of the weak coloring number  $\text{wcol}_{2r}(G)$  can be viewed as a compact representation of a (very particular) winning strategy for the cops in the Cops and Robber game with radius  $r$ . Indeed, the following yields a winning strategy for the cops: if the robber is at a vertex  $v$ , then the cops are placed on  $v$  and the vertices  $w < v$  that are  $2r$ -weakly reachable from  $v$ . To see that this strategy is winning for the cops, consider the paths  $\pi_1, \pi_2, \dots$  in  $G$ , where  $\pi_i$  is the path of length at most  $r$  along which robber traversed from his  $i$ th position  $v_i$  to his  $(i+1)$ -st position  $v_{i+1}$ . If  $m_i$  denotes the  $<$ -minimal vertex of the path  $\pi_i$ , then we have that  $m_{i+1} > m_i$ . Otherwise,  $m_{i+1}$  is  $2r$ -weakly reachable from  $v_i$  (as witnessed by the path  $\pi_i; \pi_{i+1}$ , truncated at  $m_{i+1}$ ), so at the time the robber traversed the path  $\pi_{i+1}$ , the vertex  $m_{i+1}$  that was occupied by a cop, which is impossible. Therefore, we have  $m_1 < m_2 < \dots$ , so the cops win after at most  $|G|$  rounds. This gives the upper bound in Theorem IV.5.

For the lower bound, we use the well-known (and straightforward) min-max characterization of admissibility: a graph  $G$  has  $r$ -admissibility number at least  $d$  if there is a set of vertices  $X \subseteq V(G)$  such that for every vertex  $v \in X$  there is a set of  $d$  paths of length at most  $r$  that start at  $v$  and end in some vertex of  $X - v$ , such that any two paths only share  $v$  as a common vertex.

A set  $X \subseteq V(G)$  that witnesses that  $\text{adm}_r(G) \geq d$  can be used as a haven for the robber, to elude  $d$  cops forever, similarly as in the case of degeneracy. If the robber is occupying a vertex  $v$  in  $X$  and the cops are moving to a set  $S$  of at most  $d$  new positions, then either  $v \notin S$ , or there is some path from  $v$  to a vertex in  $X - S$  of length  $\leq r$ , and the robber moves along this path. This proves  $\text{copwidth}_r(G) \geq d + 1$ .

Therefore, the equivalence of the weak coloring numbers and the admissibility numbers can again be seen as a min-max theorem for the Cops and Robber game with radius  $r$ .

Note that Theorem IV.5 does not give an exact min-max theorem, as there is a gap between the upper and lower bounds. We can get a family of parameters based on another variant of the cops and robber game, which does admit an exact min-max theorem. In this game, in each round first the cops move to some  $k$  vertices of the graph, and then the robber moves via a path of length at most  $r$ , that does not run through a cop, and loses if no such path exists. See [Tor23, Appendix A] for more details. This family of parameters again characterizes classes with bounded expansion, and does admit an exact min-max theorem that exhibits a duality between total orders describing winning strategies for the cops, and havens for the robber. Those parameters essentially

appear<sup>5</sup> in the work [RT08], [LPPT20]. However, the paper does not relate those notions with generalized coloring numbers, and with classes of bounded expansion. Curiously, the limit version of those parameters, for radius  $r = \infty$ , does not correspond to treewidth, but to a notion called  $\infty$ -admissibility [Dvo12]. Classes with bounded  $\infty$ -admissibility are characterized as clique-sums of graphs of almost bounded degree [Dvo12, Cor. 5 and Thm. 6]. Despite the appealing properties of the parameters based on this variant of the cops and robber game, they seem to be less suited for our purposes, of generalizing to dense graphs.

To summarize, our new cop-width parameters exactly characterize treewidth (for  $r = \infty$ ), degeneracy (for  $r = 1$ ), classes of bounded expansion, and nowhere dense classes. This captures an appreciable fragment of the theory of sparse graphs, while offering a new perspective on the fundamental graph parameters used for measuring sparsity, and the dualities between them. We have

$$\begin{aligned} \text{degeneracy}(G) + 1 &= \text{copwidth}_1(G) \leq \text{copwidth}_2(G) \\ &\leq \dots \leq \text{copwidth}_\infty(G) = \text{treewidth}(G) + 1, \end{aligned}$$

so for a class  $\mathcal{C}$  of graphs, if any of those parameters is bounded by a constant, then  $\mathcal{C}$  has bounded degeneracy by Theorem IV.4. So those parameters are only well suited to the study of sparse graphs: every class  $\mathcal{C}$  for which either of those parameters is bounded, is sparse, in the sense of having a bound on the edge density  $|E(G)|/|V(G)|$  for all graphs  $G$  in the class. Moreover,  $\text{copwidth}_r$  is monotone with respect to the subgraph relation: if  $H$  is a subgraph of  $G$ , then  $\text{copwidth}_r(H) \leq \text{copwidth}_r(G)$ .

## V. FLIP-WIDTH

To lift the Cops and Robber game to the setting of dense graphs, we enhance the power of the cops. Now, instead of placing cops on at most  $k$  vertices of the graph, which can be alternatively seen as removing at most  $k$  vertices, or isolating them, one player can perform *flips* on subsets of the graph  $G$ . For a fixed graph  $G$ , applying a *flip* between a pair of sets of vertices  $A, B \subseteq V(G)$  results in the graph obtained from  $G$  by inverting the adjacency between any pair of vertices  $a, b$  with  $a \in A$  and  $b \in B$ . For example, applying a flip between  $V(G)$  and  $V(G)$  in  $G$  results in the complement of  $G$ . And if  $v$  is a vertex of  $G$ , then applying a flip between  $\{v\}$  and the neighborhood  $N(v)$  of  $v$  has the same effect as *isolating*  $v$ , that is, removing all the edges adjacent to  $v$  in  $G$ . If  $G$  is a graph and  $\mathcal{P}$  is a partition of its vertex set, then call a graph  $G'$  a  $\mathcal{P}$ -*flip* of  $G$  if  $G'$  can be obtained from  $G$

<sup>5</sup>The paper [RT08] considers a variant of the game in which the robber is *lazy*, that is, does not move unless a cop is placed at his location, whereas the [LPPT20] considers a variant where the cops occupy edges instead of vertices, and the robber never remains put.

by performing a sequence of flips between pairs of parts  $A, B \in \mathcal{P}$  (possibly with  $A = B$ ). Since flips are involutive and commute with each other, such a sequence of flips can be specified by a set of at most  $\binom{|\mathcal{P}|+1}{2}$  unordered pairs of elements of  $\mathcal{P}$ . Finally, call  $G'$  a  $k$ -flip of  $G$ , if  $G'$  is a  $\mathcal{P}$ -flip of  $G$ , for some partition  $\mathcal{P}$  of  $V(G)$  with  $|\mathcal{P}| \leq k$ .

**Remark V.1.** There are many other, functionally equivalent, ways to measure the complexity of a  $k$ -flip (also called a *perturbation*)  $G'$  of  $G$ . For example, say that  $G'$  is a  $k$ -sequential-flip of  $G$  if  $G'$  is obtained from  $G$  by applying a sequence of flips between  $k$  pairs of arbitrary subsets of  $V(G)$ . If  $G'$  is a  $k$ -sequential-flip of  $G$  then  $G'$  is a  $2^k$ -flip of  $G$ , and conversely, if  $G'$  is a  $k$ -flip of  $G$ , then  $G'$  is a  $\binom{k+1}{2}$ -sequential-flip of  $G$ . We could also require that flips are only applied to pairs of the form  $(A, A)$ ; this would lead to a functionally equivalent parameter, as flipping a pair  $(A, B)$  can be obtained by flipping three pairs:  $(A \cup B, A \cup B)$ ,  $(A, A)$  and  $(B, B)$ . Other, functionally equivalent measures of the complexity of a flip  $G'$  of a graph  $G$  can be defined by considering the graph  $G' \triangle G$  with vertices  $V(G)$  and edges  $E(G') \triangle E(G)$ . Note that  $G'$  is a  $k$ -flip of  $G$  if and only if  $G' \triangle G$  is a  $k$ -flip of the edgeless graph on  $V(G)$ . This is equivalent to  $G \triangle G'$  having *neighborhood diversity* [Lam12]  $k$ . The rank of the adjacency matrix of  $G \triangle G'$  over a fixed finite field, see [NiO20], [DKo6] leads to a further, functionally equivalent complexity measure of a flip.

a) *Flipper game*: We now come to the central notions of this paper. The *flipper game* with radius  $r \in \mathbb{N} \cup \{\infty\}$  and width  $k \in \mathbb{N}$ ,  $k \geq 1$ , is played on a graph  $G$  by two players, *flipper* and *runner*. In each round  $i$  of the game, a  $k$ -flip  $G_i$  of  $G$  is declared by the flipper, and the new position  $v_i \in V(G)$  is selected by the runner, as follows. Initially,  $G_0 = G$  and  $v_0$  is a vertex of  $G$  chosen by the runner. In round  $i > 0$ , the flipper announces a new  $k$ -flip  $G_i$  of  $G$ , that will be put into effect momentarily. The runner, knowing  $G_i$ , moves to a new vertex  $v_i$  by following a path of length at most  $r$  from  $v_{i-1}$  to  $v_i$  in the *previous* graph  $G_{i-1}$ . The game terminates when the runner is trapped, that is, when  $v_i$  is isolated in  $G_i$ .

**Definition V.2.** Fix  $r \in \mathbb{N} \cup \{\infty\}$ . The radius- $r$  flip-width of a graph  $G$ , denoted  $\text{fw}_r(G)$ , is the smallest number  $k \in \mathbb{N}$  such that the flipper has a winning strategy in the flipper game of radius  $r$  and width  $k$  on  $G$ .

**Definition V.3.** A class  $\mathcal{C}$  of graphs has bounded flip-width if  $\text{fw}_r(\mathcal{C}) < \infty$  for every  $r \in \mathbb{N}$ . More explicitly: for every radius  $r \in \mathbb{N}$  there is some  $c_r \in \mathbb{N}$  such that  $\text{fw}_r(G) < c_r$  for all  $G \in \mathcal{C}$ .

**Remark V.4.** The cop-width parameters considered in Section IV are functionally equivalent (more precisely, each parameter can be bounded from above by a linear function of the other) to parameters defined by a variant of the flipper game, call it the *isolation game*, which is

played as the flipper game, but each graph  $G_i$  announced by the flipper/cops is obtained from  $G$  by isolating at most  $k$  vertices in  $G$  (see [Tor23]). The difference between the isolation game and the Cops and Robber game is that in the Cops and Robber game, the runner/robber can move through a vertex from which a cop has just departed by a helicopter, while in the isolation game, they cannot.

We now argue that the flip-width parameters are bounded in terms of the corresponding cop-width parameters.

**Lemma V.5.** For every  $r \in \mathbb{N} \cup \{\infty\}$  and graph  $G$ , we have

$$\text{fw}_r(G) \leq \text{copwidth}_r(G) + 2^{\text{copwidth}_r(G)}. \quad (3)$$

*Proof.* The main observation is that isolating a set  $S$  of at most  $k$  vertices in  $G$  can be achieved by performing a  $(k + 2^k)$ -flip: consider the partition  $\mathcal{P}_S$  that partitions  $S$  into singletons and  $V(G) - S$  according to the neighborhood in  $S$ , and flip  $\{s\}$  with every class of the partition that is complete to  $\{s\}$ . Note that  $|\mathcal{P}_S| \leq k + 2^k$ . Now, if the cops have a winning strategy in the Cops and Robber game of radius  $r$  and width  $k$  on a graph  $G$ , we can use this strategy in the flipper game of radius  $r$  and width  $k + 2^k$ , as follows: whenever the cops announce a new set  $S$  of positions of the cops in the Cops and Robber game, in the flipper game, the flipper announces the graph  $G'$  with the vertices in  $S$  isolated, which is a  $(k + 2^k)$ -flip of  $G$ . It is easy to verify that if the cops win in the Cops and Robber game, then, playing according to the above strategy, the flipper also wins in the flipper game. Inequality (3) follows.  $\square$

The following gives an improved bound. Note that if  $\text{copwidth}_1(G) \leq t$  then  $G$  excludes  $K_{t,t}$  as a subgraph, by Theorem IV.4 and the fact that  $K_{t,t}$  is not  $(t - 1)$ -degenerate. From this, one can bound the size of the partition  $\mathcal{P}_S$  considered above by  $k^t$  (see [Tor23]), and obtain:

**Theorem V.6.** Fix  $r \in \mathbb{N} \cup \{\infty\}$ . Let  $G$  be a graph and let  $t$  be the smallest number such that  $G$  excludes  $K_{t,t}$  as a subgraph; in particular  $t \leq \text{copwidth}_1(G) \leq \text{copwidth}_r(G)$ . If  $t \geq 3$  then

$$\text{fw}_r(G) \leq \text{copwidth}_r(G)^t,$$

and if  $t = 2$  then

$$\text{fw}_r(G) \leq O(\text{copwidth}_r(G)^t).$$

In particular, every class  $\mathcal{C}$  with bounded expansion has bounded flip-width.

Whereas the cop-width parameters are monotone with respect to the subgraph relation, the flip-width parameters are monotone with respect to the induced subgraph relation. This is expressed by the following, immediate lemma.

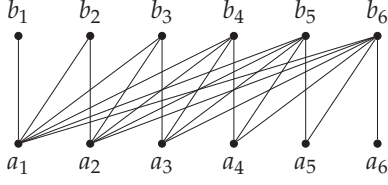


Fig. 2: A half-graph of order 6.

**Lemma V.7.** Fix  $r \in \mathbb{N} \cup \{\infty\}$ . If  $H$  is an induced subgraph of  $G$ , then  $\text{fw}_r(H) \leq \text{fw}_r(G)$ . In particular, a class  $\mathcal{C}$  has bounded flip-width if and only if its hereditary closure has bounded flip-width.

#### A. Examples

We start by giving some example classes of bounded flip-width. By Theorem V.6, every class of bounded expansion has bounded flip-width, see Example III.2 for some specific classes. Unlike cop-width, flip-width is not limited to sparse graphs, and is geared towards the study of dense graphs.

*Example V.8.* If  $\bar{G}$  is the complement of  $G$  and  $r \in \mathbb{N} \cup \{\infty\}$ , then  $\text{fw}_r(\bar{G}) = \text{fw}_r(G)$ , since any  $k$ -flip of  $G$  is also a  $k$ -flip of  $\bar{G}$  (just complement the set of flipped pairs). Therefore, if  $\mathcal{C}$  is a class with bounded expansion, then the class  $\bar{\mathcal{C}} := \{\bar{G} \mid G \in \mathcal{C}\}$ , has bounded flip-width, and if  $\mathcal{C}$  has bounded treewidth, then  $\text{fw}_\infty(\bar{\mathcal{C}}) = \text{fw}_\infty(\mathcal{C}) < \infty$ . In particular, as every edgeless graph  $G$  has  $\text{fw}_\infty(G) = 1$ , it follows that every clique  $G$  also has  $\text{fw}_\infty(G) = 1$ .

*Example V.9.* Consider the half-graph  $H_n$  of order  $n$ , as depicted in Figure 2. We show that  $\text{fw}_\infty(H_n) \leq 4$ . Observe that applying a flip between  $\{a_1, \dots, a_{i-1}\}$  and  $\{b_i, \dots, b_n\}$  breaks the half-graph into two connected components (each being a half-graph). Additionally flipping  $\{a_i\}$  and  $\{b_i, \dots, b_n\}$  makes  $a_i$  and  $b_i$  isolated. The strategy of the flipper is to perform the above two flips in the  $i$ th round, thus pushing the runner rightwards in each round. To perform those flips, partition  $V(H_n)$  into four parts:  $\{1, \dots, a_{i-1}\}$ ,  $\{a_i\}$ ,  $\{b_i, \dots, b_n\}$ , and the rest.

*Example V.10.* The comparability graph  $G$  of a rooted tree  $T$  is the graph with vertices  $V(T)$ , where two vertices are adjacent if and only if one is an ancestor of the other in  $T$ . Generalizing half-graphs, those graphs also have  $\text{fw}_\infty(G) \leq 4$ . The strategy of the flipper is, in round  $i$ , to consider the node  $u_i$  at depth  $i$  in the tree, that is the ancestor of the current position of the runner, and to isolate  $u_i$ , and remove all edges between the descendants of  $u_i$  and the ancestors of  $u_i$ . This can be done by partitioning  $V(G)$  into four parts:  $\{u_i\}$ , the ancestors of  $u_i$ , the descendants of  $u_i$ , and the rest.

*Example V.11.* Fix  $r \in \mathbb{N} \cup \{\infty\}$ . If  $G_1, \dots, G_m$  are graphs and  $G$  is their disjoint union, then the following inequality holds:

ity holds:

$$\text{fw}_r(G) \leq \max_{1 \leq i \leq m} (\text{fw}_r(G_i)) + 1.$$

The  $+1$  comes from the fact that a partition  $\mathcal{P}$  of  $V(G_i)$  into  $k$  parts induces a partition of  $V(G)$  into  $k+1$  parts, namely the  $k$  parts of  $\mathcal{P}$ , and  $V(G) - V(G_i)$ . Thus, if the flipper is hiding in the graph  $G_i$ , the runner may translate a winning strategy on  $G_i$  of width  $k := \text{fw}_r(G_i)$ , into a winning strategy on  $G$  of width  $k+1$ .

It follows that if  $\mathcal{C}$  has bounded flip-width, then the class of disjoint unions of graphs from  $\mathcal{C}$  has bounded flip-width.

A modular partition of a graph  $G$  is a partition  $\mathcal{P}$  of  $V(G)$  such that any two distinct parts are homogeneous in  $G$ . The quotient graph  $G/\mathcal{P}$  has as vertices the parts of  $\mathcal{P}$ , and as edges pairs of distinct parts that are complete in  $G$ . An extension of the idea in Example V.11 yields the following.

**Lemma V.12 (\*).** Let  $G$  be a graph and  $\mathcal{P}$  be its modular partition. Then

$$\text{fw}_r(G) \leq \max \left( \text{fw}_r(G/\mathcal{P}), \max_{A \in \mathcal{P}} \text{fw}_r(G[A]) + 2 \right).$$

The strategy for the flipper on  $G$  first follows the strategy on  $G/\mathcal{P}$ , where each  $k$ -flip of  $G/\mathcal{P}$  is lifted naturally to a  $k$ -flip of  $G$ . Once a part  $A \in \mathcal{P}$  is isolated in the game on  $G/\mathcal{P}$ , the strategy on  $G[A]$  is used. The  $+2$  in the statement is due to the fact that every partition  $\mathcal{Q}$  of  $A$  into  $k$  parts induces a partition of  $G$  into  $k+2$  parts: the  $k$  parts of  $\mathcal{Q}$ , the (common) set of neighbors of vertices in  $A$  outside of  $A$ , and the rest. See [Tor23] for details. In particular, if  $G$  is the lexicographic product of two graphs  $H$  and  $K$  (obtained by blowing up each node of  $H$  to a copy of  $K$ ) then  $\text{fw}_r(G) \leq \max(\text{fw}_r(H), \text{fw}_r(K) + 2)$ .

The substitution closure of a class of graphs  $\mathcal{C}$  is the smallest class  $\mathcal{C}^*$  containing  $\mathcal{C}$  such that if  $G$  is a graph with a modular partition  $\mathcal{P}$  into modules  $A$  satisfying  $G[A] \in \mathcal{C}^*$ , and  $G/\mathcal{P} \in \mathcal{C}$ , then  $G \in \mathcal{C}^*$ . Intuitively, a graph in  $\mathcal{C}^*$  can be obtained from a single vertex by repeatedly blowing up vertices to graphs from  $\mathcal{C}$ .

Using similar ideas as in Lemma V.12, we prove Lemma V.13.

**Lemma V.13 (\*).** For every  $r \in \mathbb{N} \cup \{\infty\}$  and graph class  $\mathcal{C}$ , we have

$$\text{fw}_r(\mathcal{C}^*) \leq \text{fw}_r(\mathcal{C}) + 2.$$

In particular, if  $\mathcal{C}$  has bounded flip-width, then  $\mathcal{C}^*$  has bounded flip-width.

*Example V.14.* The substitution closure  $\mathcal{C}^*$  of the class  $\mathcal{C}$  of subcubic graphs (graphs with maximum degree 3) has bounded flip-width. This follows from Example IV.2, Theorem V.6, and Lemma V.13. This class not among the classes that were previously known to be tame:  $\mathcal{C}^*$

has unbounded twin-width as already  $\mathcal{C}$  has unbounded twin-width [BGK<sup>+</sup>21a], and  $\mathcal{C}^*$  is not edge-stable, in particular it is not monadically stable nor structurally nowhere dense (see Section X for definitions).

More examples are given in the following sections: they include classes of bounded clique-width (see Section V-C), classes of bounded twin-width (see Section VII), and interpretations of classes of bounded expansion (see Section VIII).

### B. Hideouts

We have already seen several examples of classes of bounded flip-width. To give examples of classes of unbounded flip-width, we need a tool for proving lower bounds. We now introduce the notion of a hideout, which is a set of vertices allowing the runner to evade the flipper indefinitely, thus allowing to prove lower bounds on  $\text{fw}_r(G)$ . In Section V-D we use this notion to prove some combinatorial properties of graphs with bounded flip-width. In particular, it will follow easily that every graph of radius-one flip-width at most  $k$  has a pair of vertices whose neighborhoods differ in at most  $2k$  vertices.

Although we will use hideouts on several occasions to prove lower bounds on flip-width, we do not know whether the existence of hideouts is a necessary condition for having large flip-width. This is stated as Question XI.3 in Section XI.

**Definition V.15.** Fix  $k, d \geq 1$  and  $r \in \mathbb{N} \cup \{\infty\}$ . A  $(r, k, d)$ -hideout in a graph  $G$  is a set of vertices  $U \subseteq V(G)$  with  $|U| > d$ , satisfying the following property. For every  $k$ -flip  $G'$  of  $G$ ,

$$|\{v \in U : |B_{G'}^r(v) \cap U| \leq d\}| \leq d, \quad (4)$$

that is, there are at most  $d$  vertices  $v \in U$  such that there are at most  $d$  vertices  $u \in U$  that are connected with  $v$  by a path of length at most  $r$  in  $G'$ .

We will show that in the flipper game with radius  $r$  and width  $k$ , a runner can hide infinitely in a  $(r, k, d)$ -hideout. Intuitively, when a  $k$ -flip  $G'$  of  $G$  is announced, the runner will want to avoid all vertices  $v \in U$  with  $|B_{G'}^r(v) \cap U| \leq d$ . The condition in a hideout guarantees that there are at most  $d$  such vertices. This will allow the runner to always move to some vertex with  $|B_{G'}^r(v) \cap U| > d$ .

**Lemma V.16.** Fix  $k \geq 1$  and  $r \in \mathbb{N} \cup \{\infty\}$ . If a graph  $G$  has a  $(r, k, d)$ -hideout  $U$  for some  $d \geq 1$ , then  $\text{fw}_r(G) > k$ .

*Proof.* Let  $U \subseteq V(G)$  be a  $(r, k, d)$ -hideout. We describe a strategy for the runner in the flipper game on  $G$  with radius  $r$  and width  $k$ , which allows to elude the flipper indefinitely. The strategy is as follows: when the flipper announces a  $k$ -flip  $G'$  of  $G$ , the runner moves to some vertex  $v \in U$  such that  $|B_{G'}^r(v) \cap U| > d$ . In the first

move, pick any  $v \in U$  with  $|B_G^r(v) \cap U| > d$ . Such a vertex exists, by (4) applied to  $G' = G$ , since  $|U| > d$ .

We show it is always possible to make a move as described in the strategy. Suppose at some point in the game, the current position  $v$  of the runner is such that

$$|B_P^r(v) \cap U| > d. \quad (5)$$

where  $P$  is the previous  $k$ -flip of  $G$  announced by the flipper (in the first round,  $P = G$ ), and that the flipper now announces the next  $k$ -flip  $N$  of  $G$ . Since  $U$  is a  $(r, k, d)$ -hideout, the set  $X \subseteq U$  of vertices  $w \in U$  such that  $|B_N^r(w) \cap U| \leq d$  satisfies  $|X| \leq d$ . By (5),  $B_P^r(v)$  contains at least one vertex  $v' \in U - X$ . The runner moves from  $v$  to  $v'$  along a path of length at most  $r$  in  $P$ . As  $v' \in U - X$ , the invariant is maintained. Therefore, playing according to this strategy, the runner can elude the flipper indefinitely, so  $\text{fw}_r(G) > k$ .  $\square$

### C. Flip-width with infinite radius

As a simple case study, we first analyze the parameter  $\text{fw}_\infty$ . Recall that its sparse analogue,  $\text{copwidth}_\infty$ , corresponds to treewidth. We show that  $\text{fw}_\infty$  is functionally equivalent to the clique-width and rank-width parameters. Those parameters extend treewidth to the setting of dense graphs, and are recalled later below. To the best of our knowledge, our result is the first characterization of graph classes of bounded clique-width, in terms of a game<sup>6</sup>, analogous to the game characterization of treewidth.

**Theorem V.17 (\*).** For every graph  $G$ , we have

$$\text{rankwidth}(G) \leq 3 \text{fw}_\infty(G) + 1 \leq O(2^{\text{rankwidth}(G)}).$$

In particular, a graph class  $\mathcal{C}$  has bounded rank-width if and only if  $\text{fw}_\infty(\mathcal{C}) < \infty$ .

The rank-width parameter is functionally equivalent to clique-width, and is often more convenient to work with. We will only use rank-width. Rank-width and clique-width are functionally equivalent, as expressed below [iOS06, Prop. 6.3]:

$$\text{rankwidth}(G) \leq \text{cliquewidth}(G) < 2^{\text{rankwidth}(G)+1}.$$

A graph  $G$  has rank-width at most  $k$  if there is a tree  $T$  whose leaves are the vertices of  $G$ , and inner nodes have degree at most 3, such that for every edge  $e$  of the tree, the bipartition  $A \uplus B$  of the leaves of  $T$  into the leaves on either side of  $e$ , has cut-rank at most  $k$ . The cut-rank of a bipartition  $A \uplus B$  of the vertex set of a graph  $G$ , denoted  $\text{rk}_G(A, B)$ , is defined as the rank, over the two-element field, of the 0, 1-matrix with rows  $A$ , columns  $B$ , where the entry at row  $a \in A$  and column  $b \in B$  is 1 if  $ab \in E(G)$  and 0 otherwise.

<sup>6</sup>The radius- $\infty$  flipper game arose in a private discussion in 2018 with Michał Pilipczuk.

The upper bound in Theorem V.17 is a generalization of the bound in Example V.10, where it is shown that comparability graphs of trees have  $\text{fw}_\infty(G) \leq 4$ . We briefly sketch the argument now.

Note that a 0,1-matrix of rank at most  $k$  over the two-element field has at most  $2^k$  distinct rows and at most  $2^k$  distinct columns. It follows that if  $G$  is a graph and  $V(G) = A \uplus B$  is a bipartition of its vertex sets with cut-rank  $\text{rk}_G(A, B) \leq k$ , then  $A$  and  $B$  can be partitioned as  $A = A_1 \uplus \dots \uplus A_p$  and  $B = B_1 \uplus \dots \uplus B_q$  with  $p, q \leq 2^k$ , so that  $A_i$  and  $B_j$  are complete in  $G$ . This implies that there is a  $2^{k+1}$ -flip  $G'$  of  $G$  which has no edges with one endpoint in  $A$  and one endpoint in  $B$ .

Therefore, if  $G$  is a graph of rank-width  $k$ , then there is a subcubic tree  $T$  with leaves  $V(G)$  such that for every edge  $e$  of  $T$ , if  $V(G) = A \uplus B$  is the bi-partition induced by  $e$  (into the leaves on either side of  $e$ ), then there is a  $2^{k+1}$ -flip  $G'$  of  $G$  which has no edges with one endpoint in  $A$  and one endpoint in  $B$ . Moreover, for every inner node  $v$  of  $T$  (of degree at most three), there is a  $O(2^k)$ -flip  $G'$  of  $G$  such that for any two vertices  $a, b \in V(G)$ , if  $a$  and  $b$  are in the same connected component of  $G'$ , then  $a$  and  $b$  are connected in  $T$  by a path that avoids  $v$ . This flip can be used by the flipper in their winning strategy, to restrain the runner to the leaves of smaller and smaller subtrees of  $T$ , similarly as in Example V.10. See [Tor23] for details.

The lower bound relies on the following result characterizing obstructions to having small rank-width.

A set  $U$  of vertices of  $G$  is *well-linked* if for every bipartition  $A \uplus B$  of  $V(G)$ , the cut-rank of  $A \uplus B$  satisfies  $\text{rk}_G(A, B) \geq \min(|A \cap U|, |B \cap U|)$ . Oum and Seymour [iOSo6, Theorem 5.2] prove the following:

**Fact V.18** ([iOSo6]). *Every graph of rank-width greater than  $k$  contains a well-linked set of size  $k$ .*

Using this, the lower bound in Theorem V.17 follows from the next lemma, which is proved in [Tor23]. The proof is due to Rose McCarty (private communication).

**Lemma V.19** (\*). *Fix a graph  $G$  and number  $k \in \mathbb{N}$ . Every well-linked set  $U$  with  $|U| > 3k$  is a  $(\infty, k, k)$ -hideout.*

This implies the lower bound in Theorem V.17 as follows. Suppose  $\text{rankwidth}(G) > 3k + 1$  for some  $k \in \mathbb{N}$ . By Fact V.18,  $G$  contains a well-linked set  $U$  of size  $3k + 1$ . By Lemma V.19,  $U$  is a  $(\infty, k, k)$ -hideout. By Lemma V.16,  $\text{fw}_\infty(G) > k$ . By contrapositive, this shows that if  $\text{fw}_\infty(G) = k$  then  $\text{rankwidth}(G) \leq 3k + 1 = 3\text{fw}_\infty(G) + 1$ .

#### D. Radius-one flip-width

We move to the study of finite radii, which are our main focus, starting with the first parameter,  $\text{fw}_1$ . We have seen that its sparse analogue,  $\text{copwidth}_1$ , corresponds precisely to degeneracy (plus one), which is a

very well-understood parameter, with many good algorithmic and combinatorial properties. The parameter  $\text{fw}_1$  enjoys many useful combinatorial properties, relating it to near-twins, and to the VC-dimension.

a) *Near-twins*: We prove a first combinatorial property of graphs with small  $\text{fw}_1(G)$ , namely that such graphs have near-twins. This has several consequences. Say that two vertices  $u, v$  of a graph  $G$  are  $\delta$ -near-twins if  $|N(u) \triangle N(v)| \leq \delta$ , where  $\triangle$  denotes the symmetric difference. We show that every graph  $G$  with  $\text{fw}_1(G) \leq k$  has a pair of  $2k$ -near-twins. More generally, we prove:

**Lemma V.20**. *Let  $b, k \in \mathbb{N}$  and let  $G$  be a graph with  $\text{fw}_1(G) \leq k$  and  $|G| > bk$ . Then  $G$  contains a set of at least  $b + 1$  vertices which are mutual  $2bk$ -near-twins.*

*Proof.* Assume that  $G$  has no set containing  $b + 1$  mutual  $2bk$ -near-twins. We prove that  $V(G)$  is a  $(1, k, bk)$ -hideout in  $G$ , which implies that  $\text{fw}_1(G) > k$  by Lemma V.16.

Let  $G'$  be a  $k$ -flip of  $G$  and let  $B$  be the set of vertices of degree at most  $bk$  in  $G'$ . We show that  $|B| \leq bk$ , proving that  $V(G)$  is a  $(1, k, bk)$ -hideout in  $G$ .

Suppose that  $|B| > bk$ . Let  $\mathcal{P}$  be a partition with  $|\mathcal{P}| \leq k$  such that  $G'$  is a  $\mathcal{P}$ -flip of  $G$ . As  $|B| > bk$ , there is a set  $B_0 \subseteq B$  with  $|B_0| > b$ , such that  $B_0$  is contained in one part of  $\mathcal{P}$ . Any two vertices of  $B_0$  are  $2bk$ -near-twins in  $G'$ , as they both have degree at most  $bk$  in  $G'$ . Since  $B_0$  is contained in a single part of  $\mathcal{P}$ , it follows that any two vertices of  $B_0$  are  $2bk$ -near-twins in  $G$ , too. But  $|B_0| > b$ , so this contradicts the assumption. Hence,  $|B| \leq bk$ .  $\square$

A bipartite variant of Lemma V.20, with a very similar proof, is as follows. Recall that if  $X, Y \subseteq V(G)$  are two sets of vertices of a graph  $G$  then  $G[X, Y]$  denotes the bipartite graph semi-induced by  $X$  and  $Y$  in  $G$ .

**Lemma V.21** (\*). *Let  $b, k \in \mathbb{N}$  and let  $G$  be a graph with  $\text{fw}_1(G) \leq k$ . Then for every two sets  $X, Y \subseteq V(G)$  with  $|X|, |Y| > bk$ , there is a subset of  $X$  or a subset of  $Y$  consisting of  $b + 1$  mutual  $2bk$ -near-twins in  $G[X, Y]$ .*

Setting  $b := 1$  in Lemma V.20 we get:

**Corollary V.22**. *Let  $G$  be a graph with  $\text{fw}_1(G) \leq k$  and  $|G| > k$ . Then  $G$  has a pair of  $2k$ -near-twins.*

We can now verify that there exist graphs  $G$  with arbitrarily large  $\text{fw}_1(G)$ . It is well known that there exist graphs of arbitrarily large girth and minimum degree.

**Corollary V.23**. *A graph  $G$  with girth larger than 4 and minimum degree larger than  $k$  has  $\text{fw}_1(G) > k$ . Therefore, there exist graphs  $G$  with arbitrarily large  $\text{fw}_1(G)$ .*

*Proof.* Any two distinct vertices  $u$  and  $v$  have at most one common neighbor, so  $|N(u) \triangle N(v)| > 2k$ , so  $G$  has no pair of  $2k$ -near-twins, hence  $\text{fw}_1(G) > k$  by Corollary V.22.  $\square$

b) *VC dimension*: We now show that the VC-dimension of a graph  $G$  is bounded in terms of  $\text{fw}_1(G)$ .

Set systems and graphs of bounded VC-dimension have many useful properties, and we use one of them later for studying classes of bounded flip-width. We also consider a related parameter, called *2VC-dimension* [BT15], and denoted  $2VCdim(G)$ . This is the maximal size of a set  $X \subseteq V(G)$  such that for every two distinct  $a, b \in X$  there is a vertex  $c \in V(G)$  with  $N_G(c) \cap X = \{a, b\}$ . Clearly,  $VCdim(G) \leq 2VCdim(G)$ . We prove the following.

**Theorem V.24 (\*)**. *For every graph  $G$  we have*

$$VCdim(G) \leq 8fw_1(G), \quad (6)$$

$$2VCdim(G) \leq 8fw_2(G) + 2 \quad (7)$$

Note that  $2VCdim$  cannot be bounded in terms of  $fw_1$ , only in terms of  $fw_2$ , as witnessed by 1-subdivided cliques, which are 2-degenerate, and hence (by Theorem V.6 and Theorem IV.4), have bounded  $fw_1$ , and clearly have unbounded  $fw_2$ . Furthermore, graphs of girth larger than 4 have VC-dimension at most two, but have arbitrarily large  $fw_1$  by Corollary V.23. Hence,  $fw_1$  is not bounded in terms of  $VCdim(G)$ .

Inequality (6) in Theorem V.24 follows from Lemma V.21 (for  $b = 1$ ) and the following.

**Lemma V.25 (\*)**. *Let  $G$  be a graph with  $VCdim(G) \geq 2^m$ , for some  $m$ . Then there are two sets  $X, Y$  such that the bipartite graph  $G[X, Y]$  contains no pair of  $(2^{m-1} - 1)$ -near-twins in either of the parts  $X, Y$ .*

In the proof, the sets  $X$  and  $Y$  are two copies of the  $m$ -dimensional vector space over the two-element field, with edges connecting vectors with a nonzero dot product. See [Tor23] for details.

We prove inequality (7) in Corollary VI.8 later.

From Lemma V.25 and Lemma V.21 (for  $b = 1$ ) we conclude the following.

**Corollary V.26**. *Let  $G$  be a graph with  $VCdim(G) \geq d$ . Then  $G$  contains an induced subgraph  $H$  with  $O(d)$  vertices and with  $fw_1(H) \geq d/8$ .*

**Corollary V.27**. *If  $\mathcal{C}$  is a hereditary class of graphs such that  $fw_1(G) \leq o(|G|)$  for  $G \in \mathcal{C}$ , then  $VCdim(\mathcal{C}) < \infty$ .*

*Proof.* If  $VCdim(\mathcal{C}) = \infty$  then for every  $d$  there is a graph  $G \in \mathcal{C}$  with  $VCdim(G) \geq d$ , and by Corollary V.26 there is  $H \in \mathcal{C}$  with  $O(d)$  vertices and  $fw_1(H) = \Omega(d)$ . Since this holds for all  $d \in \mathbb{N}$ , it cannot be that  $fw_1(G) \leq o(|G|)$  for all  $G \in \mathcal{C}$ .  $\square$

*E. Relationship of radius-one flip-width with other graph parameters*

In this section, we compare radius-one flip-width with existing graph parameters. First, we mention that in Section VI-A we show that for graphs  $G$  that exclude a fixed biclique  $K_{t,t}$  as a subgraph, we have that  $fw_1(G)$  is functionally equivalent to the degeneracy of  $G$ . However, unlike degeneracy,  $fw_1$  is bounded for some graphs that contain arbitrarily large cliques or bicliques. Thus,

$fw_1(G)$  may be thought of as a dense extension of degeneracy. Two other graph parameters – *symmetric difference* and *functionality* – which could serve a similar role, are discussed below.

For a graph  $G$ , define the *symmetric difference* of  $G$ , denoted  $sd(G)$ , as the least number  $k \in \mathbb{N}$ , such that every induced subgraph  $H$  of  $G$  contains a pair  $u, v$  of  $k$ -near-twins. Classes  $\mathcal{C}$  with  $sd(\mathcal{C}) < \infty$  are called *classes with bounded symmetric difference*. For example, if  $G$  has degeneracy  $k$ , then  $sd(G) \leq 2k$ . In particular, every graph class with bounded degeneracy has bounded symmetric difference.

Since  $fw_1$  is monotone under taking induced subgraphs, Corollary V.22 implies that  $sd(G) \leq 2fw_1(G)$  for every graph  $G$ . Hence, every graph class  $\mathcal{C}$  with  $fw_1(\mathcal{C}) < \infty$  has bounded symmetric difference. We do not know whether the converse implication holds.

A more general graph parameter, called *graph functionality* and denoted  $fun(G)$ , is defined in [AAL21]. The following is a rephrasing of the original definition.

Say that a vertex  $v \in V(G)$  is a *function* of a set  $S \subseteq V(G)$  of vertices, with  $s \notin S$ , if for every  $w \in V(G) - (S \cup \{v\})$ , the adjacency of  $w$  and  $v$  in  $G$  depends only on  $N(w) \cap S$ . For instance, if  $v$  is a  $k$ -near-twin of  $u$ , then  $v$  is a function of  $\{u\} \cup (N(v) \triangle N(u)) - \{v\}$ . Define  $fun(G)$  as the least number  $k$  such that every induced subgraph  $H$  of  $G$  has a vertex  $v$  that is a function of at most  $k$  other vertices of  $H$ .

Clearly,  $fun(G) \leq sd(G) + 1$  holds for every graph  $G$ , so classes of bounded symmetric difference have bounded functionality. The converse implication is false, as witnessed by the class of permutation graphs [AAL21, Sec. 2.4], which have bounded functionality and unbounded symmetric difference.

Classes of bounded functionality have bounded VC-dimension, as shown in [AAL21, Thm. 8]. However, no effective bound is known (see [AAL21, Open problem 4]).

It follows from [ACLZ15, Thm. 2] that every hereditary graph class of bounded functionality is at most *factorial* – contains at most  $2^{O(n \log n)}$  labelled graphs on  $n$  vertices. In fact, every class that is at most factorial has bounded VC-dimension (see [AAL21, Sec. 3]).

Figure 3 summarizes the relationships among the discussed properties of graph classes.

## VI. FLIP-WIDTH IN WEAKLY SPARSE CLASSES

We have seen in Theorem V.6 radius-one flip-width is upper bounded in terms of degeneracy, and radius- $r$  flip-width is upper bounded in terms of generalized coloring numbers. In this section, we provide bounds in the other direction, in weakly sparse classes. It follows that for weakly sparse classes, having bounded flip-width is equivalent to having bounded expansion.

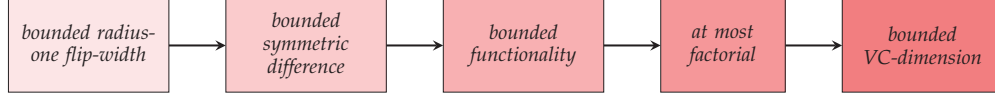


Fig. 3: Properties of graph classes and implications among them.

#### A. Radius-one flip-width and degeneracy

As  $\text{fw}_1$  is bounded in terms of  $\text{copwidth}_1$ , which is equivalent to degeneracy by Theorem IV.4, it follows that every class of bounded degeneracy has bounded  $\text{fw}_1$ . Clearly, every class of degeneracy bounded by  $t$  is weakly sparse, as it excludes  $K_{t+1,t+1}$  as a subgraph. We show that for weakly sparse classes, bounded degeneracy is equivalent to having bounded  $\text{fw}_1$ .

**Theorem VI.1.** *If  $G$  is a graph that avoids  $K_{t,t}$  as a subgraph, then*

$$\text{degeneracy}(G)/(2t^2) < \text{fw}_1(G) \leq (\text{degeneracy}(G) + 1)^t. \quad (8)$$

*As a consequence, if  $\mathcal{C}$  is a weakly sparse class of graphs then  $\text{fw}_1(\mathcal{C}) < \infty$  if and only if  $\mathcal{C}$  has bounded degeneracy.*

*Proof.* The second inequality is by Theorem V.6 and Theorem IV.4. To prove the first inequality, we show that  $\text{degeneracy}(G) < 2kt^2$ , where  $k := \text{fw}_1(G)$ .

Recall that  $G$  is  $d$ -degenerate if and only if every induced subgraph of  $G$  contains a vertex of degree at most  $d$ . Therefore, to prove  $\text{degeneracy}(G) < 2kt^2$  it is enough to show that  $G$  contains some vertex of degree less than  $2kt^2$  (since the same holds for every induced subgraph  $H$  of  $G$ , as  $\text{fw}_1(H) \leq \text{fw}_1(G) \leq k$ ).

Setting  $b := t - 1$  in Lemma V.20 we get that  $G$  contains a set  $U$  of  $t$  mutual  $2k(t-1)$ -near-twins. Pick any  $v \in U$ . Then all  $u \in U$  are  $2k(t-1)$ -near-twins of  $v$ , so  $|N(v) \Delta N(u)| \leq 2k(t-1)$  for all  $u \in U$ .

We show that  $|N(v)| < 2k(t-1)^2 + 2t$ . Suppose that  $|N(v)| \geq 2k(t-1)^2 + 2t$ . Then the set  $W := \bigcap_{u \in U} N(u)$  has at least  $2t$  elements, and so  $W - U$  has at least  $t$  elements. As every  $u \in U$  is adjacent to every  $w \in W$  and  $U$  and  $W$  are disjoint, we have a copy of  $K_{t,t}$  as a subgraph of  $G$ , a contradiction. Hence,  $|N(v)| < 2k(t-1)^2 + 2t < 2kt^2$ .  $\square$

#### B. Radius- $r$ flip-width and generalized coloring numbers

We have seen that every weakly sparse class  $\mathcal{C}$  has bounded degeneracy if and only if  $\text{fw}_1(\mathcal{C}) < \infty$ . It is known that a weakly sparse class  $\mathcal{C}$  has bounded clique-width if and only if  $\mathcal{C}$  has bounded tree-width, as clique-width and treewidth are functionally equivalent in weakly sparse classes [GW00]. This proves the following.

**Corollary VI.2.** *A graph class  $\mathcal{C}$  has bounded treewidth if and only if  $\mathcal{C}$  is weakly sparse and  $\text{fw}_\infty(\mathcal{C}) < \infty$ .*

A general theme in structural graph theory is that a dense graph parameter is often functionally equivalent

to its sparse counterpart in weakly sparse classes. We now show that for weakly sparse classes, bounded flip-width is indeed equivalent to bounded expansion:

**Theorem VI.3.** *A class  $\mathcal{C}$  has bounded expansion if and only if  $\mathcal{C}$  is weakly sparse and has bounded flip-width.*

We prove a more precise result, from which Theorem VI.3 follows.

**Theorem VI.4.** *For every  $r \geq 1$  and graph  $G$  we have*

$$\tilde{\nabla}_{r-1}(G) \leq O(r \cdot \text{fw}_r(G) \cdot \text{degeneracy}(G))^{36}.$$

We first show how Theorem VI.4 implies Theorem VI.3.

*Proof of Theorem VI.3.* For the forward direction, assume  $\mathcal{C}$  has bounded expansion. Then  $\mathcal{C}$  has bounded flip-width by Theorem V.6. Also, as remarked before Theorem V.6, every class with bounded expansion is weakly sparse.

Conversely, suppose  $\mathcal{C}$  is weakly sparse and has bounded flip-width. In particular,  $\text{fw}_1(\mathcal{C}) < \infty$ , and by Theorem VI.1,  $\mathcal{C}$  has degeneracy bounded by some constant  $d$ . By Theorem VI.4, for all  $r \geq 1$  we have that  $\tilde{\nabla}_{r-1}(\mathcal{C}) \leq O(r \cdot \text{fw}_r(\mathcal{C}) \cdot d)^{36} < \infty$ . Hence,  $\mathcal{C}$  has bounded expansion.  $\square$

In the remainder of Section VI-B we prove Theorem VI.4. Our proof relies on a result of Dvořák [Dvo18], extending a result of Kühn and Osthus [KO04], which we now recall.

An *exact  $r$ -subdivision* of a graph  $G$  is the graph obtained by replacing every edge of  $G$  by a path of length  $r+1$ . If every edge is replaced by a path of length at most  $r+1$ , the resulting graph is an  $\leq r$ -subdivision of  $G$ . For a graph  $G$ , let  $\tilde{\nabla}_r^e(G)$  denote the maximum average degree of all graphs  $H$  whose exact  $r$ -subdivision is an induced subgraph of  $G$ .

The following is [Dvo18, Lemma 9].

**Lemma VI.5.** *For every  $r, k, d \geq 1$  there is a number  $s = s(r, k, d) \leq O(rdk)^{12}$  such that for every graph  $G$ , if  $\tilde{\nabla}_0(G) \leq d$  and  $\tilde{\nabla}_r^e(G) < k$ , then  $\tilde{\nabla}_r(G) < \tilde{\nabla}_{r-1}(G) + s$ .*

An easy induction on  $r$  yields the following.

**Corollary VI.6.** *For every  $r, d \geq 1$  and  $d$ -degenerate graph  $G$ ,*

$$\tilde{\nabla}_r(G) \leq O(dr \cdot \sum_{t=1}^r \tilde{\nabla}_t^e(G))^{12}.$$

It is well known that every graph of average degree at least  $d$  contains a subgraph with minimum degree

at least  $d/2$ . The following proposition yields Theorem VI.4, using Corollary VI.6 and Fact III.5.

**Proposition VI.7 (\*)**. *Fix  $r \geq 2, k \geq 1$ . Let  $G$  be the exact  $(r-1)$ -subdivision of some graph  $H$  with minimum degree at least  $2rk$ . Then  $\text{fw}_r(G) > k$ .*

Proposition VI.7 is proved in [Tor23, Appendix F], by showing that the vertices of  $G$  that correspond to the vertices of  $H$  (those of degree larger than 2) form a  $(r, k, k)$ -hideout in  $G$ . We now show how this proves Theorem VI.4.

*Proof of Theorem VI.4.* Denote  $k = \text{fw}_r(G)$ . Since  $\text{fw}_r(G)$  is monotone in  $r$ , by Proposition VI.7, for all  $t < r$ , the graph  $G$  does not contain an induced  $t$ -subdivision of a graph of minimum degree at least  $2rk$ . Hence,  $\tilde{\nabla}_t^e(G) \leq 4rk$  for all  $t < r$ .

If  $G$  is  $d$ -degenerate then by Corollary VI.6 we have:

$$\tilde{\nabla}_{r-1}^e(G) \leq O(dr \sum_{1 \leq t < r} \tilde{\nabla}_t^e(G))^{12} \leq O(dr^3k)^{12} \leq O(drk)^{36},$$

as required.  $\square$

Proposition VI.7 easily implies the following corollary (see [Tor23] for a proof), yielding inequality (7) in Theorem V.24.

**Corollary VI.8 (\*)**. *If  $G$  is the exact 1-subdivision of an  $n$ -clique, then  $\text{fw}_2(G) > (n-1)/4$ . Furthermore, for every graph  $G$ ,  $2\text{VCdim}(G) \leq 8\text{fw}_2(G) + 2$ .*

Hence, weakly sparse classes of bounded flip-width, being exactly the classes with bounded expansion, are by now very well-understood and characterized in multiple ways. For instance, the model checking problem for first-order logic is fixed-parameter tractable for such classes, by the result of Dvořák, Král, and Thomas [DKT13].

## VII. FLIP-WIDTH OF ORDERED GRAPHS AND TWIN-WIDTH

Let us return to dense graph classes, which are our main focus. As follows from Theorem V.17, radius- $\infty$  flip-width is functionally equivalent to clique-width, so classes of bounded clique-width are examples of dense graph classes of bounded flip-width. In this section we show that classes of bounded twin-width have bounded flip-width, but the converse does not hold. To characterize twin-width in terms of flip-width, we study flip-width of graphs equipped with a total order. We start with recalling the definition of twin-width.

An *uncontraction sequence* of a graph  $G$  is a sequence  $\mathcal{P}_1, \dots, \mathcal{P}_n$  of partitions of  $V(G)$  that starts with the partition  $\mathcal{P}_1$  with one part, ends with the partition of  $V(G)$  into singletons, and such that every partition  $\mathcal{P}_{i+1}$ , for  $i < n$ , is obtained from the previous partition  $\mathcal{P}_i$  by splitting one of the parts into two. The *red graph* of a partition  $\mathcal{P}$  is the graph whose nodes are the parts of  $\mathcal{P}$ , with *red edges* connecting two distinct nodes  $A, B \in \mathcal{P}$  if

$A$  and  $B$  are not homogeneous in  $G$ . A graph has twin-width at most  $d$  if it has an uncontraction sequence such that at every time  $i$ , the red graph of the partition  $\mathcal{P}_i$  has maximum degree at most  $d$ .

First, we prove that  $\text{fw}_r(G)$  is bounded in terms of  $r$  and the twin-width  $\text{tw}(G)$  of  $G$ .

**Theorem VII.1 (\*)**. *Fix  $r \in \mathbb{N}$ . For every graph  $G$  of twin-width  $d$  we have:*

$$\text{fw}_r(G) \leq 2^d \cdot d^{O(r)}.$$

*In particular, every class of bounded twin-width has bounded flip-width.*

In the proof, the flipper uses an uncontraction sequence in order to vanquish the runner by constraining them, in round  $i$  to some ball of radius  $r$  in the red-graph of the partition  $\mathcal{P}_i$ . In round  $i = n$ , as every ball of radius  $r$  in the red graph of  $\mathcal{P}_n$  comprises a single vertex of  $G$ , the runner is trapped. We use the bounds from [BFLP24] to optimize the bounds. See [Tor23] for details.

Therefore, bounded twin-width implies bounded flip-width, but the converse does not hold: the class of subcubic graphs has bounded expansion, and hence bounded flip-width, but does not have bounded twin-width [BGK<sup>+</sup>21a]. So can twin-width be exactly characterized in terms of flip-width?

The twin-width parameter is defined not only for graphs, but also for structures equipped with one or more binary relations. As argued in [BGODM<sup>+</sup>22], twin-width may – and perhaps even should – be seen as a parameter of *ordered* graphs, rather than graphs. An ordered graph  $G = (V, E, <)$  is equipped with a (symmetric, irreflexive) edge relation  $E$  and a total order relation  $<$ . Every graph can be equipped with some total order without increasing the twin-width, so we may assume the order is present (also such an order can be easily computed from an uncontraction sequence, but the problem of finding it efficiently given the graph  $G$  only, remains open). Similarly as twin-width, the notion of flip-width extends to structures with several binary relations, as defined below. This, in particular implies a notion of flip-width for ordered graphs, but in the case of ordered graphs, we also given an alternative, more convenient definition.

a) *Flip-width of binary structures*: Fix a signature  $\Sigma$  consisting of several binary relations. For a  $\Sigma$ -structure  $A$  and partition  $\mathcal{P}$  of  $V(A)$ , a  $\mathcal{P}$ -flip of  $A$  is specified by a  $\Sigma$ -structure  $F$  with domain  $\mathcal{P}$ . Applying the flip  $F$  results in a  $\Sigma$ -structure  $A'$  with domain  $V(A)$  and relations

$$R_{A'} := R_A \triangle \bigcup_{(P,Q) \in R_F} P \times Q,$$

for each binary relation symbol  $R \in \Sigma$ . A  $k$ -flip of  $A$  is any structure  $A'$  obtained in this way, for some partition  $\mathcal{P}$  with  $|\mathcal{P}| \leq k$ . Using this notion of a  $k$ -flip, the flipper

game and the flip-width parameters  $\text{fw}_r(A)$  are defined just as for graphs (see [Tor23] for details), where in each round, the runner can traverse at most  $r$  edges of the Gaifman graph of the previously announced  $k$ -flip. Theorem VII.1 holds also in the case when  $G$  is a binary structure, rather than a graph, with the same proof.

*b) Ordered flip-width:* In the case of ordered graphs, it is convenient to work with the following variant of flip-width which takes into account that one of the binary relations is a total order. Fix an ordered graph  $G = (V, E, <)$ . A  $k$ -cut-flip of  $G$  is a triple  $G' = (V, E', S)$ , where  $(V, E')$  is a (usual) graph that is a  $k$ -flip of the graph  $(V, E)$ , and  $S \subseteq V$  is a set with  $|S| \leq k$ . Intuitively, the runner will be able to instantaneously move between any two points that are not separated by an element of  $S$ . Let  $\sim_S$  denote the equivalence relation on  $S$  such that  $u \sim_S v$  if and only if  $u = v$  or  $u < v$  and there is no  $s \in S$  with  $u \leq s \leq v$ . The *weighted graph* associated to  $G' = (V, E', S)$  is the graph with vertices  $V$  and edges  $uv$  such that  $uv \in E'$  or  $u \sim_S v$ , where each edge  $uv$  with  $u \sim_S v$  has weight 0, and the remaining edges have weight 1. Fix a radius  $r \in \mathbb{N} \cup \{\infty\}$ , a width parameter  $k$ , and an ordered graph  $G = (V, E, <)$ . In the *ordered flipper game* with radius  $r$  and width  $k$  on an ordered graph  $G = (V, E, <)$ , in round  $i$ , the flipper announces a  $k$ -cut-flip  $G_i = (V, E', S)$  of the ordered graph  $G = (V, E, <)$ , and the runner moves from his previous position  $v_{i-1}$  to a new position  $v_i$  by following a path of total weight at most  $r$  in the weighted graph associated with the previous  $k$ -cut-flip  $G_{i-1}$  (in round  $i = 1$ , as the flipper announces  $G_1$ , the runner picks  $v_1 \in V$  arbitrarily). The flipper wins if  $v_i$  is isolated in the weighted graph associated to  $G_i$ , that is, there is no  $w \neq v$  with  $vw \in E'$  or  $v \sim_S w$ . The *radius- $r$  ordered flip-width* of an ordered graph  $G = (V, E, <)$ , denoted  $\text{fw}_r^<(G)$ , is the smallest number  $k$  such that the flipper has a winning strategy in the ordered flipper game with radius  $r$  and width  $k$  on  $G$ .

For an ordered graph  $G = (V, E, <)$ , the parameter  $\text{fw}_r^<(G)$  relates to the flip-width of  $G$ , treated as a binary structure, as follows (see [Tor23]).

**Lemma VII.2 (\*).** *Fix  $r \in \mathbb{N} \cup \{\infty\}$  and an ordered graph  $G = (V, E, <)$ . Then*

$$\sqrt{\text{fw}_r(G) + 1} \leq \text{fw}_r^<(G) + 1 \leq \text{fw}_{3r+2}(G) + 1.$$

*c) Equivalence of flip-width and twin-width for ordered graphs:* We now prove the main result of Section VII, which says that for ordered graphs  $G = (V, E, <)$ , the parameters  $\text{tww}(G)$  and  $\text{fw}_1^<(G)$  are functionally equivalent. In other words, a class of ordered graphs has bounded twin-width if and only if it has bounded radius-one ordered flip-width.

**Theorem VII.3.** *The following conditions are equivalent for a class  $\mathcal{C}$  of ordered graphs:*

- 1)  $\mathcal{C}$  has bounded twin-width,
- 2)  $\mathcal{C}$  has bounded flip-width, that is,  $\text{fw}_r(\mathcal{C}) < \infty$  for every  $r \in \mathbb{N}$ ,
- 3)  $\mathcal{C}$  has bounded radius-five flip-width (as a class of binary structures), that is,  $\text{fw}_5(\mathcal{C}) < \infty$ ,
- 4)  $\mathcal{C}$  has bounded ordered flip-width, that is,  $\text{fw}_r^<(\mathcal{C}) < \infty$  for every  $r \in \mathbb{N}$ ,
- 5)  $\mathcal{C}$  has bounded radius-one ordered flip-width, that is,  $\text{fw}_1^<(\mathcal{C}) < \infty$ .

So for ordered graphs, among the parameters  $\text{fw}_r^<$ , only the parameters  $\text{fw}_\infty^<$  (characterizing bounded clique-width) and  $\text{fw}_1^<$  (characterizing bounded twin-width) are relevant, since it follows that  $\text{fw}_r^<(G)$  is bounded in terms of  $\text{fw}_1^<(G)$  and  $r$ , for every  $r \in \mathbb{N}$  and ordered graph  $G$ .

We obtain the following characterization of twin-width of usual, unordered graphs, in terms of flip-width. Recall that every graph  $G$  can be equipped with some total order without increasing the twin-width, and conversely, forgetting a total order of an ordered graph does not increase the twin-width.

**Corollary VII.4.** *A class  $\mathcal{C}$  of graphs has bounded twin-width if and only if every graph in  $\mathcal{C}$  can be equipped with a total order, so that the resulting class of ordered graphs has bounded flip-width.*

*Proof of Theorem VII.3.* The implication  $1 \rightarrow 2$  is by Theorem VII.1 (stated for binary structures); the implications  $2 \rightarrow 3$  and  $4 \rightarrow 5$  are immediate; the implications  $2 \rightarrow 4$  and  $3 \rightarrow 5$  are by Lemma VII.2. It remains to prove the implication  $5 \rightarrow 1$ . This is done in Lemma VII.5 below, which proves that for ordered graphs, twin-width is bounded in terms of  $\text{fw}_1^<$ .  $\square$

We show that ordered graphs of large twin-width have large  $\text{fw}_1^<$ . To this end, we use a core result of [BGdM<sup>+</sup>22], which states that an ordered graph has large twin-width if and only if it contains a  $k$ -rich division, for a large number  $k$ . A  $k$ -rich division of an ordered graph  $G$  is a pair of partitions  $\mathcal{L}, \mathcal{R}$  of  $V(G)$ , whose parts are intervals with respect to the order, such that for every interval  $A \in \mathcal{L}$  and  $k$  intervals  $B_1, \dots, B_k \in \mathcal{R}$ , there are at least  $k$  vertices in  $A$  with pairwise distinct neighborhoods in  $V(G) - (B_1 \cup \dots \cup B_k)$ , and symmetrically, for every interval  $B \in \mathcal{R}$  and  $k$  intervals  $A_1, \dots, A_k \in \mathcal{L}$ , there are at least  $k$  vertices in  $B$  with pairwise distinct neighborhoods in  $V(G) - (A_1 \cup \dots \cup A_k)$ . It is shown in [BGdM<sup>+</sup>22, Theorem 21] that if  $G$  has no  $k$ -rich division, then  $\text{tww}(G) \leq 2^{O(k^2)}$ . We now show that a  $(k+1)$ -rich division can be employed by the runner to evade the flipper in the ordered flipper game of radius 1 and width  $k$ . The following lemma immediately yields the implication  $4 \rightarrow 1$  in Theorem VII.3, and finishes its proof.

**Lemma VII.5.** *Let  $G$  be an ordered graph with  $\text{fw}_1^<(G) \leq k$ . Then  $G$  does not have a  $(k+1)$ -rich division  $\mathcal{L}, \mathcal{R}$ . In particular,  $\text{tw}_w(G) \leq 2^{O(k^2)}$ .*

*Proof.* Suppose  $G$  has a  $(k+1)$ -rich division  $\mathcal{L}, \mathcal{R}$ . We show that  $\text{fw}_1^<(G) > k$ , by describing a winning strategy for the runner in the ordered flipper game with radius one and width  $k$ .

The strategy is as follows: in round  $i$ , when the flipper announces a new  $k$ -cut-flip  $G_i = (V, E_i, S_i)$  of  $G$ , the runner always moves to any reachable vertex in one of the parts of  $\mathcal{L}$  (in even-numbered rounds) or of  $\mathcal{R}$  (in odd-numbered rounds) that does not contain any element of  $S_i$ . We show that the runner can always reach such a vertex  $v_i$  from his previous position  $v_{i-1}$ , by following a path of weight 1 in the previous  $k$ -cut-flip  $G_{i-1}$  of  $G$ .

By inductive assumption, suppose that  $v_{i-1}$  belongs to a part  $A$  of  $\mathcal{L}$  or  $\mathcal{R}$  that does not contain any element of  $S_{i-1}$ . Suppose, by symmetry, that  $A$  is a part of  $\mathcal{L}$ . Since  $|S_i| \leq k$ , there are at most  $k$  parts of  $\mathcal{R}$  that contain some element of  $S_i$ ; denote those parts by  $B_1, \dots, B_k$ . By the condition of a  $(k+1)$ -rich division there are  $k+1$  vertices in  $A$  with pairwise distinct neighborhoods in  $B := V(G) - (B_1 \cup \dots \cup B_k)$ . Then in the previous  $k$ -cut-flip  $G_{i-1}$  of  $G$ , there is an edge joining some vertex  $a \in A$  with some vertex  $b \in B$ . Indeed, let  $\mathcal{P}$  denote the partition of  $V(G)$  underlying the flip  $G_{i-1}$  of  $G$ , with  $|\mathcal{P}| \leq k$ . By the pigeonhole principle, two vertices  $a_1, a_2$  of  $A$  with distinct neighborhoods in  $B$ , ie.  $N_G(a_1) \cap B \neq N_G(a_2) \cap B$  belong to the same part of  $\mathcal{P}$ . Therefore also  $N_{G_{i-1}}(a_1) \cap B \neq N_{G_{i-1}}(a_2) \cap B$ , so one of those sets must be nonempty.

Thus, there is  $a \in A$  and  $b \in B$  such that  $ab \in E(G_{i-1})$ . The runner thus moves to  $b$ , by the path  $v_{i-1} - a' - b$  of weight 1 in  $G_{i-1}$ , maintaining the invariant.  $\square$

Reassuming, the duality result of [BGOdM<sup>+</sup>22] proving the equivalence of unbounded twin-width and having  $k$ -rich divisions for all  $k$ , can be seen as a min-max theorem for the flipper game of radius 1, for ordered graphs. Moreover, we now see that degeneracy and twin-width are two flip sides of the same coin:  $\text{fw}_1(G)$  corresponds to the degeneracy of  $G$  for weakly sparse graphs (see Theorem IV.4 and Theorem VI.3), and  $\text{fw}_1^<(G)$  corresponds to the twin-width of  $G$  for ordered graphs (see Theorem VII.3). Similarly, classes of bounded flip-width coincide with classes of bounded expansion in the weakly sparse case, and with classes of bounded twin-width in the ordered case. Both of those cases is by now well-understood from an algorithmic, combinatorial, and logical perspective. In particular, the model checking problem for first-order logic is fixed-parameter tractable in each of the two special cases.

## VIII. CLOSURE UNDER TRANSDUCTIONS

As we have seen, classes of bounded flip-width include all classes of bounded expansion and all classes of bounded twin-width, and characterize those notions in the weakly sparse and totally ordered settings. We argue that classes of bounded flip-width enjoy many good closure properties. For instance, if two classes  $\mathcal{C}$  and  $\mathcal{D}$  have bounded flip-width, then their union  $\mathcal{C} \cup \mathcal{D}$  also has bounded flip-width. Other such properties include: closure under disjoint unions (see Example V.11), and closure under substitution (see Lemma V.13).

An entire family of closure properties is provided by the notion of first-order *interpretations* or *transductions*. As a very special instance, we saw in Example V.8 that if  $\mathcal{C}$  has bounded flip-width, then the class of edge-complements of graphs from  $\mathcal{C}$  also has bounded flip-width. What about, say, the class of squares of graphs from  $\mathcal{C}$ ? (The square of a graph  $G$  has vertices  $V(G)$  and edges  $uv$  such that  $u$  and  $v$  have a common neighbor in  $G$ .) We show that this class also has bounded flip-width, and a similar result holds for every operation that can be defined by a first-order formula, as we now describe. We phrase our result in greater generality for colored graphs. This result has multiple corollaries, and generalizes previous results.

### A. Preservation of flip-width under transductions

We start with defining interpretations and transductions, and then state the main result of this section.

a) *Colored graphs:* Recall that a  $c$ -colored graph is a graph together with an assignment of colors from  $\{1, \dots, c\}$  to its vertices. For a  $c$ -colored graph  $G$ , its radius- $r$  flip-width  $\text{fw}_r(G)$ , for  $r \in \mathbb{N} \cup \{\infty\}$ , is defined as the same parameter for the underlying uncolored graph. Say that a class  $\mathcal{C}$  of  $c$ -colored graphs has *bounded flip-width* if the underlying class of uncolored graphs has bounded flip-width. A  $c$ -colored graph is seen as a structure over the signature consisting of the binary relation  $E(x, y)$  denoting adjacency, as well as unary predicates  $C_1(x), \dots, C_c(x)$  denoting the respective colors.

b) *Interpretations:* The following notion is a special case of a (simple, domain-preserving) first-order interpretation. Let  $G$  be a  $c$ -colored graph and  $\varphi(x, y)$  be a first-order formula in the signature of  $c$ -colored graphs. Define the graph  $\varphi(G)$  with vertices  $V(G)$  and edges  $uv$  such that  $u \neq v$  and  $\varphi(u, v) \vee \varphi(v, u)$  holds in  $G$ . For a class  $\mathcal{C}$  of  $c$ -colored graphs, denote  $\varphi(\mathcal{C}) := \{\varphi(G) \mid G \in \mathcal{C}\}$ . The class  $\varphi(\mathcal{C})$  is called an *interpretation* of  $\mathcal{C}$ , via  $\varphi$ . For example, for the formula  $\varphi(x, y) = \neg E(x, y)$  and a graph  $G$ , the graph  $\varphi(G)$  is the complement  $\bar{G}$  of  $G$ . And for the formula  $\varphi(x, y) = \exists z.[E(x, z) \wedge E(z, y)]$ , the graph  $\varphi(G)$  is the square of  $G$ .

c) *Transductions:* Say that a graph class  $\mathcal{C}$  *transduces* a graph class  $\mathcal{D}$ , or that  $\mathcal{D}$  is a *transduction* of  $\mathcal{C}$ , if there is some  $c \geq 1$  and class  $\hat{\mathcal{C}}$  of  $c$ -colored graphs which is a  $c$ -coloring of  $\mathcal{C}$ , and some first-order formula  $\varphi(x, y)$  in

the signature of  $c$ -colored graphs, such that every graph in  $\mathcal{D}$  is an induced subgraph of some graph in  $\varphi(\widehat{\mathcal{C}})$  (that is,  $\mathcal{D}$  is contained in the hereditary closure of  $\varphi(\widehat{\mathcal{C}})$ ).

*Example VIII.1.* Let  $\mathcal{C}$  be the class of all half-graphs (see Figure 2), where the half-graph  $H_n$  of order  $n$  has vertices  $a_1, \dots, a_n$  and  $b_1, \dots, b_n$ , and edges  $a_i b_j$  for  $1 \leq i < j \leq n$ . We show that  $\mathcal{C}$  transduces the class  $\mathcal{D}$  consisting of disjoint unions of cliques.

We use two colors. Let  $\widehat{\mathcal{C}}$  be the class of all 2-colored half-graphs. Consider the formula  $\varphi(x, y)$  expressing that there is no vertex of color 2 which is adjacent to one of  $x, y$ , and not the other:

$$\varphi(x, y) \equiv \forall z. C_2(z) \rightarrow (E(x, z) \leftrightarrow E(y, z)).$$

We argue that the hereditary closure of  $\varphi(\widehat{\mathcal{C}})$  contains  $\mathcal{D}$ , implying that  $\mathcal{D}$  transduces in  $\mathcal{C}$ .

Let  $F \in \mathcal{D}$  be a disjoint union of cliques. Let  $1, \dots, n$  be the vertices of  $F$ , for some  $n \geq 0$ , and assume that every connected component of  $F$  consists of consecutive vertices in the usual order  $1 < \dots < n$ . Consider the half-graph  $H_n$  with vertices  $a_1, \dots, a_n$  and  $b_1, \dots, b_n$ , and color a vertex  $b_i$  with color 2 if  $i$  is the largest element of its connected component in  $F$ , and with color 1 otherwise. Now, for all  $1 \leq i, j \leq n$ ,  $H_n \models \varphi(a_i, a_j)$  if and only if  $i$  and  $j$  are adjacent in  $F$ . Hence,  $\varphi(H_n)[\{a_1, \dots, a_n\}]$  is isomorphic to  $F$ , and therefore  $F$  is an induced subgraph of some graph in  $\varphi(\widehat{\mathcal{C}})$ .

*d) Transductions and flip-width:* We prove the following theorem.

**Theorem VIII.2 (\*).** Fix  $q \geq 0$ . There is a computable function  $T_q: \mathbb{N} \rightarrow \mathbb{N}$  with the following property. Fix numbers  $r, c \geq 1$  and a first-order formula  $\varphi(x, y)$  of quantifier rank  $q$  in the signature of  $c$ -colored graphs. Set  $r' := 2^q \cdot r$ . Then for every  $c$ -colored graph  $G$  we have

$$\text{fw}_r(\varphi(G)) \leq T_q(\text{fw}_{r'}(G) \cdot c). \quad (9)$$

In particular, if  $\mathcal{C}$  has bounded flip-width, then  $\varphi(\mathcal{C})$  has bounded flip-width. Moreover,  $T_0(k) = k$  for every  $k$ .

Before discussing the proof of Theorem VIII.2, we discuss its consequences.

The following is an immediate consequence of Theorem VIII.2, and the fact that radius- $r$  flip-width is monotone with respect to induced subgraphs.

**Corollary VIII.3.** If a class  $\mathcal{C}$  has bounded flip-width and transduces a class  $\mathcal{D}$ , then  $\mathcal{D}$  has bounded flip-width.

Since the class of all graphs has unbounded flip-width by Corollary V.23, we get the following.

**Corollary VIII.4.** If  $\mathcal{C}$  is a class of bounded flip-width, then  $\mathcal{C}$  does not transduce the class of all graphs.

In the language of model theory, Corollary VIII.4 says that classes of bounded flip-width are *monadically dependent*, see Section X. In the phrasing of [GPT22],

[BNdMS22b], Corollary VIII.3 says that classes of bounded flip-width form a *transduction ideal*. By Theorem VI.3, the weakly sparse classes in this transduction ideal are exactly the classes of bounded expansion. Therefore, this transduction ideal is contained in the *dense analogue* of bounded expansion classes [GPT22].

A graph class has *structurally bounded expansion* [GKN<sup>+</sup>20] if it is a transduction of a class with bounded expansion. Corollary VIII.3 immediately implies that those classes have bounded flip-width.

**Corollary VIII.5.** Every class of structurally bounded expansion has bounded flip-width.

We remark that an extension of Theorem VIII.2 and of Corollary VIII.3 also hold, with the same proof (but a slightly different bound in (9)), for input structures over a binary signature  $\Sigma$ , rather than graphs  $G$ , and where instead of one formula  $\varphi(x, y)$  defining a new edge relation of the output graph, we have a tuple of formulas  $\bar{\varphi} = (\varphi_1(x, y), \dots, \varphi_k(x, y))$  defining  $k$  binary relations of the output structure  $\bar{\varphi}(G)$  with domain  $V(G)$  and  $i$ th relation  $R_i$ , for  $i = 1, \dots, k$ , interpreted as

$$R_i := \{(a, b) \mid a, b \in V(G), G \models \varphi_i(a, b)\}.$$

We then obtain that if  $\mathcal{C}$  is a class of  $\Sigma$ -structures of bounded flip-width (see Sec. VII and [Tor23]), then the class  $\bar{\varphi}(\mathcal{C}) := \{\bar{\varphi}(G) \mid G \in \mathcal{C}\}$  also has bounded flip-width.

This implies (using Corollary VII.4) the result of [BKTW20], that transductions preserve classes of bounded twin-width.

**Corollary VIII.6** ([BKTW20]). If  $\mathcal{C}$  has bounded twin-width and  $\varphi(x, y)$  is a first-order formula, then  $\varphi(\mathcal{C})$  has bounded twin-width.

Indeed, let  $\mathcal{C}$  be a class of bounded twin-width and  $\varphi(x, y)$  a symmetric formula in the signature of graphs. By Corollary VII.4 there is a class  $\widehat{\mathcal{C}}$  of ordered graphs of bounded flip-width such that  $\mathcal{C}$  is obtained from  $\widehat{\mathcal{C}}$  by forgetting the order. Let  $\bar{\varphi}$  be the pair consisting of the formula  $\varphi(x, y)$  and the formula  $x < y$  defining the order. Then  $\bar{\varphi}(\widehat{\mathcal{C}})$  is a class of ordered graphs which has bounded flip-width, since  $\widehat{\mathcal{C}}$  has bounded flip-width. Moreover, the graph class  $\varphi(\mathcal{C})$  is obtained from the class of ordered graphs  $\bar{\varphi}(\widehat{\mathcal{C}})$  by forgetting the order. Hence,  $\varphi(\mathcal{C})$  has bounded twin-width, by Corollary VII.4.

Together with Theorem VI.3, Corollary VIII.3 implies the following consequence of [GKN<sup>+</sup>18].

**Corollary VIII.7.** Every weakly sparse class with structurally bounded expansion has bounded expansion.

The proof of Theorem VIII.2, which is sketched below, relies on locality of first-order logic, a central notion for analysing first-order formulas on sparse graphs [See96],

which, in some form, also plays a key role in understanding first-order formulas on classes of bounded twin-width [BKTW20], [GPPT22]. The function  $T_q(k)$ , although computable, is astronomical:

$$T_q(k) := \underbrace{2^{2^{\cdot^{\cdot^{\cdot^m}}}}}_{\text{height } q}$$

where  $m$  is the number of distinct  $k$ -colored graphs with vertex set  $\{1, \dots, q+1\}$ . However, for  $q = 0$  we have  $T_0(k) = k$ , so the bound (9) becomes  $\text{fw}_r(\varphi(G)) \leq c \cdot \text{fw}_r(G)$  in the case when  $\varphi(x, y)$  is a quantifier-free formula and  $G$  is a  $c$ -colored graph with underlying graph  $G_0$ .

e) *CMSO transductions and radius- $\infty$  flip-width*: For the case of radius  $r = \infty$ , we prove that classes  $\mathcal{C}$  with  $\text{fw}_\infty(\mathcal{C}) < \infty$  are preserved under transductions expressed in the more powerful logic called CMSO. We remark that this result is known for classes of bounded clique-width, so our result follows from Theorem V.17, which states that  $\mathcal{C}$  has bounded clique-width if and only if  $\text{fw}_\infty(\mathcal{C}) < \infty$ . However, we provide a separate proof here, as it is very analogous to the proof of Theorem VIII.2, and nicely illustrates the parallels between the bounded flip-width case, and the limit case of bounded radius- $\infty$  flip-width. We also show how the closure under CMSO transductions can be used to obtain an alternative proof of Theorem V.17. Let us first define the logic, and state the result.

*Counting Monadic Second Order Logic* (CMSO) is the extension of first-order logic, where apart from first-order quantifiers  $\exists x, \forall x$ , that range over vertices  $x$  of a graph, we have second-order quantifiers  $\exists X, \forall X$ , that range over sets  $X$  of vertices of the graph. We additionally have the atomic predicate  $x \in X$  that allows to check whether a given vertex belongs to a given set, and the divisibility predicates  $\text{div}_k(X)$ , for  $k \geq 1$ , where  $\text{div}_k(X)$  holds for a given set of vertices  $X$  if and only if  $|X|$  is divisible by  $k$ . The usual constructs of first-order logic (boolean connectives and relation symbols, such as adjacency in a graph) are also included.

CMSO is able to express non-local properties. For instance, the formula

$$\begin{aligned} \varphi(x, y) = & \neg \exists X. [(x \in X) \wedge (y \notin X) \\ & \wedge \forall z. \forall t. [(z \in X) \wedge E(z, t) \rightarrow (t \in X)]] \end{aligned}$$

expresses that  $x$  and  $y$  lie in the same connected component.

Still, CMSO enjoys property similar to locality of first-order logic, called compositionality (or a Feferman-Vaught-Mostowski type result), which is an analogue of locality for the radius  $r = \infty$ . Using this instead of locality, with the same proof as in Theorem VIII.2, we get the following.

**Theorem VIII.8 (\*).** *Let  $\mathcal{C}$  be a class of  $c$ -colored graphs of bounded  $\infty$ -flip-width and let  $\varphi(x, y)$  be a formula of CMSO. Then  $\varphi(\mathcal{C})$  has bounded  $\infty$ -flip-width.*

Say that a class  $\mathcal{C}$  CMSO-transduces a class  $\mathcal{D}$ , or that  $\mathcal{D}$  is a CMSO-transduction of  $\mathcal{C}$ , if there is a  $c$ -coloring  $\hat{\mathcal{C}}$  of  $\mathcal{C}$ , for some  $c \geq 1$ , and some CMSO formula  $\varphi(x, y)$  in the signature of  $c$ -colored graphs, such that every graph in  $\mathcal{D}$  is an induced subgraph of some graph in  $\varphi(\hat{\mathcal{C}})$  (that is,  $\mathcal{D}$  is contained in the hereditary closure of  $\varphi(\hat{\mathcal{C}})$ ).

**Corollary VIII.9.** *If  $\mathcal{C}$  is a class with  $\text{fw}_\infty(\mathcal{C}) < \infty$  that CMSO-transduces a class  $\mathcal{D}$ , then  $\text{fw}_\infty(\mathcal{D}) < \infty$ .*

We now show that, together with a result of Courcelle and Oum, Corollary VIII.9 yields an alternative proof of the backwards implication in Theorem V.17, that classes of bounded radius- $\infty$  flip-width have bounded clique-width.

Courcelle and Oum [CiO07, Corollary 7.5] proved the following result<sup>7</sup>.

**Theorem VIII.10.** *Every graph class of unbounded clique-width CMSO-transduces the class of all graphs.*

In particular, as the class of all graphs has unbounded radius-one flip-width by Corollary V.23, it follows from Corollary VIII.9 and Theorem VIII.10 that every class  $\mathcal{C}$  of unbounded clique-width has  $\text{fw}_\infty(\mathcal{C}) = \infty$ , proving the backward implication in Theorem V.17.

## B. Transferring strategies

Before giving the details about the proof of Theorem VIII.2, we define a tool that will be also useful in other contexts. The following notion allows to transfer a winning strategy of the flipper from a graph  $G$  to a graph  $H$ . Let  $G$  and  $H$  be two graphs with  $V(H) \subseteq V(G)$ . Fix  $r_G, r_H \in \mathbb{N} \cup \{\infty\}$  and  $k, \ell \in \mathbb{N}$ . Suppose furthermore that the following are given:

- a mapping  $F$  that maps each  $k$ -flip  $G'$  of  $G$  to an  $\ell$ -flip  $H' = F(G')$  of  $H$ ,
- a strategy of the flipper in the game on the graph  $G$  with radius  $r_G$  and width  $k$ .

This induces the following strategy of the flipper in the flipper game of radius  $r_H$  and width  $\ell$  on the graph  $H$ ; call this game the  $H$ -game. Consider a play of this game the  $H$ -game. Simultaneously, initiate the  $G$ -game on  $G$ , with radius  $r_G$  and width  $k$ , in which we will copy runner's moves from the  $H$ -game. Whenever the flipper announces a  $k$ -flip  $G'$  of  $G$  in the  $G$ -game, then in the  $H$ -game the flipper announces the  $\ell$ -flip  $H' = F(G')$  of  $H$ . Whenever the runner moves to a vertex  $v$  in the  $H$ -game, we also move the runner to  $v$  in the  $G$ -game. Note that this move might not be a valid move in the

<sup>7</sup>In fact, they proved an equivalent statement, but with 'all square grids' instead of 'all graphs'. However, it is easy to see that the class of all square grids CMSO-transduces the class of all graphs, and transductions can be composed.

*H*-game. In this case, the flipper continues the *G*-game by playing arbitrarily (e.g. always announcing *G* as the next flip) until the end of the game. We say the resulting strategy of the flipper (in the *G*-game) is *transferred* from the considered strategy in the *H*-game, according to the mapping *F*.

The following lemma gives a condition which implies that if the original strategy on *G* is winning, then the transferred strategy on *H* is winning.

**Lemma VIII.11.** *Fix  $r \in \mathbb{N} \cup \{\infty\}$  and  $k, \ell, s \geq 1$ . Let *H, G* be two graphs with  $V(H) \subseteq V(G)$ . Suppose that for all  $k \geq 1$  and every *k*-flip  $G'$  of *G* there is some  $\ell$ -flip  $H' = F(G')$  of *H*, such that*

$$\text{dist}_{G'}(u, v) \leq s \quad \text{for all } uv \in E(H'). \quad (10)$$

*Then transferring a winning strategy of the flipper from the flipper game on *G* with radius  $rs$  and width  $k$ , according to the mapping *F*, results in a winning strategy of the flipper in the flipper game on *H* with radius  $r$  and width  $\ell$ . In particular,*

$$\text{fw}_{rs}(G) \leq k \quad \text{implies} \quad \text{fw}_r(H) \leq \ell.$$

In the case  $r = \infty$ , the conclusion reads “ $\text{fw}_\infty(G) \leq k$  implies  $\text{fw}_\infty(H) \leq \ell$ ”. Moreover we can then replace  $\leq s$  by  $< \infty$  in (10), since for  $s := |V(G)|$  we have that  $\text{dist}_{G'}(u, v) \leq s \iff \text{dist}_{G'}(u, v) < \infty$ . We state this explicitly below.

**Corollary VIII.12.** *Fix  $k, \ell \geq 1$ . Let *H, G* be two graphs with  $V(H) \subseteq V(G)$ . Suppose that for all  $k \geq 1$  and every *k*-flip  $G'$  of *G* there is some  $\ell$ -flip  $H' = F(G')$  of *H*, such that*

$$\text{dist}_{G'}(u, v) < \infty \quad \text{for all } uv \in E(H'). \quad (11)$$

*Then*

$$\text{fw}_\infty(G) \leq k \quad \text{implies} \quad \text{fw}_\infty(H) \leq \ell.$$

*Proof of Lemma VIII.11.* Consider a play in the flipper game on *H* of radius  $r$  and width  $\ell$ , according to the strategy transferred from *G*, as described above. Suppose that, in some round,  $G'$  is the announced flip in the *G*-game and  $H' = F(G')$  is the announced flip in the *H*-game, and that  $v$  is the new vertex chosen by the runner in the *H*-game. By (10), the following holds:

$$B_{H'}^r(v) \subseteq B_{G'}^{rs}(v).$$

In the case  $r = \infty$ , each side of the inclusion should be interpreted as the connected component of  $v$  in the appropriate graph.

In particular, in the next round, every valid move of runner in the *H*-game will also be a valid move of runner in the *G*-game (this is trivially satisfied in the first round), and if runner is trapped in the *G*-game, that is,  $|B_{G'}^{rs}(v)| = 1$ , then also  $|B_{H'}^r(v)| = 1$ , so he is also trapped in the *H*-game. Thus, this describes a winning strategy for the flipper in the flipper game on *H* with radius  $r$  and width  $\ell$ . In particular,  $\text{fw}_r(H) \leq \ell$ .  $\square$

### C. Proof of Theorem VIII.2

We now sketch the proof of Theorem VIII.2. The details are presented in [Tor23], along with the proof of Theorem VIII.8.

For simplicity, we assume the case  $c = 1$ , that is, when the considered graphs *G* have no colors. The general case proceeds analogously.

Our aim is to transfer a winning strategy of the flipper from *G* to  $H = \varphi(G)$  (with appropriate radii), by applying Lemma VIII.11. So for every flip  $G'$  of *G* we need to produce a flip  $\varphi(G)'$  of  $\varphi(G)$  such that adjacent vertices in  $\varphi(G)'$  are not too far in  $G'$ . We first show how to achieve this in the case when  $G' = G$ , using locality, a well known tool from finite model theory, which we now recall.

a) *Locality*: Fix a number  $s \in \mathbb{N}$ . Say that a formula  $\varphi(x, y)$  is *s*-local, if for any graph *G* there is a labelling of the vertices of *G* using a bounded number of labels (depending only on  $\varphi$ , and not on *G*) such that for any two vertices  $u, v$  of *G* with  $\text{dist}_G(u, v) > s$ , whether or not  $G \models \varphi(u, v)$  depends only on the label of  $u$  and the label of  $v$ . It is well-known (and follows for instance from Gaifman’s locality theorem) that every formula  $\varphi(x, y)$  of first-order logic is *s*-local for some radius  $s$  depending only on the quantifier rank  $q$  of  $\varphi$ . Namely, one can take  $s := 2^q$ . The label assigned to a vertex  $v$  of *G* as above, is essentially the set of formulas  $\alpha(x)$  of quantifier rank at most  $q$ , such that  $\alpha(v)$  holds in *G*. The number of such formulas, up to equivalence, is finite, and is bounded by  $T_q(1)$ , where  $T_q$  is the function described above.

b) *Flipping  $\varphi(G)$* : We now show how to obtain a flip  $\varphi(G)'$  of  $\varphi(G)$  such that vertices that are adjacent in  $\varphi(G)'$  are not too far in *G*. Let  $\mathcal{P}$  be the partition of  $V(G) = V(\varphi(G))$  such that two vertices of *G* are in the same part if they get the same label. In particular,  $|\mathcal{P}| \leq T_q(1)$ . Now, in the graph  $\varphi(G)$  flip a pair of parts  $A, B$  of  $\mathcal{P}$  if and only if there is a pair of vertices  $u \in A$  and  $v \in B$ , such that  $\text{dist}_G(u, v) > s$  and  $G \models \varphi(u, v)$ , equivalently,  $uv \in E(\varphi(G))$ . The statement above implies that whether or not we flip  $A$  and  $B$ , does not depend on the choice of  $u \in A$  and  $v \in B$  such that  $\text{dist}_G(u, v) > s$ . This yields a  $\mathcal{P}$ -flip of  $\varphi(G)$ , which we denote  $\varphi(G)'$ . Then the following holds for all  $u, v \in V(G)$ :

$$uv \in E(\varphi(G)') \quad \text{implies} \quad \text{dist}_G(u, v) \leq s.$$

We now generalize the above reasoning to the case when  $G'$  is a *k*-flip of *G*. Again, the goal is to construct a flip  $\varphi(G)'$  of  $\varphi(G)$  such that vertices that are adjacent in  $\varphi(G)'$  are not too far in  $G'$ .

We treat  $G'$  as a *k*-colored graph, by adding colors that mark parts of the partition that is used to produce the *k*-flip  $G'$  of *G*. The key observation is that we can write a formula  $\varphi'(x, y)$  that makes use of those colors, and such that

$$G' \models \varphi'(u, v) \iff G \models \varphi(u, v) \quad \text{for all } u, v \in V(G).$$

This is because we can write a formula  $\varepsilon(x, y)$  such that  $G' \models \varepsilon(u, v)$  if and only if  $G \models E(u, v)$  (the formula  $\varepsilon(x, y)$  checks the colors of  $x$  and  $y$ , whether  $x$  and  $y$  are adjacent in  $G'$ , and inverts the flip). The formula  $\varphi'(u, v)$  is obtained by replacing each atomic formula  $E(z, t)$  with  $\varepsilon(z, t)$ . In particular,  $\varphi'$  has the same quantifier rank as  $\varphi$ , and is therefore also  $s$ -local. Applying the same argumentation as above, but this time to the formula  $\varphi'$  and the  $k$ -colored graph  $G'$ , we obtain a  $\mathcal{P}$ -flip  $\varphi(G)'$  such that the following holds for all  $u, v \in V(G)$ :

$$uv \in E(\varphi(G)') \quad \text{implies} \quad \text{dist}_{G'}(u, v) \leq s.$$

Moreover, the size of  $\mathcal{P}$  can be bounded by a number  $\ell$  depending on the formula  $\varphi$  and the number  $k$ . Since this holds for every  $k$ -flip  $G'$  of  $G$ , we can now apply Lemma VIII.11 and conclude that  $\text{fw}_r(\varphi(G)) \leq \ell$ .

#### IX. DEFINABLE FLIP-WIDTH

Determining the flip-width of radius  $r$  of a given graph  $G$  seems computationally difficult. The space of all configurations in the flipper game of radius  $r$  and width  $k$  (consisting of a  $k$ -flip of  $G$  and the runner's position) has size exponential in  $|G|$ , and the naive algorithm for determining whether  $\text{fw}_r(G) \leq k$ , which explores the space of all configurations, therefore runs in time exponential in  $|G|$ . We expect that for possible algorithmic applications, an algorithm which approximates  $\text{fw}_r(G)$ , instead of computing it exactly, should suffice. This is what happens in the case of weak coloring numbers, and of twin-width, where such an approximation algorithm is not known in general, but is known in various special cases.

In this section, we introduce the *definable flipper game* of radius  $r$  and width  $k$ , in which the moves allowed for the flipper are parameterized by tuples of vertices of the graph, rather than by partitions of the vertex set. Among other things, this reduces (comparing to the flipper game) the number of configurations, and the computational complexity of determining the definable flip-width of a given graph. The main result of this section is Theorem IX.4 that says that the definable flip-width at radius  $r$  can be bounded in terms of the flip-width at radius  $5r$ . This allows to obtain, in Theorem IX.7, an algorithm for approximating flip-width, which runs in time  $|G|^{O(k)} \cdot O_k(1)$ , where  $k$  is the flip-width. We use tools related to VC-dimension to accomplish this.

*a) Atomic types and definable flips:* Let  $S$  be a set of vertices of a graph  $G$ . Consider the partition  $\mathcal{P}$  of  $V(G)$  such that two vertices  $u, v \in V(G)$  are in the same part if  $N(u) \cap S = N(v) \cap S$ . The equivalence classes of the partition  $\mathcal{P}$  are called *S-types*. In particular,  $|\mathcal{P}| \leq 2^{|S|}$ .

Say that a graph  $G'$  is an *S-definable flip* of  $G$  if  $G'$  is a  $\mathcal{P}$ -flip of  $G$ , where  $\mathcal{P}$  is the partition of  $V(G)$  into  $S$ -types. Say that  $G'$  is a *k-definable flip* of  $G$ , if  $G'$  is an  $S$ -definable flip of  $G$  for some  $S \subseteq V(G)$  with  $|S| \leq k$ .

Note that a  $k$ -definable flip of  $G$  is a  $2^k$ -flip of  $G$ . However, there is no function  $f$  such every  $k$ -flip of a

graph  $G$  is a  $f(k)$ -definable flip of  $G$ . For instance, the graph  $G_n$  obtained from an  $n$ -clique  $K_n$  by adding  $n$  isolated vertices, is a 2-flip of  $K_{2n}$ , but is not an  $(n-1)$ -definable flip of  $K_{2n}$ .

*b) Definable flipper game:* Fix  $r \in \mathbb{N} \cup \{\infty\}$ . The *definable flipper game* of radius  $r$  and width  $k$  is defined in the same way as the flipper game of radius  $r$ , but now in each round the flipper is allowed to announce a  $k$ -definable flip  $G'$  of  $G$ , rather than a  $k$ -flip of  $G$ .

**Definition IX.1.** Fix  $r \in \mathbb{N} \cup \{\infty\}$ . The *radius- $r$  definable flip-width* of a graph  $G$ , denoted  $\text{dfw}_r(G)$ , is the smallest number  $k$  such that the flipper has a winning strategy in the definable flipper game of radius  $r$  and width  $k$  on  $G$ .

As every  $k$ -definable flip of  $G$  is a  $2^k$ -flip of  $G$ , it follows that for every  $r \in \mathbb{N} \cup \{\infty\}$  and graph  $G$  we have:

$$\text{fw}_r(G) \leq 2^{\text{dfw}_r(G)}. \quad (12)$$

One advantage of the definable version of the flipper game is that it has far fewer configurations than the original flipper game. As there are only  $O(|G|^{k+1} \cdot 2^k)$  configurations in the definable flipper game of width  $k$ , we get that it can be decided in time  $|G|^{O(k)} \cdot 2^{O(2^k)}$  whether a given graph  $G$  has  $\text{dfw}_r(G) \leq k$  (see [Tor23] for a proof).

**Lemma IX.2 (\*)**. There is an algorithm that, given a graph  $G$  and numbers  $k \in \mathbb{N}$  and  $r \in \mathbb{N} \cup \{\infty\}$ , determines whether  $\text{dfw}_r(G) \leq k$  in time  $n^{O(k)} \cdot 2^{O(2^k)}$ .

The following lemma is an immediate consequence of the Sauer-Shelah-Perles lemma (Lemma III.8).

**Lemma IX.3 (\*)**. Fix  $r \in \mathbb{N} \cup \{\infty\}$ . For every graph  $G$  we have:

$$\text{fw}_r(G) \leq O(\text{dfw}_r(G)^{\text{VCdim}(G)}). \quad (13)$$

The following is the main result of Section IX. It says that the definable flip-width can be bounded in terms of the flip-width, at the cost of increasing the radius. Below,  $5 \cdot \infty$  is interpreted as  $\infty$ .

**Theorem IX.4.** Fix  $r \in \mathbb{N} \cup \{\infty\}$ . For every graph  $G$  we have:

$$\text{dfw}_r(G) \leq O(\text{fw}_{5r}(G)^3). \quad (14)$$

We prove Theorem IX.4 below. We first observe some consequences. The bounds (12) and (14) give the following.

**Corollary IX.5.** The following conditions are equivalent for a graph class  $\mathcal{C}$ :

- 1)  $\mathcal{C}$  has bounded flip-width, that is,  $\text{fw}_r(\mathcal{C}) < \infty$  for all  $r \geq 1$ ,
- 2)  $\text{dfw}_r(\mathcal{C}) < \infty$  for all  $r \geq 1$ .

Similarly, for the case  $r = \infty$ , using Theorem V.17 we get the following characterization of classes of bounded

clique-width in terms of the definable flipper game with radius  $\infty$ :

**Corollary IX.6.** *The following conditions are equivalent for a graph class  $\mathcal{C}$ :*

- 1)  $\mathcal{C}$  has bounded clique-width,
- 2)  $\text{dfw}_\infty(\mathcal{C}) < \infty$ .

Finally, we get an algorithm for approximating the flip-width of a given graph. The algorithm is an approximation algorithm: unlike the algorithm in Lemma IX.2, it does not allow to exactly determine whether the radius- $r$  flip-width of a given graph  $G$  is smaller than a given number  $k$ . Rather, it recognises one of two, non-exclusive, cases: whether  $\text{fw}_r(G)$  is small comparing to  $k$ , and whether  $\text{fw}_{5r}(G)$  is large comparing to  $k$ . Note that there is a gap in the radii,  $r$  and  $5r$ . The running time of the algorithm is  $O_k(1) \cdot n^{O(k)}$ , which is called an XP algorithm (parameterized by  $k$ ) in the language of parameterized complexity.

**Theorem IX.7.** *There is a constant  $C > 0$  and an algorithm that inputs a graph  $G$  and numbers  $r, k \in \mathbb{N}$ , runs in time  $n^{O(k)} \cdot 2^{O(2^k)}$ , and either concludes that  $\text{fw}_r(G) \leq 2^k$ , or concludes that  $\text{fw}_{5r}(G) \geq C \cdot k^{1/3}$ .*

*Proof.* The algorithm tests whether  $\text{dfw}_r(G) \leq k$  in time  $n^{O(k)} \cdot 2^{O(2^k)}$ , using Lemma IX.2. If  $\text{dfw}_r(G) \leq k$  it concludes that  $\text{fw}_r(G) \leq 2^k$ , by (12). If  $\text{dfw}_r(G) > k$ , it concludes that  $\text{fw}_{5r}(G) \geq Ck^{1/3}$  by (14), where  $C > 0$  is some fixed constant.  $\square$

c) *Proof of Theorem IX.4:* We now turn to the proof of Theorem IX.4. We use a result concerning graphs of small VC-dimension. Recall that  $\text{VCdim}(G) \leq O(\text{fw}_1(G))$  by Theorem V.24. The following result from [BDG<sup>+</sup>22], relies on the  $(p, q)$ -theorem of Alon-Kleitman-Matoušek [Mat04] (see [Tor23]).

**Lemma IX.8 (\*).** *Fix  $k, d \in \mathbb{N}$ . Let  $V$  be a set equipped with:*

- a binary relation  $E \subseteq V \times V$  of VC-dimension at most  $d$ ,
- a pseudometric  $\text{dist}: V \times V \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\}$  (that is, a function satisfying the triangle inequality),
- and a partition  $\mathcal{P}$  of size at most  $k$ ,

*such that  $E(u, v)$  depends only on the  $\mathcal{P}$ -class of  $u$  and the  $\mathcal{P}$ -class of  $v$  whenever  $\text{dist}(u, v) > 1$ . Then there is a set  $S \subseteq V$  of size  $O(dk^2)$ , such that  $E(u, v)$  depends only on the  $S$ -types of  $u$  and of  $v$ , whenever  $\text{dist}(u, v) > 5$ .*

We reformulate Lemma IX.8 in terms of flips, as follows.

**Corollary IX.9.** *Let  $G$  be a graph and  $d = \text{VCdim}(G)$ , and let  $G'$  be a  $k$ -flip of  $G$ . Then there is a  $O(dk^2)$ -definable flip  $H'$  of  $G$  such that*

$$\text{dist}_{G'}(u, v) \leq 5 \quad \text{for all } u, v \in E(H') \quad (15)$$

*Proof.* Apply Lemma IX.8 to  $V = V(G)$ ,  $E = E(G)$ ,  $\text{dist}: V \times V \rightarrow \mathbb{N} \cup \{\infty\}$  denoting the shortest path metric in  $G'$ , and the partition  $\mathcal{P}$  with  $|\mathcal{P}| \leq k$  such that  $G'$  is a  $\mathcal{P}$ -flip of  $G$ . Note that for any two vertices  $u, v \in V$ , whether or not  $uv \in E(G)$  holds, can be determined basing only on the  $\mathcal{P}$ -class of  $u$ , the  $\mathcal{P}$ -class of  $v$ , and on the information whether  $uv \in E(G')$  holds. In particular,  $uv \in E(G)$  depends only on the  $\mathcal{P}$ -class of  $u$  and the  $\mathcal{P}$ -class of  $v$ , for all  $u, v \in V$  that are not adjacent in  $G'$ , equivalently, with  $\text{dist}(u, v) > 1$ . Hence, the assumption of Lemma IX.8 is satisfied.

Let  $S$  with  $|S| \leq O(dk^2)$  be as in the conclusion of the lemma, so that whether or not  $uv \in E(G)$ , depends only on the  $S$ -type of  $u$  and the  $S$ -type of  $v$ , for all  $u, v \in V$  with  $\text{dist}(u, v) > 5$ .

Let  $\mathcal{P}_S$  denote the partition of  $V(G)$  into  $S$ -types. Let  $H'$  be the  $\mathcal{P}_S$ -flip of  $G$  that flips between two  $S$ -types  $A$  and  $B$  if and only if there are some  $u \in A, v \in B$  with  $\text{dist}(u, v) > 5$  and  $uv \in E(G)$ . The conclusion follows.  $\square$

Theorem IX.4 easily follows from Corollary IX.9.

*Proof of Theorem IX.4.* Let  $G$  be a graph and let  $k = \text{fw}_{5r}(G)$ . In particular, by Theorem V.24, we have that  $\text{VCdim}(G) \leq O(k)$ . By Corollary IX.9 applied to  $H := G$ , the assumptions of Lemma VIII.11 are satisfied, where the  $\ell$ -flip  $H'$  of  $H = G$ , for  $\ell = 2^{O(k^3)}$  is a  $O(k^3)$ -definable flip, as provided by Corollary IX.9. By transferring the winning strategy of the flipper in the flipper game of radius  $5r$  and width  $k$  on  $G$ , according to this mapping  $G' \mapsto H'$ , we get a strategy for the flipper in flipper game of radius  $r$  on  $G$ , which uses only  $O(dk^2)$ -definable flips. By Lemma VIII.11, this yields a winning strategy of the flipper that will use only  $O(k^3)$ -definable flips. Hence,  $\text{dfw}_r(G) \leq O(k^3)$ .  $\square$

## X. ALMOST BOUNDED FLIP-WIDTH

Recall from Fact III.7 that a hereditary graph class  $\mathcal{C}$  is nowhere dense if and only if for every  $r \geq 1$  and  $G \in \mathcal{C}$  we have  $\text{wcol}_r(G) = |G|^{o(1)}$ . Inspired by this characterization, we extend the notion of bounded flip-width as follows.

**Definition X.1.** *A graph class  $\mathcal{C}$  has almost bounded flip-width if for every  $r \geq 1$  and real  $\varepsilon > 0$  we have  $\text{fw}_r(G) \leq O_{\varepsilon, r}(|G|^\varepsilon)$  for every graph  $G$  in the hereditary closure of  $\mathcal{C}$ .*

Note that we consider all graphs  $G$  from the hereditary closure of  $\mathcal{C}$ . Otherwise, the class consisting of every graph  $G$  with  $2^{|G|}$  isolated vertices added to it, would have almost bounded flip-width, while according to the above definition, it does not. Indeed, we have the following lemma, which is an immediate consequence of Lemma V.26.

**Lemma X.2.** *Every graph class with almost bounded flip-width has bounded VC-dimension.*

Clearly, every class with bounded flip-width has almost bounded flip-width. As we conjecture (see Conjecture X.7), classes of almost bounded flip-width coincide with monadically dependent classes (see definition below), analogously to the characterization of nowhere dense classes in Fact III.7.

In this section, we provide some evidence towards this conjecture. In Theorem X.9, we prove that a weakly sparse class has almost bounded flip-width if and only if it is nowhere dense, if and only if it is monadically dependent. In Theorem X.12 we prove that structurally nowhere dense classes have almost bounded flip-width. In Theorem X.15 we prove that edge-stable classes of almost bounded flip-width are monadically dependent. In Theorem X.17 we prove that classes of ordered graphs of almost bounded flip-width coincide with classes of bounded twin-width, and with classes of bounded flip-width. We start with recalling the discussed notions.

#### A. Monadic dependence and monadic stability

The following notion, due to Shelah [She86] (see also [BL21]), originates in model theory.

**Definition X.3.** A graph class  $\mathcal{C}$  is monadically dependent (or monadically NIP) if and only if it does not transduce the class of all graphs.

Monadically dependent classes have recently attracted attention in areas of structural and algorithmic graph theory [AA14], [NdMP<sup>+</sup>21], [BGOdM<sup>+</sup>22], [GPT22], as it is conjectured (see Conjecture I.1) that monadically dependent classes are precisely those for which model checking first-order logic is fixed-parameter tractable.

Monadically dependent classes include all nowhere dense classes, and in fact, among weakly sparse classes, they provide an exact characterization:

**Fact X.4** (Consequence of [Dvo18]+[AA14]). Let  $\mathcal{C}$  be a weakly sparse graph class. Then  $\mathcal{C}$  is nowhere dense if and only if  $\mathcal{C}$  is monadically dependent.

Monadically dependent classes also include all classes of bounded flip-width, by Corollary VIII.4.

An important subfamily of monadically dependent classes consists of monadically stable classes. A graph class is *monadically stable* if it does not transduce the class of all half-graphs (see Fig. 2). A graph class is *edge-stable* if it excludes some half-graph as a semi-induced bipartite graph. More precisely, there is  $k \geq 1$  such that there do not exist  $G \in \mathcal{C}$  and vertices  $a_1, \dots, a_k, b_1, \dots, b_k$  of  $G$  such that  $a_i b_j \in E(G) \iff i < j$  for all  $i, j \in \{1, \dots, k\}$ .

The following result is proved in [NdMP<sup>+</sup>21, Theorem 1.3], see also [BL22, Theorem 3.20].

**Fact X.5.** Let  $\mathcal{C}$  be an edge-stable graph class. Then  $\mathcal{C}$  is monadically stable if and only if it is monadically dependent.

Monadically stable classes include all nowhere dense classes [AA14], as well as transductions of nowhere

dense classes, called *structurally nowhere dense* classes. It is not known whether all monadically stable classes are structurally nowhere dense (this has been conjectured in [NdMP<sup>+</sup>21, Conjecture 6.1]). The class of half-graphs is clearly not monadically stable, and has bounded (linear) clique-width. Hence, monadically stable classes are incomparable with classes of bounded clique-width. They are also incomparable with classes of bounded flip-width, as witnessed by the class of half-graphs on one side, and any nowhere dense class which does not have bounded expansion on the other side.

Monadic dependence can be defined not only for graph classes, but for arbitrary classes of structures, e.g. classes of ordered graphs. A class of ordered graphs is monadically dependent if it does not transduce the class of all graphs, where now the transduction may involve the edge relation symbol, as well as the total order  $<$  (and the color predicates). The following result is proved in [BGOdM<sup>+</sup>22].

**Fact X.6.** Let  $\mathcal{C}$  be a class of ordered graphs. Then  $\mathcal{C}$  is monadically dependent if and only if  $\mathcal{C}$  has bounded twin-width.

In this section, we study classes of almost bounded flip-width in the three settings discussed above: of weakly sparse classes, of edge-stable classes, and of classes of ordered graphs. We show that in the first and last settings, those classes coincide with monadically dependent classes, and that in the edge-stable case they also coincide, assuming all monadically stable classes are structurally nowhere dense.

We conjecture that the property of having almost bounded flip-width coincides exactly with monadic dependence.

**Conjecture X.7.** A graph class has almost bounded flip-width if and only if it is monadically dependent.

Currently, we are able to prove neither of the two implications in this conjecture. However, in the rest of Section X, we provide evidence towards this conjecture, by confirming it in restricted settings.

Another conjecture that is supported by the evidence presented below, predicts a collapse result: that a bound  $o(n^{1/2})$  on the flip-width parameters implies a bound  $o(n^\epsilon)$ , for every fixed  $\epsilon > 0$ .

**Conjecture X.8.** The following conditions are equivalent for a hereditary graph class  $\mathcal{C}$ :

- 1)  $\mathcal{C}$  has almost bounded flip-width, that is, for every fixed  $r \in \mathbb{N}$  and  $\epsilon > 0$  we have that  $\text{fw}_r(G) = o(|G|^\epsilon)$  holds for all  $G \in \mathcal{C}$ ,
- 2) for every fixed  $r \in \mathbb{N}$  we have that  $\text{fw}_r(G) = o(|G|^{1/2})$  holds for all  $G \in \mathcal{C}$ .

A similar collapse occurs in the sparse case [NdM12].

### B. Weakly sparse classes of almost bounded flip-width

Analogously to Theorem VI.3, which characterizes classes with bounded expansion as exactly the weakly sparse classes of bounded flip-width, we get a characterization of nowhere dense classes in terms of almost bounded flip-width.

**Theorem X.9.** *Let  $\mathcal{C}$  be a weakly sparse graph class. Then the following conditions are equivalent:*

- 1)  $\mathcal{C}$  is nowhere dense,
- 2)  $\mathcal{C}$  has almost bounded flip-width,
- 3)  $\mathcal{C}$  is monadically dependent.

The equivalence of the first and last condition is by Fact X.4, so we only prove the equivalence of the first two.

*Proof.* We first show that every nowhere dense class has almost bounded flip-width. Every nowhere dense class  $\mathcal{C}$  is weakly sparse, so excludes some  $K_{t,t}$  as a subgraph. By Theorem V.6, Theorem IV.5, and the forward implication in Fact III.7, we have that  $\text{fw}_r(G) \leq O_{r,\varepsilon}(n^{t\varepsilon})$  for every  $n$ -vertex graph  $G \in \mathcal{C}$ . Since this holds for every  $\varepsilon > 0$  and  $t$  is fixed, the conclusion follows by rescaling  $\varepsilon$ .

Conversely, suppose that  $\mathcal{C}$  has almost bounded flip-width, and excludes  $K_{t,t}$  as a subgraph. Without loss of generality, we may assume that  $\mathcal{C}$  is hereditary. We have  $\text{fw}_1(G) \leq O_\varepsilon(n^\varepsilon)$  for every  $n$ -vertex graph  $G \in \mathcal{C}$ , and by Theorem VI.1,  $\text{degeneracy}(G) \leq O_\varepsilon(n^\varepsilon \cdot 2t^2) \leq O_\varepsilon(n^\varepsilon)$ . By Theorem VI.4, for every  $r \geq 1$  and  $\varepsilon > 0$  we have

$$\begin{aligned} \tilde{\nabla}_{r-1}(G) &\leq (r \cdot \text{fw}_r(G) \cdot \text{degeneracy}(G))^{O(1)} \\ &\leq (r \cdot O_{r,\varepsilon}(|G|^\varepsilon) \cdot O_\varepsilon(|G|^\varepsilon))^{O(1)} \leq O_{r,\varepsilon}(|G|^{O(\varepsilon)}), \end{aligned}$$

for all  $G \in \mathcal{C}$ . By Facts III.3 and III.5 we have  $\text{wcol}_r(G) \leq O_{r,\varepsilon}(|G|^{O(\varepsilon)})$  for every  $r \geq 1$  and graph  $G \in \mathcal{C}$ . Therefore,  $\mathcal{C}$  is nowhere dense, by the backwards implication in Fact III.7.  $\square$

As there exist nowhere dense classes of unbounded degeneracy, we get the following.

**Corollary X.10.** *There is a class that has almost bounded flip-width, but does not have bounded flip-width.*

By taking the substitution closure (see Section V-A) of the class from the corollary above, we obtain a class which has almost bounded flip-width (by Lemma V.13), but is not nowhere dense (not even edge-stable), and has unbounded flip-width.

### C. Structurally nowhere dense classes

We are unable to determine whether classes of almost bounded flip-width are closed under transductions. Observe that the bound  $\text{fw}_r(\varphi(G)) \leq T_q(\text{fw}_{r'}(G))$  in Theorem VIII.2 is not polynomial in  $\text{fw}_{r'}(G)$ . It is, however, linear in the case when  $\varphi$  is a quantifier-free formula  $\varphi(x, y)$ , so we get the following.

**Corollary X.11.** *Let  $\mathcal{C}$  be a class of  $k$ -colored colored graphs of almost bounded flip-width, and let  $\varphi(x, y)$  be a quantifier-free formula. Then the class  $\varphi(\mathcal{C})$  has almost bounded flip-width.*

Even though we do not know whether classes of almost bounded flip-width are closed under transductions, we confirm that structurally nowhere dense classes (transductions of nowhere dense classes) have almost bounded flip-width.

**Theorem X.12.** *Every structurally nowhere dense class has almost bounded flip-width.*

To prove Theorem X.12, we use the main result of [DGK<sup>+</sup>22a], which essentially implies that for every structurally nowhere dense class  $\mathcal{C}$  there is an *almost nowhere dense* class  $\mathcal{B}$  of structures equipped with functions, and a quantifier-free formula  $\varphi(x, y)$  involving function symbols, such that  $\mathcal{C} \subseteq \varphi(\mathcal{B})$ . This is made precise below.

Say that a class  $\mathcal{C}$  of graphs is *almost nowhere dense* if for every  $r \geq 1$  and  $\varepsilon > 0$  we have  $\text{wcol}_r(G) \leq O_{r,\varepsilon}(|G|^\varepsilon)$ , for all  $G \in \mathcal{C}$ . Crucially,  $\mathcal{C}$  does not need to be hereditary, otherwise this notion would coincide with nowhere denseness by Fact III.7.

Fix a signature  $\Sigma$  consisting of unary relation symbols, binary relation symbols, and unary function symbols. The VC-dimension of a  $\Sigma$ -structure  $B$ , denoted  $\text{VCdim}(B)$ , is the maximum of the VC-dimensions of the binary relations of  $B$  (see Section III-C). Here, the functions of  $B$  are ignored.

The following result, apart from the ‘moreover’ part, is a straightforward consequence of [DGK<sup>+</sup>22a, Theorem 3]. It says that every structurally nowhere dense class  $\mathcal{C}$  interprets in an almost nowhere dense class  $\mathcal{B}$  of structures, via a quantifier-free interpretation using a unary function symbol. Additionally, every  $G \in \mathcal{C}$  interprets in some  $B \in \mathcal{B}$  with  $|B| \leq O(|G|)$ . Moreover, the binary relations of the binary structures in  $\mathcal{B}$  have bounded VC-dimension, which will be important in the next lemma, for controlling the bounds on the flip-width of  $G$ .

**Theorem X.13 (\*).** *Let  $\mathcal{C}$  be a structurally nowhere dense graph class. There is a signature  $\Sigma$  consisting of unary and binary relation symbols and one function symbol, a class  $\mathcal{B}$  of  $\Sigma$ -structures which is almost nowhere dense, and a quantifier-free symmetric formula  $\varphi(x, y)$  with the following property. For every graph  $G \in \mathcal{C}$  there is some  $B \in \mathcal{B}$  with  $|B| \leq O(|G|)$ , such that  $G$  is an induced subgraph of  $\varphi(B)$ . Moreover,  $\text{VCdim}(\mathcal{B}) < \infty$ .*

The ‘moreover’ part is shown by analysing the construction from [DGK<sup>+</sup>22a] (see [DGK<sup>+</sup>22b] for the full version), and observing that the constructed quasi-bushes interpret (via a  $d$ -dimensional interpretation, for some fixed  $d$ ) in graphs from a nowhere dense classes, and thus have bounded VC-dimension by the results of [AA14].

The following lemma extends the ideas used in the proof of Theorem VIII.2, specifically, of the special case of quantifier-free interpretations considered in Corollary X.11. The lemma considers quantifier-free interpretations that use function symbols, and says that under some technical conditions, such interpretations map graphs of small cop-width to graphs of small flip-width.

**Lemma X.14** (\*). *Let  $\Sigma$  be a signature consisting of unary and binary relation symbols, and unary function symbols. Fix  $k, r \geq 0$ , and a symmetric quantifier-free  $\Sigma$ -formula  $\varphi(x, y)$ . There are numbers  $p \leq O_\varphi(k)$  and  $r' \leq O_\varphi(r)$  such that the following holds. Let  $B$  be a  $\Sigma$ -structure with  $\text{VCdim}(B) \leq k$  and  $G_B$  be its Gaifman graph. Then*

$$\text{fw}_r(\varphi(B)) \leq O(\text{copwidth}_{r'}(G_B))^p.$$

The key insight is that a bound  $k$  on the VC-dimension implies that if the cops in the Cops and Robber game on  $G$  occupy a set  $S$  of vertices of a graph  $G$ , then, by the Sauer-Shelah-Perles lemma (Lemma III.8), the partition of  $V(G)$  into  $S$ -types has size  $O(|S|^k)$ , and this partition is used by the flipper in the flipper game.

The two statements above are proved in [Tor23]. Theorem X.12 follows, as we now show.

*Proof of Theorem X.12.* Let  $\mathcal{C}$  be a structurally nowhere dense class. Without loss of generality,  $\mathcal{C}$  is hereditary. Let  $\mathcal{B}$  and  $\varphi(x, y)$  be as in Theorem X.13, and  $k \in \mathbb{N}$  be such that  $\text{VCdim}(B) < k$  for  $B \in \mathcal{B}$ .

Let  $G \in \mathcal{C}$  and  $B \in \mathcal{B}$  be such that  $G$  is an induced subgraph of  $\varphi(B)$  and  $|B| \leq O(|G|)$ , and let  $G_B$  be the Gaifman graph of  $B$ . Let  $p$  and  $r'$  be as in Lemma X.14. Fix  $\varepsilon > 0$ . Then we have:

$$\begin{aligned} \text{fw}_r(G) &\leq \text{fw}_r(\varphi(B)) \leq O(\text{copwidth}_{r'}(G_B))^p \\ &\leq O_{r', \varepsilon}(|B|^{\varepsilon p}) \leq O_{r', \varepsilon}(|G|^{\varepsilon p}). \end{aligned}$$

Since  $r'$  depends only on  $r$  and  $\varphi$ , and  $p$  is a constant, and  $\varepsilon > 0$  is arbitrary, this proves that  $\mathcal{C}$  has almost bounded flip-width.  $\square$

#### D. Edge-stable classes of almost bounded flip-width

Conjecture X.7 predicts that a graph class has almost bounded flip-width if and only if it is monadically dependent. Currently, we are able to prove neither of the two implications. However, we prove the forward implication under the assumption that  $\mathcal{C}$  is edge-stable (that is, excludes some half-graph as a semi-induced bipartite graph).

**Theorem X.15.** *Let  $\mathcal{C}$  be an edge-stable, hereditary graph class such that for all  $r \in \mathbb{N}$ ,*

$$\text{fw}_r(G) = o(|G|^{1/2}) \quad \text{for } G \in \mathcal{C}.$$

*Then  $\mathcal{C}$  is monadically stable. In particular, every edge-stable, hereditary graph class of almost bounded flip-width is monadically stable.*

As far as we know, every monadically stable class might be structurally nowhere dense. This is conjectured in [NdMP<sup>+</sup>21, Conjecture 6.1]. If this were true, then Theorem X.15 and Theorem X.12 would imply that among edge-stable graph classes, almost bounded flip-width coincides with monadically stable class (and thus with monadically dependent classes, by Fact X.5). Moreover, this would imply a collapse result, confirming Conjecture X.8.

To prove Theorem X.15, we use the following result of [GMM<sup>+</sup>23, Thm. 1.4].

**Fact X.16** ([GMM<sup>+</sup>23]). *Let  $\mathcal{C}$  be a hereditary, edge-stable class of graphs. If  $\mathcal{C}$  is not monadically stable then there are  $r, k \geq 1$ , a  $k$ -coloring  $\hat{\mathcal{C}}$  of  $\mathcal{C}$  and a quantifier-free formula  $\varphi(x, y)$  such that  $\varphi(\hat{\mathcal{C}})$  contains the exact  $r$ -subdivision of every graph.*

*Proof of Theorem X.15.* Let  $\mathcal{C}$  be a hereditary, edge-stable graph class, such that for every fixed  $r \in \mathbb{N}$ , we have that  $\text{fw}_r(G) = o(|G|^{1/2})$  holds for all  $G \in \mathcal{C}$ . We prove that  $\mathcal{C}$  is monadically stable. Suppose otherwise. Let  $r, k, \hat{\mathcal{C}}$  and  $\varphi(x, y)$  be as in Fact X.16.

Pick a number  $n \geq 1$ . Let  $K_n^{(r)}$  denote the exact  $r$ -subdivision of the clique  $K_n$ . Then  $\text{fw}_{r+1}(K_n^{(r)}) \geq \Omega_r(n)$  by Proposition VI.7. By Fact X.16, there is some  $G_n \in \hat{\mathcal{C}}$  such that  $\varphi(G_n)$  is isomorphic to  $K_n^{(r)}$ . In particular,  $|V(G_n)| = |V(K_n^{(r)})| \leq O(rn^2)$ .

As  $\varphi$  is quantifier-free and involves  $k$  colors, by Theorem VIII.2, we have  $\text{fw}_{r+1}(\varphi(G_n)) \leq k \cdot \text{fw}_{r+1}(G_n)$ , and altogether:

$$\begin{aligned} \Omega_r(n) &\leq \text{fw}_{r+1}(K_n^{(r)}) \leq \text{fw}_{r+1}(\varphi(G_n)) \\ &\leq k \cdot \text{fw}_{r+1}(G_n) = o((n^2)^{1/2}) \leq o(n), \end{aligned}$$

which is impossible. Hence,  $\mathcal{C}$  is monadically stable.  $\square$

#### E. Classes of ordered graphs of almost bounded flip-width

Recall that flip-width is defined for arbitrary binary relational structures (see [Tor23]), and there is a variant of flip-width tailored to ordered graphs, defined in Section VII. It follows from Lemma VII.2, that a class  $\mathcal{C}$  of ordered graphs has almost bounded flip-width if and only if for every  $r$ , we have  $\text{fw}_r^<(\mathcal{C}) < \infty$ .

In this section, we prove that for hereditary classes of ordered graphs, almost bounded flip-width coincides with bounded twin-width, as well as with bounded flip-width (by Theorem VII.3). Moreover, in the setting of ordered graphs, a very strong form of the collapse predicted by Conjecture X.8, holds: from  $o(n^{1/2})$  all the way down to  $O(1)$  (as opposed to just  $n^{o(1)}$ , as predicted by the conjecture). This total collapse is, ultimately, a consequence of the Marcus-Tardos theorem/Stanley-Wilf conjecture.

**Theorem X.17.** *The following conditions are equivalent for a hereditary class  $\mathcal{C}$  of ordered graphs:*

- 1)  $\mathcal{C}$  is monadically dependent,
- 2)  $\mathcal{C}$  has bounded twin-width,
- 3)  $\mathcal{C}$  has bounded flip-width,
- 4)  $\mathcal{C}$  has almost bounded flip-width,
- 5)  $\text{fw}_1^<(G) = o(|G|^{1/2})$  for all  $G \in \mathcal{C}$ .

The equivalence  $1 \leftrightarrow 2$  is proved in [BGdM<sup>+</sup>22]. The equivalence  $2 \leftrightarrow 3$  is by Theorem VII.3. The implications  $3 \rightarrow 4$  and  $4 \rightarrow 5$  are immediate. We prove the implication  $5 \rightarrow 2$ . For this, we use the following consequence of the main result of [BGdM<sup>+</sup>22], which itself is a manifestation of the Marcus-Tardos theorem/Stanley-Wilf conjecture [MTo4].

Denote  $[n] := \{1, \dots, n\}$ . Fix a symbol  $s \in \{=, \neq, \leq_l, \geq_l, \leq_r, \geq_r\}$  and let  $G = (V, E, <)$  be an ordered graph and  $A, B \subseteq V(G)$ . Say that  $A$  and  $B$  form a  $s$ -pattern of order  $n^2$  if  $|A| = |B| = n^2$ , and the following condition holds (see Fig. 4). Let  $\alpha: [n]^2 \rightarrow A$  and  $\beta: [n]^2 \rightarrow B$  be order-preserving bijections, where  $[n]^2$  is ordered by the lexicographic order  $<_{\text{lex}}$ , while  $A$  and  $B$  are ordered by  $<$ . Then for  $(i, j), (i', j') \in [n]^2$ , we have that  $\alpha(i, j)\beta(i', j') \in E$  if and only if the following condition holds:

- $(i, j) = (j', i')$ , if  $s$  is  $=$ ,
- $(i, j) \neq (j', i')$ , if  $s$  is  $\neq$ ,
- $(j, i) \leq_{\text{lex}} (i', j')$ , if  $s$  is  $\leq_l$ ,
- $(j, i) \geq_{\text{lex}} (i', j')$ , if  $s$  is  $\geq_l$ ,
- $(i, j) \leq_{\text{lex}} (j', i')$ , if  $s$  is  $\leq_r$ ,
- $(i, j) \geq_{\text{lex}} (j', i')$ , if  $s$  is  $\geq_r$ .

**Fact X.18** ([BGdM<sup>+</sup>22]). *Let  $\mathcal{C}$  be a class of ordered graphs of unbounded twin-width. Then there is  $s \in \{=, \neq, \leq_l, \geq_l, \leq_r, \geq_r\}$  such that for every  $n \geq 1$ , there is some graph  $G \in \mathcal{C}$  and sets  $A$  and  $B$  that form an  $s$ -pattern of order  $n^2$ .*

*Proof.* Follows from [BGdM<sup>+</sup>21, Lemma 40], by considering the  $(s, \sigma)$ -matching (in their notation), where  $\sigma$  is the permutation of  $[n]^2$  (ordered lexicographically) such that  $\sigma((i, j)) = (j, i)$  for  $(i, j) \in [n]^2$ .  $\square$

Recall the definition of an  $n$ -rich division in an ordered graph  $G$ , and that the existence of such a division implies  $\text{fw}_1^<(G) \geq n$  (see Lemma VII.5 and definition before it).

**Lemma X.19.** *Let  $n$  be even, and  $G$  be an ordered graph containing an  $s$ -pattern of order  $n^2$ . Then  $G$  has an  $\frac{n}{2}$ -rich division. In particular,  $\text{fw}_1^<(G) \geq n/2$ .*

*Proof.* Let  $A$  and  $B$  form an  $s$ -pattern of order  $n^2$ . Let  $\mathcal{L} = \{I_1, \dots, I_n\}$  be the partition of  $V(G)$  into  $n$  intervals, with respect to the order of  $G$  restricted to  $A$ , each containing exactly  $n$  elements of  $A$ . Similarly, let  $\mathcal{R} = \{J_1, \dots, J_n\}$  be the partition of  $V(G)$  into  $n$  intervals, each containing exactly  $n$  elements of  $B$ . We show that  $\mathcal{L}$  and  $\mathcal{R}$  form an  $\frac{n}{2}$ -rich division of  $G$ .

Let  $I_i$  be an interval of  $\mathcal{L}$  and pick a family  $\mathcal{R}' \subseteq \mathcal{R}$  of  $n/2$  intervals of  $\mathcal{R}$ . Let  $\{j_1, \dots, j_{n/2}\}$  be the indices of the  $n/2$  intervals  $J_j$  in  $\mathcal{R}'$  and let  $J$  be their union. We show that there are  $n/2$  vertices in  $A \cap I_i$  with pairwise distinct

neighborhoods in  $B \cap J$ . This (by symmetry) implies that  $\mathcal{L}$  and  $\mathcal{R}$  form an  $\frac{n}{2}$ -rich division of  $G$ .

Let  $\alpha$  and  $\beta$  be as in the definition of an  $s$ -pattern of order  $n^2$ . Then, for  $p, q \in [n/2]$  we have that  $\alpha(i, j_p) \in A \cap I_i$  and  $\beta(j_q, i) \in B \cap J$ . Furthermore,  $\alpha(i, j_p)\beta(j_q, i) \in E(G)$  if and only if:

- $p = q$ , if  $s$  is  $=$ ,
- $p \neq q$ , if  $s$  is  $\neq$ ,
- $p \leq q$ , if  $s$  is  $\leq_l$  or  $\leq_r$ ,
- $p \geq q$ , if  $s$  is  $\geq_l$  or  $\geq_r$ .

In any case, the vertices  $\alpha(i, j_1), \dots, \alpha(i, j_{n/2}) \in A \cap I_i$  have distinct neighborhoods in  $B \cap J$ . This (and a symmetric argument, exchanging the roles of  $\mathcal{L}$  and  $\mathcal{R}$ ) demonstrates that  $\mathcal{L}$  and  $\mathcal{R}$  form an  $\frac{n}{2}$ -rich division. By Lemma VII.5, we have that  $\text{fw}_1^<(G) \geq n/2$ .  $\square$

*Proof of Theorem X.17.* As argued, it remains to prove the implication  $5 \rightarrow 2$ . Let  $\mathcal{C}$  be a hereditary class of ordered graphs with unbounded twin-width. We prove that there are infinitely many ordered graphs  $G \in \mathcal{C}$  with  $\text{fw}_1^<(G) \geq \Omega(|G|)^{1/2}$ , which contradicts condition 5.

Let  $s \in \{=, \neq, \leq_l, \geq_l, \leq_r, \geq_r\}$  be as in Fact X.18.

Fix an even  $n \geq 2$ . By Fact X.18 there is an ordered graph  $G_n \in \mathcal{C}$  and sets  $A, B \subseteq V(G_n)$  of size  $n^2$ , that form an  $s$ -pattern of order  $n^2$  in  $G$ . As  $\mathcal{C}$  is hereditary, we may assume that  $V(G_n) = A \cup B$ , so in particular  $|G_n| \leq 2n^2$ . By Lemma X.19,  $\text{fw}_1^<(G_n) \geq n/2$ . Altogether, we have that

$$\text{fw}_1^<(G_n) \geq \frac{1}{2}(|G_n|/2)^{1/2} = \Omega(|G_n|^{1/2}).$$

Since  $G_n \in \mathcal{C}$  for all even  $n \geq 2$ , it cannot be that  $\text{fw}_1^<(G) \leq o(|G|)^{1/2}$  holds for all  $G \in \mathcal{C}$ .  $\square$

*a) Summary:* In Section X we have defined graph classes of almost bounded flip-width. We conjecture that they coincide with monadically dependent graph classes. We provide the following evidence towards this conjecture. We have shown that, when restricted to weakly sparse graph classes, almost bounded flip-width coincides with nowhere denseness, and with monadic dependence. And when restricted to edge-stable graph classes, almost bounded flip-width generalizes structurally nowhere denseness, and is generalized by monadic stability. As it is conjectured that structurally nowhere dense classes coincide with monadically stable classes, this would imply that for edge-stable graph classes, almost bounded flip-width coincides with monadic stability, and with monadic dependence. We have shown that for classes of ordered graphs, almost bounded flip-width coincides with bounded flip-width, and with bounded twin-width, and therefore, with monadic dependence.

In all the special cases studied above – of weakly sparse graph classes, of edge-stable graph classes, and of classes of ordered graphs – the model-checking problem is known to be fixed-parameter tractable [GKS17],

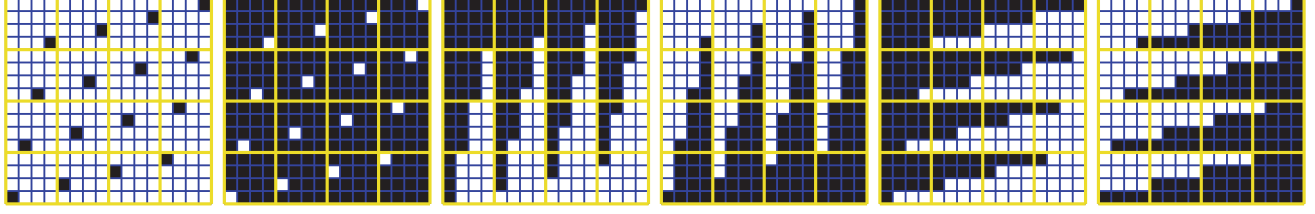


Fig. 4: An  $s$ -pattern  $A, B$  of order  $n^2$  for  $n = 4$  and  $s \in \{=, \neq, \leq_l, \geq_l, \leq_r, \geq_r\}$ , represented as incidence matrices of  $G[A, B]$ . The origin is at the lower-left corner and black entries denote incidence.

[DMS23], [BGdM<sup>+</sup>22] (in the edge-stable case, this assumes that monadic stability coincides with structural nowhere denseness). We therefore obtain the following corollary.

**Corollary X.20.** *Let  $\mathcal{C}$  be a class of almost bounded flip-width that is either weakly sparse, or is a class of ordered graphs, or is structurally nowhere dense. Then the model checking problem for first-order logic is fpt on  $\mathcal{C}$ .*

We remark that, up to best of our knowledge, all hereditary graph classes for which the model-checking problem is currently known to be fixed-parameter tractable – apart from one family of examples – have almost bounded flip-width. Those includes the listed cases above, as well as some classes of bounded twin-width [BCK<sup>+</sup>22], [GT23]. However, we do not know whether classes of structurally bounded local clique-width [BDG<sup>+</sup>22] have almost bounded flip-width (they are monadically dependent, and have tractable model-checking).

## XI. DISCUSSION

In this section, we discuss some conjectured relationships between various notions defined in this paper, and other known notions. We speculate on possible routes towards proving some conjectures.

### A. Obstructions to small flip-width

The results of this paper strongly suggest that the flip-width parameters are the sought dense analogues of generalized coloring numbers, that classes of bounded flip-width are the sought dense analogues of classes with bounded expansion, and almost bounded flip-width is a dense counterpart of nowhere dense classes. What is currently missing to complete this picture, is an analogue of the key result of Sparsity theory, which is a min-max theorem that relates, on one hand, explicit descriptions of winning strategies of the robber, and on the other hand, explicit descriptions of winning strategies for the cops in the Cops and Robber game (see Section IV). In the sparse case, the former are obstructions to having small  $r$ -admissibility numbers, or (by Fact III.5),  $\leq_r$ -subdivisions of graphs with large minimum degree. Finding an analogous notion in the dense case seems to be a major challenge. It seems plausible that finding such notions is related to the question of efficiently

approximating the flip-width parameters. This goal can be formalized as follows.

**Goal XI.1.** *Obtain an fpt approximation algorithm for radius- $r$  flip-width: an algorithm running in time  $f(r, k) \cdot n^c$ , for some constant  $c$  and function  $f: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ , which given an  $n$ -vertex graph  $G$  and numbers  $r, k$ , either concludes that  $\text{fw}_r(G) > k$ , or that  $\text{fw}_r(G) < f(r, k)$ .*

A first step in this direction is achieved by Theorem IX.7, which achieves an XP approximation algorithm, rather than an fpt algorithm.

A related, but less concrete goal is the following.

**Goal XI.2.** *Describe explicit forbidden weak obstructions at depth  $r$  and density  $\ell$ , such that there are functions  $f, g: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$  such that for all  $r, \ell \geq 1$ , every graph  $G$  with  $\text{fw}_r(G) \geq f(r, \ell)$  contains such an obstruction of density  $\ell$  as an induced subgraph, and conversely, no graph  $G$  which contains such an obstruction of density  $g(r, \ell)$  as an induced subgraph satisfies  $\text{fw}_r(G) \geq \ell$ .*

One attempt at formalizing what we mean by “explicit”, is by requiring that the following holds: for every fixed  $r$  there is a formula  $\varphi_r(x, y)$  (possibly involving colors) and a function  $f_r: \mathbb{N} \rightarrow \mathbb{N}$  such that if  $G$  is a forbidden weak obstruction at depth  $r$  and density at least  $f_r(\ell)$  then there is some  $k$ -coloring of  $G$  such that  $\varphi_r(G)$  contains an  $r$ -subdivision of a graph with minimum degree at least  $\ell$  as an induced subgraph.

The following question suggests a path towards achieving Goal XI.2.

**Question XI.3.** *Fix  $r \geq 1$ . Is it the case that  $\text{fw}_r(\mathcal{C}) < \infty$  if and only if there is a  $k \in \mathbb{N}$  such that no graph  $G \in \mathcal{C}$  contains a  $(r, k, k)$ -hideout (see Definition V.15)?*

Hideouts are not induced subgraphs, as posited in Goal XI.2, but this could serve as a starting point.

The following notions are defined in [GPT21]. A *transduction ideal* is a property of graph classes that is preserved under first-order transductions: if a class  $\mathcal{C}$  has the considered property, and  $\mathcal{D}$  transduces in  $\mathcal{C}$ , then  $\mathcal{D}$  has the property, too. The *dense analogue of bounded expansion* is the largest transduction ideal such that all weakly sparse classes that belong to it have bounded expansion.

By Corollary VIII.3, classes of bounded flip-width form a transduction ideal, and by Theorem VI.3, they are contained in the dense analogue of bounded expansion. We conjecture that bounded flip-width is exactly the dense analogue of bounded expansion, according to the above definition. This is equivalently stated as follows.

**Conjecture XI.4.** *Let  $\mathcal{C}$  be a class that does not have bounded flip-width. Then  $\mathcal{C}$  transduces a weakly sparse class  $\mathcal{D}$  which does not have bounded expansion.*

A road towards proving Conjecture XI.4 is through Goal XI.2. Indeed, if  $\mathcal{C}$  contains obstructions at a fixed depth  $r \geq 1$  and of arbitrarily large density  $\ell$ , then according to the requirement given below Goal XI.2,  $\mathcal{C}$  transduces a class  $\mathcal{D}$  which contains  $r$ -subdivisions of arbitrarily dense graphs, and therefore  $\mathcal{D}$  does not have bounded expansion.

#### B. Model checking

Recall that Conjecture I.1 predicts that for a hereditary class  $\mathcal{C}$ , first-order model-checking is fixed-parameter tractable (fpt) on  $\mathcal{C}$  if and only if  $\mathcal{C}$  is monadically dependent. Both implications are open. We now discuss approaching the backward implication, giving the upper bound: that model-checking is fpt on every monadically dependent graph class.

The question whether the model checking problem is fpt on every class  $\mathcal{C}$  of graphs that has bounded twin-width, remains open. (The result of [BKTW20] says that this is the case, if the input graph  $G$  is given together with its contraction sequence. In some special cases [BGdM<sup>+</sup>22], [BCK<sup>+</sup>22], [GT23], such a contraction sequence can be efficiently computed.) As bounded twin-width implies bounded flip-width, we do not know whether the model checking problem is fpt on every class  $\mathcal{C}$  of graphs that has bounded flip-width. However, we believe that approximating the radius- $r$  flip-width of a given graph  $G$ , might be easier than approximating its twin-width. This is indicated for example by Theorem IX.7, which gives an XP approximation algorithm for flip-width, while no such algorithm for twin-width is known. Still, an analogue of the model-checking result of [BKTW20] is missing, and is posed as the following goal.

**Goal XI.5.** *Devise an efficient representation of winning strategies for the flipper such that for a fixed formula  $\varphi$  there is a number  $r$ , such that given an efficient representation of a winning strategy for the flipper in the radius- $r$  flipper game on a given graph  $G$ , one can efficiently check whether  $\varphi$  holds in  $G$ .*

In particular, Goal XI.5 combined with Goal XI.1 could allow to solve the model-checking problem on classes of bounded flip-width (and also on classes of bounded twin-width, overcoming the lack of an fpt approximation algorithm for twin-width).

An extension of Goal XI.5 to monadically dependent classes, with the hope of confirming Conjecture I.1, could lead through Conjecture X.7, which characterizes the hereditary monadically dependent classes as those with almost bounded flip-width. This, however, remains very speculative, and most likely requiring further insights, on top of the ones needed to achieve Goal XI.5 and to solve Conjecture X.7.

#### C. Stable classes of bounded flip-width

It is conjectured [GPT21, Conjecture 6] that the edge-stable classes in the dense analogue of bounded expansion have structurally bounded expansion, i.e., are transductions of classes with bounded expansion. As classes of bounded flip-width are contained in the dense analogue of bounded expansion, this would imply the following.

**Conjecture XI.6.** *Every edge-stable class with bounded flip-width has structurally bounded expansion.*

Conjecture XI.6 would generalize the main result of [GPT22], [GPT21], which states that every edge-stable class of bounded twin-width has structurally bounded expansion. The conjunction of Conjecture XI.4 and Conjecture XI.6 implies the following duality statement for edge-stable graph classes.

**Conjecture XI.7.** *For every edge-stable graph class  $\mathcal{C}$ , either  $\mathcal{C}$  is a transduction of a class with bounded expansion, or  $\mathcal{C}$  transduces a weakly sparse class with unbounded expansion.*

Conjecture XI.4 and Conjecture XI.6 imply Conjecture XI.7: If  $\mathcal{C}$  is not monadically stable, then  $\mathcal{C}$  is not monadically dependent (by Fact X.5), and hence transduces every weakly sparse class of graphs. If  $\mathcal{C}$  is monadically stable and has unbounded flip-width, then  $\mathcal{C}$  transduces a weakly sparse class that does not have bounded expansion by Conjecture XI.4. Finally, if  $\mathcal{C}$  is monadically stable and has bounded flip-width, then  $\mathcal{C}$  transduces in a class of bounded expansion, by Conjecture XI.6.

#### D. Restrictions of the flipper game

Restricted classes of bounded flip-width can be defined by imposing additional constraints on the flipper game. We consider the following restrictions<sup>8</sup> on the strategy of the flipper in the flipper game. We say that the strategy of the flipper is:

- blind** if their move does not depend on the current position of the runner (but may depend on the graph and on the round number);
- positional** if their move is only based on the current position of the runner (and not on the past moves, nor the round number);

<sup>8</sup>I thank Rose McCarty and Pierre Ohlmann for suggesting the positional and bounded depth variants.

**of bounded depth** if there is a bound  $\ell$  such that the flipper wins within  $\ell$  rounds;

**branching-blind** if in each round move, the flipper proposes a partition  $\mathcal{P}$  of the vertex set, and bases his move only on the part of  $\mathcal{P}$  which is occupied by the runner, and furthermore, ensure that the runner will remain in that part until the end of the game.

Each of those variants of the game defines a variant of the flip-width parameter, and the related classes. Thus, a class  $\mathcal{C}$  has bounded  $X$  flip-width, for  $X \in \{\text{blind}, \text{branching blind}, \text{positional}\}$ , if for every radius  $r$ , there is a  $k$  such that the flipper wins the flipper game of radius  $r$  and width  $k$  on every graph  $G \in \mathcal{C}$ , using a strategy with property  $X$ . Similarly,  $\mathcal{C}$  has *bounded flip-depth* if for every  $r \in \mathbb{N}$  there are  $k, \ell$  such that for every  $G \in \mathcal{C}$ , the flipper has a strategy in the flipper game of radius  $r$  and width  $k$ , that wins in at most  $\ell$  rounds.

We may also consider the limit case of the above games, for  $r = \infty$ , thus defining classes with *blind  $\infty$ -flip-width*, *positional  $\infty$ -flip-width*,  *$\infty$ -flip-depth*, *branching-blind  $\infty$ -flip-width*, analogously as above.

It turns out that these notions relate to notions and conjectures that have been studied earlier, as we discuss below. We start with the following observation.

**Proposition XI.8.** *All the above properties are preserved under first-order transductions, and under CMSO transductions for the  $\infty$  variants.*

Proposition XI.8 follows by observing that in the proof of Theorem VIII.2 and Theorem VIII.8, the strategy on  $\varphi(G)$  is obtained by transferring a winning strategy from  $G$  (see Section VIII-B). And transferring a strategy with one of the listed properties, results in a strategy with the same property.

a) *Infinite radius:* In the limit case of radius  $\infty$ , the picture is quite well-understood, thanks to the following.

**Theorem XI.9.** *Let  $\mathcal{C}$  be a class of graphs. Then:*

- 1)  $\mathcal{C}$  has *branching-blind  $\infty$ -flip-width* if and only if  $\mathcal{C}$  has *bounded clique-width*;
- 2)  $\mathcal{C}$  has *blind  $\infty$ -flip-width* if and only if  $\mathcal{C}$  has *bounded linear clique-width*;
- 3)  $\mathcal{C}$  has *bounded positional  $\infty$ -flip-width*, if and only if  $\mathcal{C}$  has *bounded  $\infty$ -flip-depth*, if and only if  $\mathcal{C}$  has *bounded shrubdepth*.

A class  $\mathcal{C}$  has *bounded shrubdepth* [GHN<sup>+</sup>12] if and only if it transduces in a class of trees of depth bounded by a constant.

*Sketch of proof.* (1). For the forward implication, observe that the strategy presented in the proof of Theorem V.17 (see [Tor23]) is branching blind. For the backward implication, note that if  $\mathcal{C}$  has unbounded clique-width then by Theorem V.17 it has unbounded flip-width, so in particular, it has unbounded branching-blind flip-width.

(2). For the forward implication, observe that the strategy presented in Example V.9, for half-graphs, is blind. Therefore, the class of half-graphs has bounded blind flip-width. It is well-known that every class  $\mathcal{C}$  of bounded linear clique-width is a CMSO transduction of the class of half-graphs (or of the class of finite total orders). By Proposition XI.8,  $\mathcal{C}$  has bounded blind flip-width.

Conversely, if a class  $\mathcal{C}$  has unbounded linear clique-width, then it CMSO transduces the class of all trees (this follows from [HJMW20], [DT17] and from [CiO07]). It is not difficult to verify that the class of trees does not have bounded blind  $\infty$ -flip-width. Again by Proposition XI.8, this implies that  $\mathcal{C}$  has unbounded blind  $\infty$ -flip-width.

(3). First observe that in a tree of depth at most  $d$ , the flipper has a positional winning strategy in the radius- $\infty$  flipper game, that wins in at most  $d$  rounds (this is essentially the same as the strategy in Example IV.3). Hence, every class  $\mathcal{D}$  of trees of depth bounded by a constant has bounded positional  $\infty$ -flip-width, and bounded  $\infty$ -flip-depth. By Proposition XI.8, the same holds for every class that transduces in  $\mathcal{D}$ , hence for all classes of bounded shrubdepth.

Conversely, if a class  $\mathcal{C}$  has unbounded shrubdepth, then it CMSO-transduces the class of all paths, by [KMOW21]. It is not difficult to verify that the class of all paths does not have bounded positional  $\infty$ -flip-width, and does not have bounded  $\infty$ -flip-depth. Proposition XI.8 implies that  $\mathcal{C}$  has unbounded positional  $\infty$ -flip-width, and unbounded  $\infty$ -flip-depth.  $\square$

b) *Blind case:* We now move the case of classes of bounded blind flip-depth. We start with some examples.

*Example XI.10.* Classes of bounded blind flip-width include classes of bounded pathwidth, and more generally, classes of bounded linear rank-width (or linear clique-width), by Theorem XI.9.

*Example XI.11.* On the other hand, the strategies described in Example IV.3 and in Example V.10, for trees and for comparability graphs of trees, respectively, are not blind. One can show that those classes do not have bounded blind flip-width. A more general statement is given below.

**Conjecture XI.12.** *A class  $\mathcal{C}$  has bounded blind flip-width if and only if  $\mathcal{C}$  has bounded linear clique-width.*

As bounded linear clique-width is equivalent to bounded blind  $\infty$ -flip-width by Theorem XI.9, Conjecture XI.12 states that bounded blind flip-width collapses to bounded blind  $\infty$ -flip-width.

c) *Branching-blind case:* We now discuss the branching-blind case. By Theorem XI.9, every class of bounded clique-width has bounded branching-blind flip-width.

**Conjecture XI.13.** *A class  $\mathcal{C}$  has bounded branching-blind flip-width if and only if  $\mathcal{C}$  has bounded clique-width.*

As bounded clique-width is equivalent to bounded branching-blind  $\infty$ -flip-width by Theorem XI.9, Conjecture XI.13 states that bounded branching-blind flip-width collapses to bounded branching-blind  $\infty$ -flip-width.

d) *Positional case:* Let us now look at some examples related to bounded positional flip-width.

*Example XI.14.* Every class with bounded expansion has bounded positional flip-width. This follows from the proof of the upper bound in Theorem IV.5, as the cops' strategy in the radius- $r$  Cops and Robber game is to occupy (or isolate) the vertices that are  $2r$ -weakly reachable from the vertex that is currently occupied by the robber. Clearly, this is a positional strategy.

By Proposition XI.8 we get the following.

**Corollary XI.15.** *Every class with structurally bounded expansion has bounded positional flip-width.*

On the other hand, we have:

*Example XI.16.* Half-graphs have unbounded positional flip-width.

It follows that every class  $\mathcal{C}$  with bounded positional flip-width is edge-stable. Since  $\mathcal{C}$  is also monadically dependent by Corollary VIII.4, and every monadically dependent, edge-stable class is monadically stable by Fact X.5, we get the following.

**Corollary XI.17.** *Every class with bounded positional flip-width is monadically stable.*

Conjecture XI.6 would therefore imply the following characterization of classes with bounded positional flip-width.

**Conjecture XI.18.** *A class  $\mathcal{C}$  has bounded positional flip-width if and only if  $\mathcal{C}$  has structurally bounded expansion.*

e) *Bounded flip-depth:* Finally, we look at classes of bounded flip-depth.

*Example XI.19.* Classes of bounded degree have bounded flip-depth, as seen in Example IV.2. Also, classes of bounded treedepth, or more generally, classes of bounded shrubdepth, have bounded flip-depth.

A class of examples combining the above is provided by the following notion. Fix  $d, h \geq 1$ . A *hybrid tree* of depth  $h$  and degree  $d$  is a graph  $G$  that can be obtained from a rooted tree  $T$  of depth  $h$  by:

- first adding some edges connecting siblings in  $T$ , in such a way that every vertex is adjacent in  $G$  to at most  $d$  of its siblings in  $T$ ,
- afterwards, taking a subgraph of the resulting graph.

**Proposition XI.20.** *Fix  $d, h \geq 1$ . The class  $\mathcal{H}_{d,h}$  of hybrid trees of depth  $h$  and degree  $d$  has bounded flip-depth.*

The following is a consequence of Simon's factorisation theorem [Sim90]. We omit the details.

**Theorem XI.21.** *Every class  $\mathcal{C}$  of bounded pathwidth, and more generally, every stable class of bounded linear clique-width, transduces in  $\mathcal{H}_{2,h}$ , for some fixed  $h$ .*

**Corollary XI.22.** *Every class of bounded pathwidth, and more generally, every stable class of bounded linear clique-width has bounded flip-depth.*

We now pose two conjectures characterizing classes of bounded flip-depth in complementary ways.

Define the *tree-rank* of a graph class  $\mathcal{C}$  as the largest number  $d \in \mathbb{N}$  such that  $\mathcal{C}$  transduces the class of all forests of depth  $d$ , or  $\infty$  if the largest such number does not exist. Note that classes of rank  $d \in \mathbb{N}$  are in particular monadically dependent, in fact, monadically stable, since the class of half-graphs transduces every class of forests of bounded depth.

Next, the following conjecture relates the above notion to bounded flip-depth.

**Conjecture XI.23.** *A class  $\mathcal{C}$  has bounded flip-depth if and only if  $\mathcal{C}$  has rank  $< d$ , for some  $d \in \mathbb{N}$ .*

We finish with the following conjecture, relating classes of bounded flip-width with the complexity of the model-checking problem for first-order logic.

Say that a graph class  $\mathcal{C}$  has *non-uniform elementary-fpt* first-order model-checking if there are numbers  $h, c \geq 1$  such that for every first-order formula  $\varphi$  of length  $k$  there is an algorithm which determines if a given  $n$ -vertex graph  $G \in \mathcal{C}$  satisfies  $\varphi$  in time at most

$$\underbrace{2^{2^{\cdot^{\cdot^{\cdot^2}}}}}_{\text{height } h} \cdot n^c.$$

In the *uniform* variant, there is a single algorithm, which inputs  $\varphi$  and  $G \in \mathcal{C}$ , and has the above running time.

It is known [DGKS07] that every class of bounded degree has uniform elementary-fpt model-checking. Furthermore, the class of trees does not have uniform elementary-fpt model-checking, unless  $\text{AW}[*] = \text{FPT}$ , by a result of Frick and Grohe [FG04] (see also [DGKS07]), and the class of colored linear orders does not have uniform elementary-fpt model-checking unless  $\text{P} = \text{NP}$ , also by Frick and Grohe.

Together with Michał Pilipczuk, we pose the following.

**Conjecture XI.24.** *A class  $\mathcal{C}$  of graphs has non-uniform elementary-fpt first-order model-checking if and only if  $\mathcal{C}$  does not transduce the class of forests of depth  $d$ , for some  $d \geq 1$ .*

Together with Conjecture XI.23, this would characterize classes with non-uniform elementary-fpt first-order model-checking precisely as those with bounded flip-depth.

# REFERENCES

- [AA14] Hans Adler and Isolde Adler. Interpreting nowhere dense graph classes as a classical notion of model theory. *Eur. J. Comb.*, 36:322–330, 2014. doi:[10.1016/j.ejcb.2013.06.048](https://doi.org/10.1016/j.ejcb.2013.06.048).
- [AAL21] Bogdan Alecu, Aistis Atminas, and Vadim Lozin. Graph functionality. *Journal of Combinatorial Theory, Series B*, 147:139–158, 2021. doi:<https://doi.org/10.1016/j.jctb.2020.11.002>.
- [ACLZ15] A. Atminas, A. Collins, V. Lozin, and V. Zamaraev. Implicit representations and factorial properties of graphs. *Discrete Mathematics*, 338(2):164–179, 2015. doi:<https://doi.org/10.1016/j.disc.2014.09.008>.
- [Adl08] Hans Adler. An introduction to theories without the independence property. available online, 2008.
- [AF84] M. Aigner and M. Fromme. A game of cops and robbers. *Discrete Applied Mathematics*, 8(1):1–12, 1984. doi:[https://doi.org/10.1016/0166-218X\(84\)90073-8](https://doi.org/10.1016/0166-218X(84)90073-8).
- [AM15] Noga Alon and Abbas Mehrabian. Chasing a fast robber on planar graphs and random graphs. *Journal of Graph Theory*, 78(2):81–96, 2015. doi:<https://doi.org/10.1002/jgt.21791>.
- [BCK<sup>+</sup>22] Édouard Bonnet, Dibayan Chakraborty, Eun Jung Kim, Noleen Köhler, Raul Lopes, and Stéphane Thomassé. Twin-width VIII: delineation and win-wins. In Holger Dell and Jesper Nederlof, editors, *17th International Symposium on Parameterized and Exact Computation, IPEC 2022, September 7–9, 2022, Potsdam, Germany*, volume 249 of *LIPIcs*, pages 9:1–9:18. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2022. doi:[10.4230/LIPIcs.IPEC.2022.9](https://doi.org/10.4230/LIPIcs.IPEC.2022.9).
- [BDG<sup>+</sup>22] Édouard Bonnet, Jan Dreier, Jakub Gajarský, Stephan Kreutzer, Nikolas Mählmann, Pierre Simon, and Szymon Toruńczyk. Model Checking on Interpretations of Classes of Bounded Local Cliquewidth. In *37th Annual ACM/IEEE Symposium on Logic in Computer Science, LICS 2022*. ACM, 2022. URL: <https://hal.archives-ouvertes.fr/hal-03751012>.
- [BFLP24] Édouard Bonnet, Florent Foucaud, Tuomo Lehtilä, and Aline Parreau. Neighbourhood complexity of graphs of bounded twin-width. *European Journal of Combinatorics*, 115:103772, 2024. doi:<https://doi.org/10.1016/j.ejcb.2023.103772>.
- [BGK<sup>+</sup>21a] Édouard Bonnet, Colin Geniet, Eun Jung Kim, Stéphane Thomassé, and Rémi Watrigant. Twin-width II: small classes. In Daniel Marx, editor, *Proceedings of the 2021 ACM-SIAM Symposium on Discrete Algorithms, SODA 2021, Virtual Conference, January 10 - 13, 2021*, pages 1977–1996. SIAM, 2021. doi:[10.1137/1.9781611976465.118](https://doi.org/10.1137/1.9781611976465.118).
- [BGK<sup>+</sup>21b] Édouard Bonnet, Colin Geniet, Eun Jung Kim, Stéphane Thomassé, and Rémi Watrigant. Twin-width III: Max Independent Set, Min Dominating Set, and Coloring. In Nikhil Bansal, Emanuela Merelli, and James Worrell, editors, *48th International Colloquium on Automata, Languages, and Programming (ICALP 2021)*, volume 198 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 35:1–35:20. Dagstuhl, Germany, 2021. Schloss Dagstuhl - Leibniz-Zentrum für Informatik. doi:[10.4230/LIPIcs.ICALP.2021.35](https://doi.org/10.4230/LIPIcs.ICALP.2021.35).
- [BGOdM<sup>+</sup>21] Édouard Bonnet, Ugo Giocanti, Patrice Ossona de Mendez, Pierre Simon, Stéphane Thomassé, and Szymon Toruńczyk. Twin-width iv: ordered graphs and matrices. *arXiv*, 2021. arXiv:2102.03117.
- [BGOdM<sup>+</sup>22] Édouard Bonnet, Ugo Giocanti, Patrice Ossona de Mendez, Pierre Simon, Stéphane Thomassé, and Szymon Toruńczyk. Twin-width iv: Ordered graphs and matrices. In *Proceedings of the 54th Annual ACM SIGACT Symposium on Theory of Computing, STOC 2022*, pages 924–937, New York, NY, USA, 2022. Association for Computing Machinery. doi:[10.1145/3519935.3520037](https://doi.org/10.1145/3519935.3520037).
- [BGOdMT23] Édouard Bonnet, Ugo Giocanti, Patrice Ossona de Mendez, and Stéphane Thomassé. Twin-Width V: Linear Minors, Modular Counting, and Matrix Multiplication. In Petra Berenbrink, Patricia Bouyer, Anuj Dawar, and Mamadou Moustapha Kanté, editors, *40th International Symposium on Theoretical Aspects of Computer Science (STACS 2023)*, volume 254 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 15:1–15:16. Dagstuhl, Germany, 2023. Schloss Dagstuhl - Leibniz-Zentrum für Informatik. doi:[10.4230/LIPIcs.STACS.2023.15](https://doi.org/10.4230/LIPIcs.STACS.2023.15).
- [BGTT22] Édouard Bonnet, Colin Geniet, Romain Tessera, and Stéphane Thomassé. Twin-width VII: groups. *arXiv*, abs/2204.12330, 2022. arXiv:2204.12330.
- [BKRT22] Édouard Bonnet, Eun Jung Kim, Amadeus Reinald, and Stéphane Thomassé. Twin-width VI: the lens of contraction sequences. In Joseph (Seffi) Naor and Niv Buchbinder, editors, *Proceedings of the 2022 ACM-SIAM Symposium on Discrete Algorithms, SODA 2022, Virtual Conference / Alexandria, VA, USA, January 9 - 12, 2022*, pages 1036–1056. SIAM, 2022. doi:[10.1137/1.9781611977073.45](https://doi.org/10.1137/1.9781611977073.45).
- [BKTW20] Édouard Bonnet, Eun Jung Kim, Stéphane Thomassé, and Rémi Watrigant. Twin-width I: tractable FO model checking. In Sandy Irani, editor, *61st IEEE Annual Symposium on Foundations of Computer Science, FOCS 2020, Durham, NC, USA, November 16–19, 2020*, pages 601–612. IEEE, 2020. doi:[10.1109/FOCS46700.2020.00062](https://doi.org/10.1109/FOCS46700.2020.00062).
- [BL21] Samuel Brainfeld and Michael C. Laskowski. Characterizations of monadic nlp. *Transactions of the American Mathematical Society, Series B*, 2021.
- [BL22] Samuel Brainfeld and Michael C Laskowski. Existential characterizations of monadic NIP. *arXiv preprint arXiv:2209.05120*, 2022.
- [BNdMS22a] Samuel Brainfeld, Jaroslav Nešetřil, Patrice Ossona de Mendez, and Sebastian Siebertz. Decomposition horizons: from graph sparsity to model-theoretic dividing lines, 2022. arXiv:2209.11229.
- [BNdMS22b] Samuel Brainfeld, Jaroslav Nešetřil, Patrice Ossona de Mendez, and Sebastian Siebertz. On the first-order transduction quasiover of hereditary classes of graphs. *arXiv*, abs/2208.14412, 2022. arXiv:2208.14412.
- [BS85] J.T. Baldwin and S. Shelah. Second-order quantifiers and the complexity of theories. *Notre Dame Journal of Formal Logic*, 26(3):229–303, 1985.
- [BT15] Nicolas Bousquet and Stéphane Thomassé. Vc-dimension and erdős-pósa property. *Discrete Mathematics*, 338(12):2302–2317, 2015. doi:<https://doi.org/10.1016/j.disc.2015.05.026>.
- [CER93] Bruno Courcelle, Joost Engelfriet, and Grzegorz Rozenberg. Handle-rewriting hypergraph grammars. *Journal of Computer and System Sciences*, 46(2):218–270, 1993. doi:[https://doi.org/10.1016/0022-0000\(93\)90004-G](https://doi.org/10.1016/0022-0000(93)90004-G).
- [CiO07] Bruno Courcelle and Sang il Oum. Vertex-minors, monadic second-order logic, and a conjecture by seese. *Journal of Combinatorial Theory, Series B*, 97(1):91–126, 2007. doi:<https://doi.org/10.1016/j.jctb.2006.04.003>.
- [DGK<sup>+</sup>22a] Jan Dreier, Jakub Gajarský, Sandra Kiefer, Michal Pilipczuk, and Szymon Toruńczyk. Treelike decompositions for transductions of sparse graphs. In Christel Baier and Dana Fisman, editors, *LICS '22: 37th Annual ACM/IEEE Symposium on Logic in Computer Science, Haifa, Israel, August 2 - 5, 2022*, pages 31:1–31:14. ACM, 2022. doi:[10.1145/3531130.3533349](https://doi.org/10.1145/3531130.3533349).
- [DGK<sup>+</sup>22b] Jan Dreier, Jakub Gajarský, Sandra Kiefer, Michal Pilipczuk, and Szymon Toruńczyk. Treelike decompositions for transductions of sparse graphs. *arXiv*, abs/2201.11082, 2022. arXiv:2201.11082.
- [DGKS07] Anuj Dawar, Martin Grohe, Stephan Kreutzer, and Nicole Schweikardt. Model theory makes formulas large. In Lars Arge, Christian Cachin, Tomasz Jurdzinski, and Andrzej Tarlecki, editors, *Automata, Lan-*

- languages and Programming, pages 913–924, Berlin, Heidelberg, 2007. Springer Berlin Heidelberg.
- [DKo6] Guoli Ding and Andrei Kotlov. On minimal rank over finite fields. *ELA. The Electronic Journal of Linear Algebra [electronic only]*, 15:210–214, 2006. URL: <http://dml.mathdoc.fr/item/05279950>.
- [DKT13] Zdeněk Dvořák, Daniel Král, and Robin Thomas. Testing first-order properties for subclasses of sparse graphs. *J. ACM*, 60(5):36:1–36:24, 2013. doi:10.1145/2499483.
- [DMS23] Jan Dreier, Nikolas Mählmann, and Sebastian Siebertz. First-order model checking on structurally sparse graph classes. In Barna Saha and Rocco A. Servedio, editors, *Proceedings of the 55th Annual ACM Symposium on Theory of Computing, STOC 2023, Orlando, FL, USA, June 20-23, 2023*, pages 567–580. ACM, 2023. doi:10.1145/3564246.3585186.
- [DT17] Thanh N. Dang and Robin Thomas. Minors of two-connected graphs of large path-width, 2017.
- [Dvo12] Zdeněk Dvořák. A stronger structure theorem for excluded topological minors. *ArXiv*, abs/1209.0129, 2012.
- [Dvo13] Zdeněk Dvořák. Constant-factor approximation of the domination number in sparse graphs. *European Journal of Combinatorics*, 34(5):833–840, 2013. doi:https://doi.org/10.1016/j.ejc.2012.12.004.
- [Dvo18] Zdeněk Dvořák. Induced subdivisions and bounded expansion. *Eur. J. Comb.*, 69(C):143–148, mar 2018. doi:10.1016/j.ejc.2017.10.004.
- [FGo4] Markus Frick and Martin Grohe. The complexity of first-order and monadic second-order logic revisited. *Annals of Pure and Applied Logic*, 130(1):3–31, 2004. Papers presented at the 2002 IEEE Symposium on Logic in Computer Science (LICS). doi:https://doi.org/10.1016/j.apal.2004.01.007.
- [FGK<sup>+</sup>10] Fedor V. Fomin, Petr A. Golovach, Jan Kratochvíl, Nicolas Nisse, and Karol Suchan. Pursuing a fast robber on a graph. *Theoretical Computer Science*, 411(7):1167–1181, 2010. doi:https://doi.org/10.1016/j.tcs.2009.12.010.
- [FKL12] Alan Frieze, Michael Krivelevich, and Po-Shen Loh. Variations on cops and robbers. *Journal of Graph Theory*, 69(4):383–402, 2012. doi:https://doi.org/10.1002/jgt.20591.
- [FTo8] Fedor V. Fomin and Dimitrios M. Thilikos. An annotated bibliography on guaranteed graph searching. *Theoretical Computer Science*, 399(3):236–245, 2008. Graph Searching. doi:https://doi.org/10.1016/j.tcs.2008.02.040.
- [GHN<sup>+</sup>12] Robert Ganian, Petr Hliněný, Jaroslav Nesetril, Jan Obdržálek, Patrice Ossona de Mendez, and Reshma Ramadurai. When trees grow low: Shrubs and fast MSO<sub>1</sub>. In Branislav Rován, Vladimiro Sassone, and Peter Widmayer, editors, *Mathematical Foundations of Computer Science 2012 - 37th International Symposium, MFCS 2012, Bratislava, Slovakia, August 27-31, 2012. Proceedings*, volume 7464 of *Lecture Notes in Computer Science*, pages 419–430. Springer, 2012. doi:10.1007/978-3-642-32589-2\\_38.
- [GHO<sup>+</sup>20] Jakub Gajarský, Petr Hliněný, Jan Obdržálek, Daniel Lokshtanov, and M. S. Ramanujan. A new perspective on fo model checking of dense graph classes. *ACM Trans. Comput. Logic*, 21(4), 2020. doi:10.1145/3383206.
- [GKN<sup>+</sup>18] Jakub Gajarský, Stephan Kreutzer, Jaroslav Nesetril, Patrice Ossona de Mendez, Michal Pilipczuk, Sebastian Siebertz, and Szymon Torunczyk. First-order interpretations of bounded expansion classes. In Ioannis Chatzigiannakis, Christos Kaklamani, Daniel Marx, and Donald Sannella, editors, *45th International Colloquium on Automata, Languages, and Programming, ICALP 2018, July 9-13, 2018, Prague, Czech Republic*, volume 107 of *LIPICs*, pages 126:1–126:14. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2018. doi:10.4230/LIPICs.ICALP.2018.126.
- [GKN<sup>+</sup>20] Jakub Gajarský, Stephan Kreutzer, Jaroslav Nesetril, Patrice Ossona de Mendez, Michal Pilipczuk, Sebastian Siebertz, and Szymon Torunczyk. First-order interpretations of bounded expansion classes. *ACM Trans. Comput. Log.*, 21(4):29:1–29:41, 2020. doi:10.1145/3382093.
- [GKR<sup>+</sup>18] Martin Grohe, Stephan Kreutzer, Roman Rabinovich, Sebastian Siebertz, and Konstantinos Stavropoulos. Coloring and covering nowhere dense graphs. *SIAM Journal on Discrete Mathematics*, 32(4):2467–2481, 2018. doi:10.1137/18M1168753.
- [GKS17] Martin Grohe, Stephan Kreutzer, and Sebastian Siebertz. Deciding first-order properties of nowhere dense graphs. *J. ACM*, 64(3):17:1–17:32, 2017. doi:10.1145/3051095.
- [GMM<sup>+</sup>23] Jakub Gajarský, Nikolas Mählmann, Rose McCarty, Pierre Ohlmann, Michal Pilipczuk, Wojciech Przybyszewski, Sebastian Siebertz, Marek Sokolowski, and Szymon Torunczyk. Flipper games for monadically stable graph classes. In Kousha Etessami, Uriel Feige, and Gabriele Puppis, editors, *50th International Colloquium on Automata, Languages, and Programming, ICALP 2023, July 10-14, 2023, Paderborn, Germany*, volume 261 of *LIPICs*, pages 128:1–128:16. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2023. doi:10.4230/LIPICs.ICALP.2023.128.
- [GPPT22] Jakub Gajarský, Michal Pilipczuk, Wojciech Przybyszewski, and Szymon Torunczyk. Twin-width and types. In Mikolaj Bojanczyk, Emanuela Merelli, and David P. Woodruff, editors, *49th International Colloquium on Automata, Languages, and Programming, ICALP 2022, July 4-8, 2022, Paris, France*, volume 229 of *LIPICs*, pages 123:1–123:21. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2022. doi:10.4230/LIPICs.ICALP.2022.123.
- [GPT21] Jakub Gajarský, Michał Pilipczuk, and Szymon Toruńczyk. Stable graphs of bounded twin-width. *arXiv*, 2021. arXiv:2107.03711.
- [GPT22] Jakub Gajarský, Michał Pilipczuk, and Szymon Toruńczyk. Stable graphs of bounded twin-width. In *Proceedings of the 37th Annual ACM/IEEE Symposium on Logic in Computer Science, LICS '22, New York, NY, USA, 2022*. Association for Computing Machinery. doi:10.1145/3531130.3533356.
- [Gro07] Martin Grohe. Logic, graphs, and algorithms. *Electron. Colloquium Comput. Complex.*, TR07-091, 2007. URL: <https://eccc.weizmann.ac.il/eccc-reports/2007/TR07-091/index.html>.
- [GT23] Colin Geniet and Stéphan Thomassé. First order logic and twin-width in tournaments. In Inge Li Gørtz, Martin Farach-Colton, Simon J. Puglisi, and Grzegorz Herman, editors, *31st Annual European Symposium on Algorithms, ESA 2023, September 4-6, 2023, Amsterdam, The Netherlands*, volume 274 of *LIPICs*, pages 53:1–53:14. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2023. doi:10.4230/LIPICs.ESA.2023.53.
- [GWoo] Frank Gurski and Egon Wanke. The tree-width of clique-width bounded graphs without kn,n. In Ulrik Brandes and Dorothea Wagner, editors, *Graph-Theoretic Concepts in Computer Science*, pages 196–205, Berlin, Heidelberg, 2000. Springer Berlin Heidelberg.
- [HJMW20] Tony Huynh, Gwenaél Joret, Piotr Micek, and David Wood. Seymour’s conjecture on 2-connected graphs of large pathwidth. *Combinatorica*, 40:839–868, 2020. doi:10.1007/s00493-020-3941-3.
- [iOs06] Sang il Oum and Paul Seymour. Approximating clique-width and branch-width. *Journal of Combinatorial Theory, Series B*, 96(4):514–528, 2006. doi:https://doi.org/10.1016/j.jctb.2005.10.006.
- [KMOW21] O-joung Kwon, Rose McCarty, Sang-il Oum, and Paul Wollan. Obstructions for bounded shrub-depth and rank-depth. *Journal of Combinatorial Theory, Series B*, 149:76–91, 2021. doi:https://doi.org/10.1016/j.jctb.2021.01.005.

- [KOo4] Daniela Kühn and Deryk Osthus. Induced subdivisions in  $k$ s, $s$ -free graphs of large average degree. *Combinatorica*, 24(2):287–304, 2004. doi:[10.1007/s00493-004-0017-8](https://doi.org/10.1007/s00493-004-0017-8).
- [KPS17] O-joung Kwon, Michał Pilipczuk, and Sebastian Siebertz. On low rank-width colorings. In Hans L. Bodlaender and Gerhard J. Woeginger, editors, *Graph-Theoretic Concepts in Computer Science*, pages 372–385, Cham, 2017. Springer International Publishing.
- [Lam12] Michael Lampis. Algorithmic meta-theorems for restrictions of treewidth. *Algorithmica*, 64(1):19–37, 2012. doi:[10.1007/s00453-011-9554-x](https://doi.org/10.1007/s00453-011-9554-x).
- [LPPT20] Stratis Limnios, Christophe Paul, Joanny Perret, and Dimitrios M. Thilikos. Edge degeneracy: Algorithmic and structural results. *Theoretical Computer Science*, 839:164–175, 2020. doi:<https://doi.org/10.1016/j.tcs.2020.06.006>.
- [Mat04] Jiří Matoušek. Bounded VC-dimension implies a fractional Helly theorem. *Discrete Comput. Geom.*, 31(2):251–255, 2004.
- [MT04] Adam Marcus and Gábor Tardos. Excluded permutation matrices and the stanley–wilf conjecture. *Journal of Combinatorial Theory, Series A*, 107(1):153–160, 2004. doi:<https://doi.org/10.1016/j.jcta.2004.04.002>.
- [NdM12] Jaroslav Nešetřil and Patrice Ossona de Mendez. *Sparcity - Graphs, Structures, and Algorithms*, volume 28 of *Algorithms and combinatorics*. Springer, 2012. doi:[10.1007/978-3-642-27875-4](https://doi.org/10.1007/978-3-642-27875-4).
- [NdMP<sup>+</sup>21] Jaroslav Nešetřil, Patrice Ossona de Mendez, Michał Pilipczuk, Roman Rabinovich, and Sebastian Siebertz. Rankwidth meets stability. In *Proceedings of the Thirty-Second Annual ACM-SIAM Symposium on Discrete Algorithms, SODA '21*, pages 2014–2033, USA, 2021. Society for Industrial and Applied Mathematics.
- [NiO20] Huy-Tung Nguyen and Sang il Oum. The average cut-rank of graphs. *European Journal of Combinatorics*, 90:103183, 2020. doi:<https://doi.org/10.1016/j.ejc.2020.103183>.
- [NO08] Jaroslav Nešetřil and Patrice Ossona de Mendez. Grad and classes with bounded expansion i. decompositions. *European Journal of Combinatorics*, 29(3):760–776, 2008. doi:<https://doi.org/10.1016/j.ejc.2006.07.013>.
- [NO11] Jaroslav Nešetřil and Patrice Ossona de Mendez. On nowhere dense graphs. *Eur. J. Comb.*, 32(4):600–617, 2011. doi:[10.1016/j.ejc.2011.01.006](https://doi.org/10.1016/j.ejc.2011.01.006).
- [NORS21] Jaroslav Nešetřil, Patrice Ossona de Mendez, Roman Rabinovich, and Sebastian Siebertz. Classes of graphs with low complexity: The case of classes with bounded linear rankwidth. *European Journal of Combinatorics*, 91:103223, 2021. Colorings and structural graph theory in context (a tribute to Xuding Zhu). doi:<https://doi.org/10.1016/j.ejc.2020.103223>.
- [PST18] Michał Pilipczuk, Sebastian Siebertz, and Szymon Torunczyk. On the number of types in sparse graphs. In Anuj Dawar and Erich Grädel, editors, *Proceedings of the 33rd Annual ACM/IEEE Symposium on Logic in Computer Science, LICS 2018, Oxford, UK, July 09–12, 2018*, pages 799–808. ACM, 2018. doi:[10.1145/3209108.3209178](https://doi.org/10.1145/3209108.3209178).
- [PZ78] Klaus-Peter Podewski and Martin Ziegler. Stable graphs. *Fundamenta Mathematicae*, 100(2):101–107, 1978.
- [RS84] Neil Robertson and P.D Seymour. Graph minors. iii. planar tree-width. *Journal of Combinatorial Theory, Series B*, 36(1):49–64, 1984. doi:[https://doi.org/10.1016/0095-8956\(84\)90013-3](https://doi.org/10.1016/0095-8956(84)90013-3).
- [RT08] David Richerby and Dimitrios M. Thilikos. Searching for a visible, lazy fugitive. In Hajo Broersma, Thomas Erlebach, Tom Friedetzky, and Daniël Paulusma, editors, *Graph-Theoretic Concepts in Computer Science, 34th International Workshop, WG 2008, Durham, UK, June 30 - July 2, 2008. Revised Papers*, volume 5344 of *Lecture Notes in Computer Science*, pages 348–359, 2008. doi:[10.1007/978-3-540-92248-3\\_31](https://doi.org/10.1007/978-3-540-92248-3_31).
- [Sau72] N Sauer. On the density of families of sets. *Journal of Combinatorial Theory, Series A*, 13(1):145–147, 1972. doi:[https://doi.org/10.1016/0097-3165\(72\)90019-2](https://doi.org/10.1016/0097-3165(72)90019-2).
- [See96] Detlef Seese. Linear time computable problems and first-order descriptions. *Mathematical Structures in Computer Science*, 6(6):505–526, 1996. doi:[10.1017/S0960129500070079](https://doi.org/10.1017/S0960129500070079).
- [She72] Saharon Shelah. A combinatorial problem; stability and order for models and theories in infinitary languages. *Pacific Journal of Mathematics*, 41(1):247–261, 1972. doi:[10.2307/2371432](https://doi.org/10.2307/2371432).
- [She86] Saharon Shelah. *Monadic logic: Hanf Numbers*, pages 203–223. Springer Berlin Heidelberg, Berlin, Heidelberg, 1986. doi:[10.1007/BFb0098511](https://doi.org/10.1007/BFb0098511).
- [Sim90] Imre Simon. Factorization forests of finite height. *Theoretical Computer Science*, 72(1):65–94, 1990. doi:[https://doi.org/10.1016/0304-3975\(90\)90047-L](https://doi.org/10.1016/0304-3975(90)90047-L).
- [SS71] Saharon Shelah and Saharon Shelah. Stability, the f.c.p., and superstability; model theoretic properties of formulas in first order theory. *Annals of Mathematical Logic*, 3:271–362, 1971.
- [ST93] P.D Seymour and R. Thomas. Graph searching and a min-max theorem for tree-width. *Journal of Combinatorial Theory, Series B*, 58(1):22–33, 1993. doi:<https://doi.org/10.1006/jctb.1993.1027>.
- [Tor23] Szymon Torunczyk. Flip-width: Cops and robber on dense graphs. *CoRR*, abs/2302.00352, 2023. arXiv:2302.00352.
- [war16] Algorithms, logic and structure workshop in warwick, open problem session. [https://warwick.ac.uk/fac/sci/maths/people/staff/daniel\\_kral/alglogstr/openproblems.pdf](https://warwick.ac.uk/fac/sci/maths/people/staff/daniel_kral/alglogstr/openproblems.pdf), 2016. [Online; accessed 23-Jan-2023].
- [Zhu09] Xuding Zhu. Colouring graphs with bounded generalized colouring number. *Discrete Mathematics*, 309(18):5562–5568, 2009.